

anko

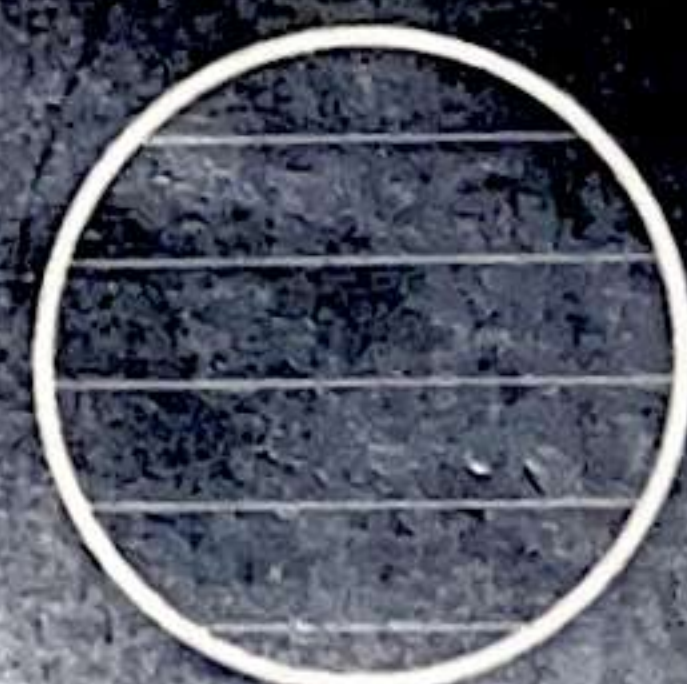
Methods
十一年級 笔记本

Name Wing yee
Subject Yr 11 Method Notebook
Year _____ School _____

120

Page

A4 Spiral Notebook



7mm
RULES

Linear Equation

Standard form:

$$Ax + By = C$$

Slope-intercept form:

$$y = mx + c \quad \text{y-intercept}$$

$$\text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-slope form:

$$y - y_1 = m(x - x_1)$$

Sketch linear equation:

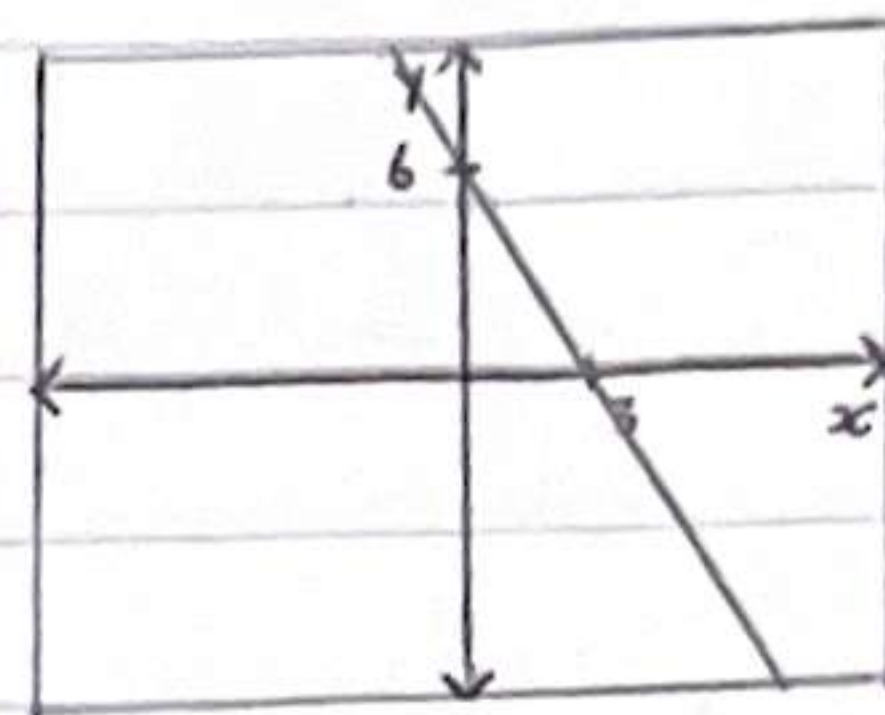
Example: $2x + y = 6$

$\uparrow$ x-in: set $y=0$, $2x+0=6$

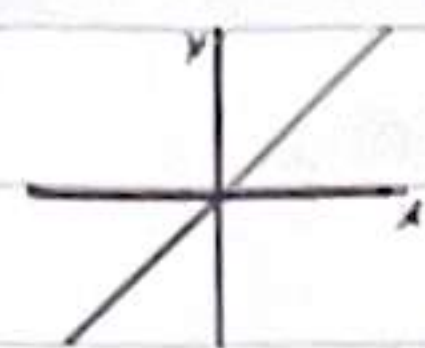
$$x=3, (3,0)$$

$\uparrow$ y-in: set $x=0$, $2x+y=6$

$$y=6, (0,6)$$



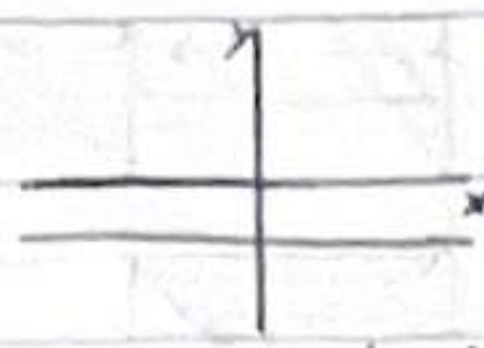
∞



positive gradient



negative gradient



zero gradient



undefined gradient

Form equation:

Example: a line pass two points $(4, 8)$ and $(3, 2)$, determine the line's equation

$$\text{Find gradient: } \frac{2-8}{3-4} = 6$$

sub one point into the eqn to find c $y = mx + c$

$$8 = 6 \times 4 + c$$

$$c = -16$$

summarise known info: $y = 6x - 16$

Distance between two points

$$\text{Distance} = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$$

Midpoint

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Parallel lines

If AB is parallel to CD, $AB \parallel CD$, then they would:

- Have the same gradient
- Never cross

Collinear points ~~to see a line straight or not~~

If two lines have the same gradient and share a point, they're co-linear.

Perpendicular Lines

- Two lines are perpendicular if and only if: $m_1 \times m_2 = -1$

∞ A perpendicular bisector is a line that cross another line at right angles and cuts the line into two equal lengths.

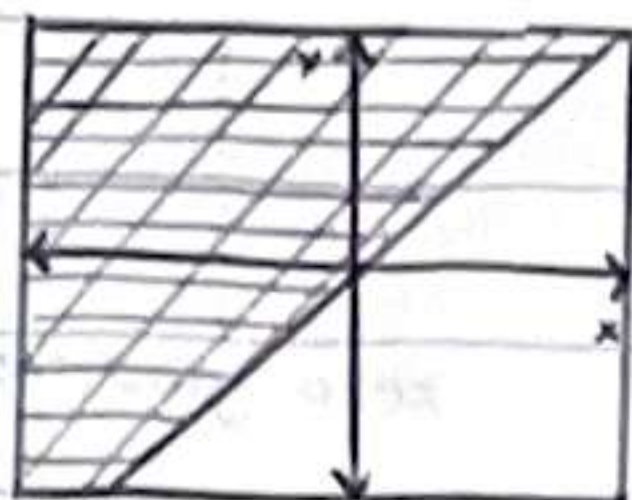
Linear Inequations

Graphing

Example:

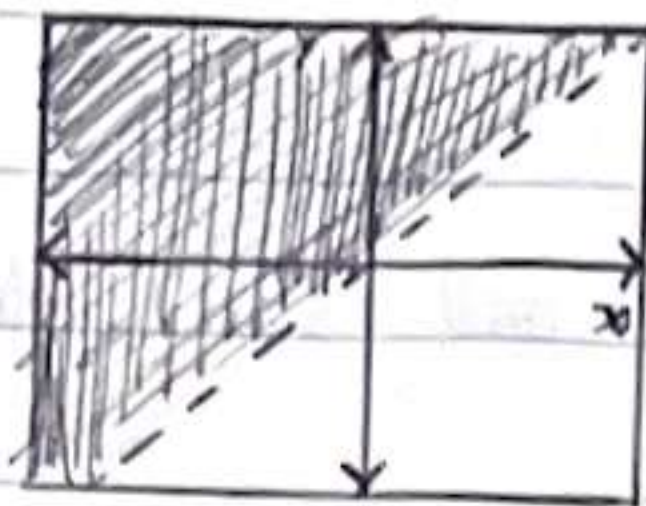
$$y \geq 2x$$

因为有等号, 所以是实线



$$y > 2x$$

虚线



Indices

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{m \times n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(ab)^n = a^n b^n$$

$$a^0 = 1$$

$$x^{-n} = \frac{1}{x^n} \text{ 反推 } \frac{1}{x^n} = x^{-n} \text{ example: } \frac{1}{3^4} = 3^{-4}$$

$$x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

$$\sqrt{x} = x^{\frac{1}{2}}$$

Calculator tips:

$$\text{example: } 3^n = 9, \text{ find } n$$

$$\log(9) \div \log(3) = 2, n=2$$

Example question

$$\textcircled{1} \frac{3^4 \times 4^2}{12^3} = \frac{3^4 \times 4^2}{(3 \times 4)^3} = \frac{3^4 \times 4^2}{3^3 \times 4^3}$$

$$\textcircled{b} \frac{8p^{-3}}{7p^{-5}} = \frac{8p^5}{7p^3} = \frac{8p^2}{7}$$

$$= \frac{3}{4}$$

$$\textcircled{2} x^{\frac{1}{3}} \div x^{\frac{1}{4}} = x^{\frac{1}{3} - \frac{1}{4}} = x^{\frac{1}{12}}$$

$$\textcircled{3} 4^x - 4^{x-1} = 48, \text{ solve for } x$$

$$4^x - 4^x \times 4^{-1} = 48$$

$$\text{let } y = 4^x$$

$$y - y \times 4^{-1} = 48$$

$$y - \frac{y}{4} = 48$$

$$y = 64$$

$$64 = 4^x$$

$$x = 3$$

Simplify

$$\textcircled{a} \frac{3c^{-5}}{7c^{-3}} = 3c^{-5} \times \frac{1}{7c^{-3}}$$

$$\frac{1}{c^{-3}} = c^3 \text{ as } x^{-n} = \frac{1}{x^n}$$

$$3c^{-5} \times c^3 \times \frac{1}{7} = \frac{3c^{-2}}{7} = \frac{1}{7c^2}$$

Surd

Features

- A surd is an irrational number
- Numbers that are not surds can be simplified to rational numbers. For example: $\sqrt{9} = 3$

Factorize

$$a\sqrt{c} + b\sqrt{c} = (a+b)\sqrt{c}$$

$$a\sqrt{c} - b\sqrt{c} = (a-b)\sqrt{c}$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

$$\cancel{a\sqrt{c} + a\sqrt{d}} = \cancel{a^2} \quad \frac{\sqrt{c}}{\sqrt{d}} = \sqrt{\frac{c}{d}}$$

$$\sqrt{c} \times \sqrt{c} = c$$

Expansions

$$\sqrt{a}(\sqrt{b} + \sqrt{c}) = \sqrt{ab} + \sqrt{ac}$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{c} + \sqrt{d}) = \sqrt{ac} + \sqrt{ad} + \sqrt{bc} + \sqrt{bd}$$

Rationalising denominators

$$\frac{a}{b\sqrt{c}} \times \frac{\sqrt{c}}{\sqrt{c}} = \frac{a\sqrt{c}}{b(\sqrt{c} \times \sqrt{c})} = \frac{a\sqrt{c}}{bc}$$

$$\frac{1}{\sqrt{a} + \sqrt{b}} \times \frac{(\sqrt{a} - \sqrt{b})}{(\sqrt{a} - \sqrt{b})} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$$

Example Questions ★:

① Simplify

$$(a) 3\sqrt{405}$$

$$= 3\sqrt{81} \times \sqrt{5}$$

$$= 27\sqrt{5}$$

$$(b) 5\sqrt{180x^3y^5}$$

$$= 5\sqrt{36 \times 5 \times x^2 \times x \times y^4 \times y}$$

$$= 5 \times \sqrt{36x^2y^4} \times \sqrt{5xy}$$

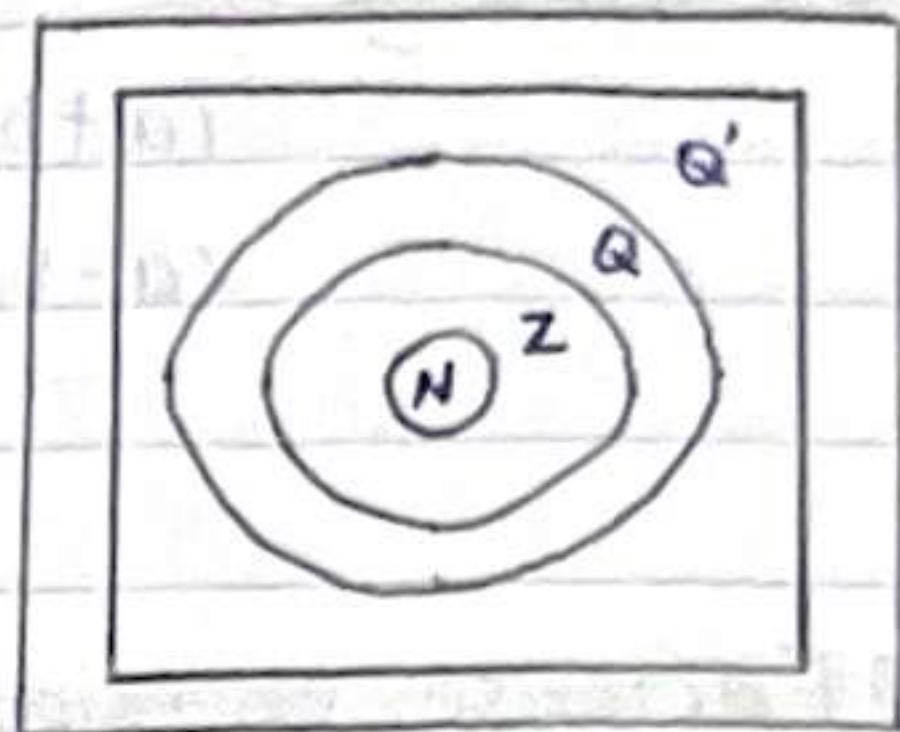
$$= 5 \times 6 \times xy^2 \times \sqrt{5xy}$$

$$= 30xy^2\sqrt{5xy}$$

Sets of real numbers

Types

Name	Symbol	Example
Natural numbers	N	$1, 2, 3, 4, \dots$
Integers	Z	$-3, -2, -1, 0, 1, \dots$
Rational numbers	Q	$-\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \dots$
Irrational numbers	Q'	π



$\infty \text{ and } Q' = \emptyset$: means they do not intersect

Set notation

$$A = \{1, 2, 3, 4, 5\} \quad B = \{1, 3, 5\}$$

- $2 \in A$ means 2 is an element of set A

- $2 \notin B$

- $B \subset A$ means B is a subset of set A

- $A \not\subset B$

- $A \cup B$ contains element either in A or B or Both

- $A \cap B$ contains element must be in both A & B

- $A \setminus B$ removes any element of B from A

Example questions ★

① Determine values of x

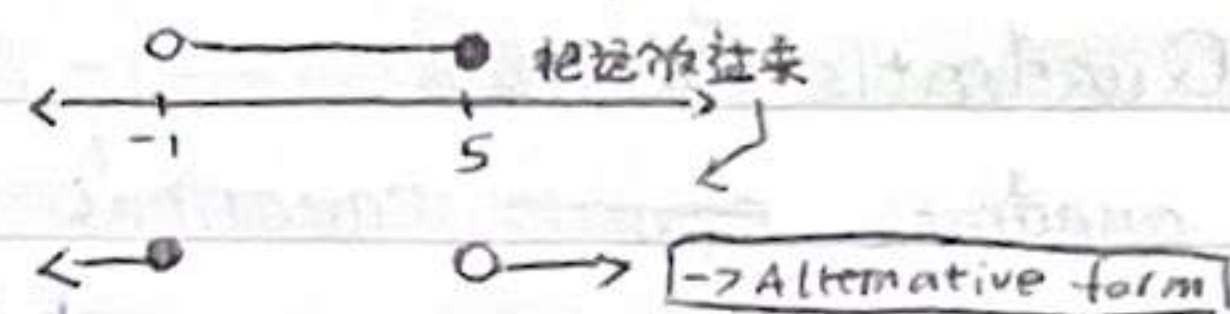
(a) $\frac{x+2}{x-2}$ 分母不能为0

$$x-2=0$$

$$x \neq 2 \text{ 所以 } x \text{ 不能是 } 2$$

$$\text{ans: } R \setminus \{2\}$$

② Use an alternative form of notation to describe the following set: $R \setminus (-1, 5]$



$$(-\infty, -1] \cup (5, \infty)$$

(b) $\frac{4}{x^2-4x}$

$$x(x-4)$$

$$x \neq 0$$

$$x \neq 4$$

$$R \setminus \{0, 4\}$$

Algebraic Expression

Expanding

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$(a+b)(a-b) = a^2 - b^2$$

$$1004x^3 - 3x^3 = x^3$$

Factorise expressions with 3 terms

CAS : 1. go to Main $\sqrt{a}$

2. Type in the expression

3. Highlight (choose) ~~your~~ your expression

4. Interactive $\rightarrow$ Transformation $\rightarrow$ Factor $\rightarrow$ Factor

Solve Quadratic equation

Example

$$\textcircled{1} x(x+3)(5x-2) = 0$$

$$x+3=0$$

$$5x-2=0$$

$$x=0$$

$$\text{Ans: } 0, -3, \frac{2}{5}$$

$$\textcircled{2} 15x - 12x^2 = 0$$

$$3x(5-4x) = 0$$

$$3x = 0$$

$$5-4x=0$$

$$\text{Ans: } 0, \frac{5}{4}$$

The Quadratic Formula

For quadratic equations: $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant Δ

$$- \Delta = b^2 - 4ac$$

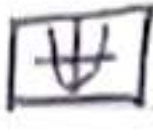
- $\Delta = 0$, only one rational solution

- $\Delta < 0$, no real solution

- $\Delta > 0$ $\rightarrow$ if Δ is a perfect square, the solutions are rational

$\rightarrow$ if Δ is not a perfect square, the solutions are irrational.

0 CAS

- Type equation $\rightarrow$ select it $\rightarrow$ Interactive $\rightarrow$ Advanced $\rightarrow$ solve $\rightarrow$ select 'solve'
- $\rightarrow$ Equation: the equa $\rightarrow$ Variable: the unknown variable $\rightarrow$ 'OK'
- Solve graphically on CAS
 - ① Menu $\rightarrow$ Graph Tables
 - ② Type the equation
 - ③ Select the equa by tick the box
 - ④ Click the graph icon on left-top  click like this

0 Example Question

① Zero of $x = -5$

zero of $x = 0$

* Zero of $x = -5$

$(x+5)$ is a factor

* Zero of $x = 0$

x is a factor

* $x(x+5)$

* Expand: $x^2 + 5x$

② Solve $9x^4 + 17x^2 - 2 = 0$

* $x^2 = a$, $9a^2 + 17a - 2 = 0$

* $(9a-1)(a+2)$

$x^2 = \frac{1}{9}$, $x^2 = -2$

* $x^2 = \frac{1}{9}$ or $x^2 = -2$, no real solu so reject

* $x = \pm \sqrt{\frac{1}{9}}$

$= \pm \sqrt{\frac{1}{3}}$

③ use 'completing the square' to factorise where possible, the following over $\mathbb{R}$

$x^2 - 2x + 6$

$(x-1)^2 + 5$ 是加号不能

The sum of two squares cannot

be factorised over $\mathbb{R}$.

④ Without actually solving the equations, determine the number and the nature of the roots of the following equations.

$12x - 1 - 36x^2 = 0$

$-36x^2 + 12x - 1$

$12^2 - 4 \times (-1) \times (-36)$

$144 - 144 = 0$

$\Delta = 0$, one real solution, one rational root.

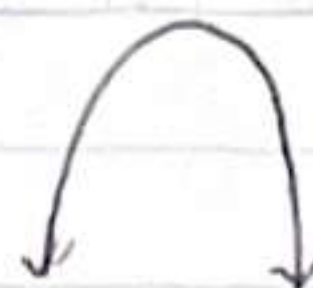
Parabolas 抛物线

Definition

- The graph of all quadratic relationships are called parabola
- Parabolas are symmetrical; they have an axis of symmetry 对称轴. 对称轴 equation
- Axis of symmetry always equal to x -value of the Turning point $x = -\frac{b}{2a}$
- Parabola shapes:



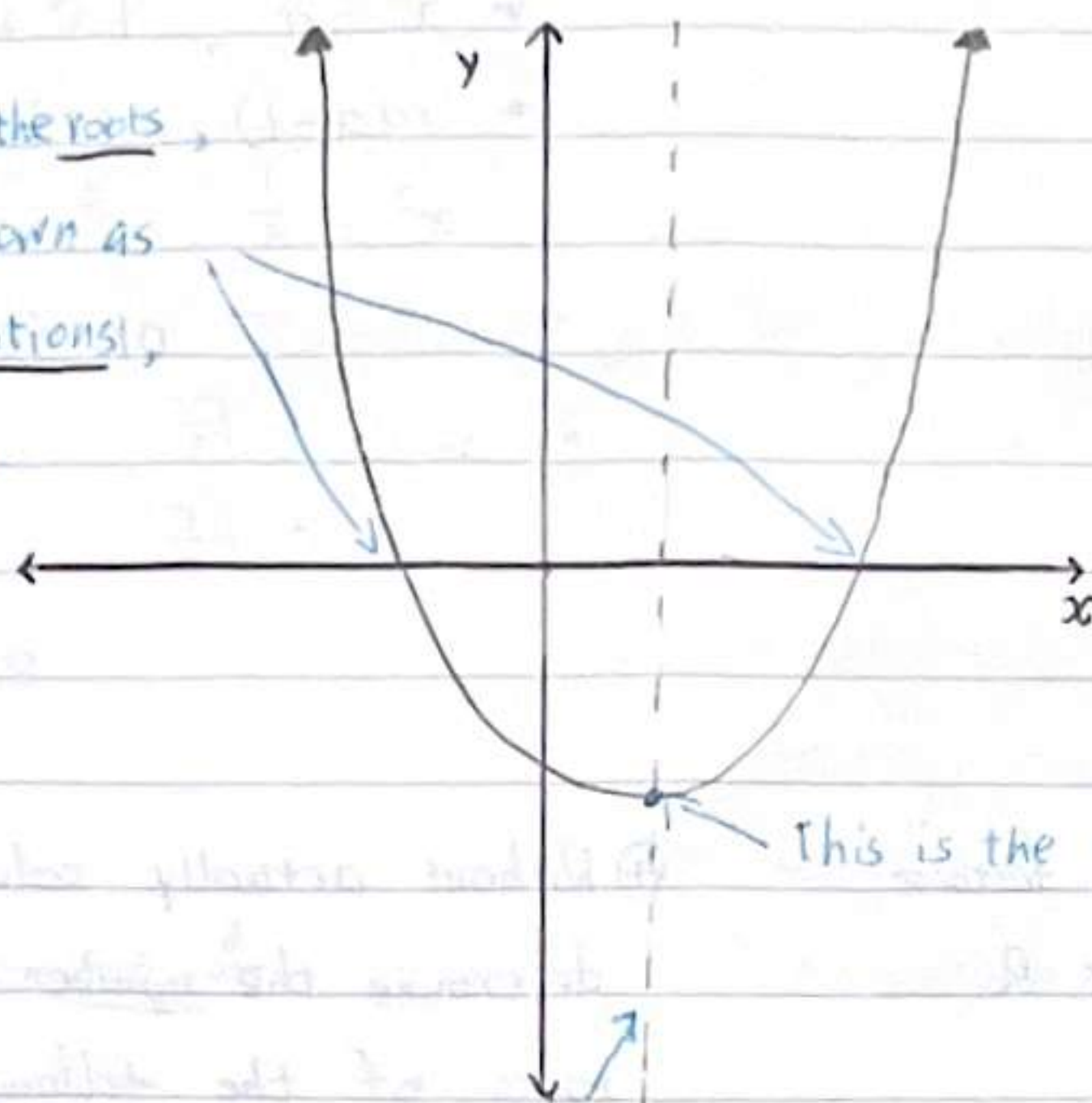
- To be 'concave up'
- have a minimum tp



- said to be 'concave down'
- have a maximum tp

Features

These are the roots,
they also known as
zeros, solutions,
x-intercepts



TP form:

$$y = a(x-h)^2 + k$$

This is the vertex, [TP] (h, k)

This is called the axis of symmetry

Sketch

- ① Find the x -intercepts (factorise)
- ② Find the y -intercepts
- ③ Find the TP
- ④ Plot the points and link them and label the coordinates & equation

Example: ① $y = (x-5)(x-2)$

$$x\text{-in: } (5, 0) (2, 0)$$

$$y\text{-in: } (0-5)(0-2) = 10 \quad (0, 10)$$

$$x \text{ of TP: } \frac{5+2}{2} = 3.5$$

$$y \text{ of TP: } (3.5-5)(3.5-2) = -2.25$$

$$\text{TP } (3.5, -2.25)$$

② $y = x^2 + 4x + 2$

complete the square
change to TP form: $x^2 + 4x + 4 - 4 + 2$
 $= (x+2)^2 - 2$

$$\text{TP } (-2, -2)$$

$$x\text{-in: } \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \approx -0.59, -3.41$$

$$y\text{-in: } y = 0^2 + 4 \times 0 + 2 = 2 \quad (0, 2)$$

Changes from basic quadratic equa: $y=x^2$

o Dilation

$$y = a(x-h)^2 + k$$

- when a is greater than 1, it become narrower
- when a is less than 1 and greater than 0, it become wider
- when a is less than 0, it become reflection

- Example:

$$y = 2x^2$$

narrower

Min TP (0,0)

$$y = -2x^2$$

narrower (dilation)

Max TP, reflected TP (0,0)

o Translation

o Vertical translation

$$y = a(x-h)^2 + k$$

- when k is positive, it go up
- when k is negative, it go down
- Example:

$$y = x^2 + 2$$

vertical translation of 2 units up

TP (0, 2)

$$y = x^2 - 2$$

vertical translation of 2 units down

TP (0, -2)

o Horizontal translation

$$y = a(x-h)^2 + k$$

- when h is positive, it go right
- when h is negative, it go left
- Example:

$$y = (x-2)^2$$

move 2 units right

TP (2, 0)

$$y = (x+2)^2$$

move 2 units left

TP (-2, 0)

Determine the equa for a parabola (quadratic polynomial)

1. general form $y = ax^2 + bx + c$ ← use when any 3 points is given
2. Turning point form $y = a(x-h)^2 + k$ ← use when TP is given
3. factor form $y = a(x-x_1)(x-x_2)$ ← use when x -intercepts are given

Example Questions ★

① Express in expanded polynomial form

TP $(-2, 0)$ Pass $(2, 2)$

$$y = a(x-h)^2 + k$$

$$2 = a(2+2)^2$$

$$a = \frac{1}{8}$$

$$y = \frac{1}{8}(x+2)^2$$

$$\text{Expand } y = \frac{1}{8}x^2 + \frac{1}{2}x + \frac{1}{2}$$

③ Find equation, Pass $(1, -4)$ $(-1, 10)$ $(3, -2)$

$$y = ax^2 + bx + c$$

$$\textcircled{1} (1, -4) \quad -4 = a + b + c$$

$$\textcircled{2} (-1, 10) \quad 10 = a - b + c$$

$$\textcircled{3} (3, -2) \quad -2 = 9a + 3b + c$$

$$\textcircled{2} - \textcircled{1} : 14 = -2b$$

$$b = -7$$

$$\textcircled{3} - \textcircled{1} : 2 = 8a + 2b$$

$$a = 2$$

$$c = 1$$

$$\text{Equa: } y = 2x^2 - 7x + 1$$

② The graph is the same shape as $y = 3x^2$
vertex at $(-1, -5)$. Express in $y = ax^2 + bx + c$

$$a = 3 \quad \text{TP } (-1, -5)$$

$$y = a(x-h)^2 + k$$

$$y = 3(x+1)^2 - 5$$

$$y = 3(x^2 + 2x + 1) - 5$$

$$y = 3x^2 + 6x - 2$$

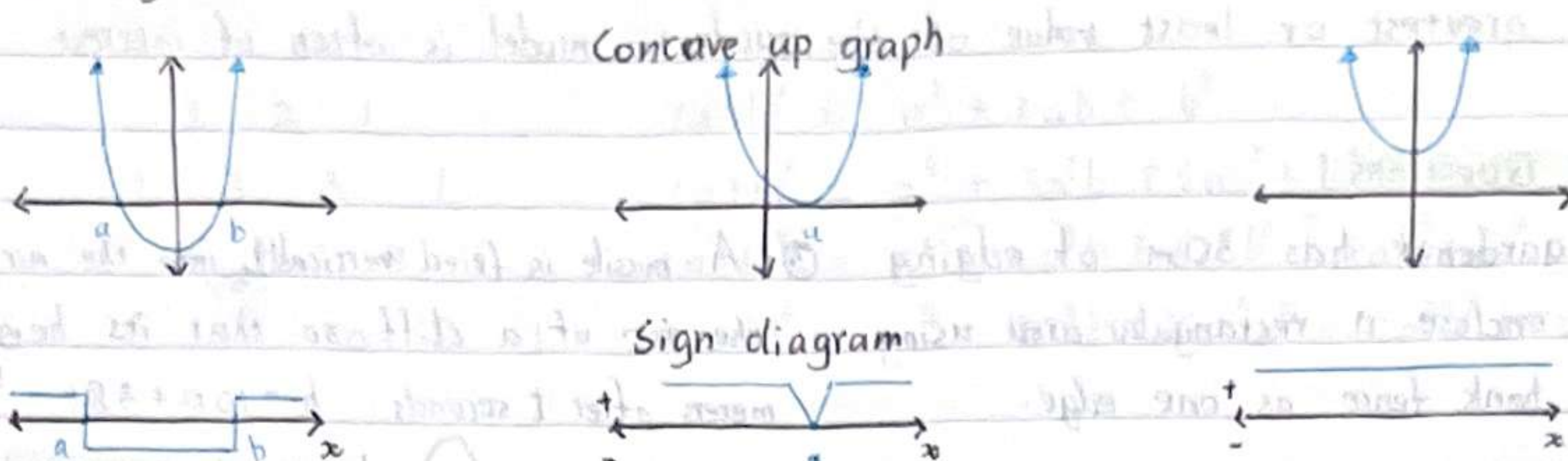
④ x -in: $(3, 0)$ $(8, 0)$ Find Equa

$$y = a(x-x_1)(x-x_2)$$

$$y = (x-3)(x-8) \quad \text{如果有别的 point 套进去找 } a$$

Quadratic Inequalities

0 Sign diagrams



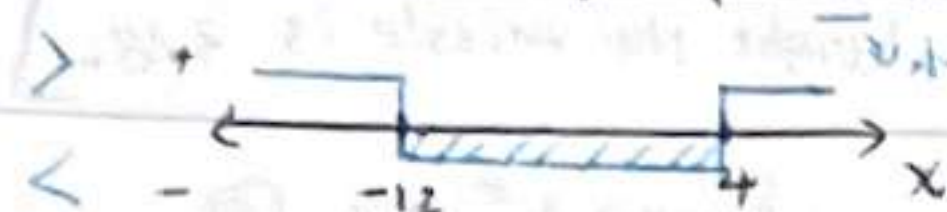
1 Example Questions

① solve $x^2 + 8x - 48 \leq 8$

$(\frac{x}{2})^2 = 16$

$(x+4)^2 - 8^2$

$(x+12)(x-4) \leq 0$



② solve pairs of simultaneous equations

$y = 5x + 2$ and $y = x^2 - 4$

$5x + 2 = x^2 - 4$

$(x+1)(x-6) = 0$ $x = -1, 6$

for $y: 6^2 - 4 = 32, -1^2 - 4 = -3$

so $x = 6, y = 32$ or $x = -1, y = -3$

③ Find the set

(a) $\{36 - 12x + x^2 > 0\}$

$(x-6)^2 > 0$



$x \in \mathbb{R} \setminus \{6\}$ 因为没等号

The solution set is $\mathbb{R} \setminus \{6\}$

④ Show line $y = 4x$ is a tangent to the parabola $y = x^2 + 4$

$4x = x^2 + 4$

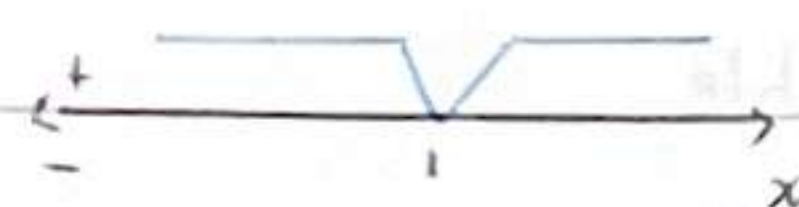
$(x-2)^2 = 0$ $x = 2$

for $y, 2^2 + 4 = 8$

when $x = 2, y = 8$ thus the point of touch is $(2, 8)$

(b) $\{x: 6x^2 - 12x + 6 \leq 0\}$

$6(x-1)^2 \leq 0$ $x = 1$



The solution set is $\{1\}$

(c) $\{x(1-10x) \leq 21\}$

$(5x-7)(2x+3)$ $x = \frac{7}{5}, -\frac{3}{2}$



⑤ For what value of m will there be at least one intersection between the line

$y = mx - 7$ and $y = 3x^2 + 6x + 5$

$3x^2 + 6x + 5 = mx - 7$

$= 3x^2 + x(6-m) + 12$

$\Delta = (6-m)^2 - 4(3)(12)$

$= (6-m)^2 - 144$

$\Delta \geq 0$

if $m \leq 6$ or $m \geq 18$, there will be at least one intersection

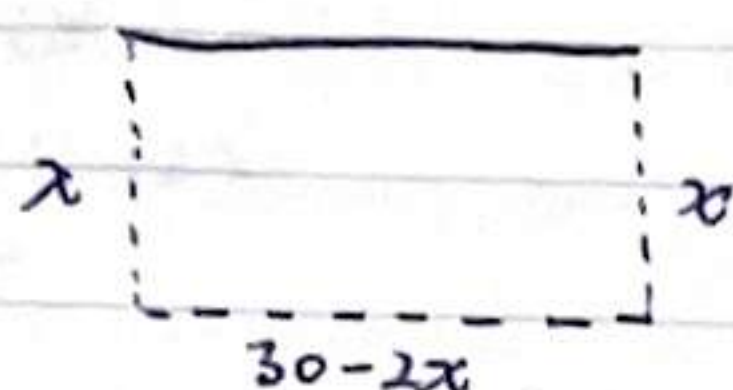
Quadratic models & Applications

Maximum and Minimum values

- The greatest or least value of the quadratic model is often of interest

Example Questions

- ① A gardener has 30m of edging to enclose a rectangular area using the bank fence as one edge.



- (a) Show $\text{Area} = 30x - 2x^2$ where x is the width.

$$A = \text{width} \times \text{length} = (30-2x)x = 30x - 2x^2$$

- (b) Calculate the dimensions of the garden bed for maximum area.

$$\text{TP: } x = \frac{15+0}{2} = 7.5$$

$$y: A = 7.5(30-15) = 112.5 \quad \text{TP}(7.5, 112.5)$$

Ans: Dimensions of garden for maximum area are width 7.5m, length 15m.

- (c) what is the maximum area that can be enclosed?

$$112.5 \text{ m}^2$$

- ③ A missile is fired vertically into the air from the top of a cliff so that its height h meters after t seconds, $h = 100 + 38t - \frac{19}{12}t^2$

what is the greatest height?

$$a = -\frac{19}{12}, b = 38, c = 100$$

$a < 0$, so the y -coordinate of TP is the greatest ~~value~~ height.

$$t = -\frac{b}{2a} = \frac{-38}{2 \times (-\frac{19}{12})} = 12$$

when $t = 12$

$$h = 100 + 38 \times 12 - \frac{19}{12} \times 12^2 = 328 \text{ m}$$

Ans: The greatest height the missile is 328m

- ④ Ball: $10h = 16t + 4 - 9t^2$

will the ball strike the foliage overhanging from a tree if the foliage is 1.6m vertically above the ball's point of projection?

$$\text{let } h = 1.6$$

$$9t^2 - 16t + 12 = 0$$

$$\Delta = -176 \quad \Delta < 0, \text{ no value for } t \text{ when } h = 1.6$$

The ball does not strike the overhanging foliage.

- ⑤ If sum of two number is 16, find the number for which their product is greatest.

$$x + y = 16$$

$$y = 16 - x \quad * P: \text{product} = xy$$

$$= x(16-x)$$

$$x = 0, 16$$

$$\text{TP: } x = \frac{16}{2} = 8 \quad y = 8 \times (16-8) = 64$$

$$(8, 64)$$

Ans: The product is

$$y: 16 - 8 = 8 \quad (8, 8)$$

greatest when both numbers are 8.

SKETCH on CAS

- Graph & Table

- Type equation

- $\boxed{\Psi}$

- 最上面 - Analysis - G-solve

TP: Max/Min y -in: y -intercept

Pascal's triangle

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1 \\
 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

$$\begin{aligned}
 (a+b)^2 &= a^2 + 2ab + b^2 \\
 (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
 (a+b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \\
 (a-b)^5 &= a^5 - 5a^4b + 10a^3b^2 - 10a^2b^3 + 5ab^4 - b^5
 \end{aligned}$$

Factorise

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

Example:

① $x^3 - 125$

$$x^3 - 5^3$$

$$(x-5)(x^2 + 5x + 25)$$

② $3(x^3 + 1)$

$$3(x+1)(x^2 - x + 1)$$

③ $27x^3 + 64y^3$

$$(3x)^3 + (4y)^3$$

$$(3x+4y)(9x^2 - 12xy + 16y^2)$$

Expand

~~$$(a+b)^3 = (a-b)(a^2 + ab + b^2)$$~~

~~$$(a-b)^3 =$$~~

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

refer to 4.2

Polynomials

Definition

- A polynomial is an algebraic expression in which the power of the variable is a positive whole number. Example: $3x^2 + 5x - 1$, $\sqrt{3}x + 5$

Features

- The degree of a polynomial is the highest power of the variable.
- The leading term is the term that containing the highest power of the variable.
- The constant term is the term that does not contain the variable.
- Example: $5x^3 + 2x^2 - 3x + 4$

Degree: 3

Coefficient of the leading term: 5

Constant term: 4

Type of coefficients: This is a polynomial over $\mathbb{Z}$, because the coefficients in polynomial are integers.

Polynomial notation

Name of polynomials variable

$$P(x) = 5x^3 + 2x^2 - 3x + 4$$

read as "P of x"

Two identical polynomials

- If two polynomials are identically equal, then the coefficients of like terms are equal.
- Example: Calculate values of a, b, c , so that $x(x-7) = a(x-1)^2 + b(x-1) + c$

$$x^2 - 7x = a(x^2 - 2x + 1) + bx - b + c$$

$$= ax^2 - 2ax + a + bx - b + c$$

$$= ax^2 + bx - 2ax + a - b + c$$

$$= ax^2 + (b-2a)x + a-b+c$$

$$x^2 = ax^2$$

$$-7x = (b-2a)x$$

$$0 = a - b + c$$

$$a = 1$$

$$b = -5$$

$$c = -b$$

Devision of polynomials

∞ CAS Active: standard mode, Action/Transformation/Fraction/PropFrac

$$\text{type } (x^3 - 4x^2 + 2x - 3) \div (2x - 5)$$

$$\text{you will get } \frac{x^2}{2} - \frac{3x}{4} - \frac{59}{8(2x-5)} - \frac{7}{8}$$

Quotient

$\hookrightarrow -\frac{59}{8} \div (2x-5)$ remainder $\hookrightarrow$ quotient

∞ Devision Example Question

$$\textcircled{1} \frac{x+7}{x+5}$$

$$= \frac{(x+5)+2}{x+5}$$

$$= 1 + \frac{2}{x+5}$$

Remainder: 2

Quotient: 1

$$\textcircled{2} \frac{x^2+x+5}{x-2} \rightarrow \text{quotient}$$

$$x-2 \overline{) x^3 - x^2 + 3x - 5}$$

$$\begin{array}{r} x^3 - 2x^2 \\ \hline x^2 + 3x \end{array}$$

$$- (x^2 - 2x)$$

$$- (5x - 10)$$

Remainder: 5

The Remainder and factor theorems

Factor theorem

if $P(a) = 0$ then $(x-a)$ is a factor of $P(x)$

Remainder theorem

if $P(x)$ is divided by $(x-a)$, then $P(a)$ is the remainder.

Example Questions

① Find the remainder when $P(x) = x^3 + 4x^2 - 3x + 5$ is divided by $(x+2)$

$$x = -2$$

$$P(-2) = (-2)^3 + 4(-2)^2 - 3(-2) + 5 = 19$$

The remainder is 19.

② When $x^2 - 4x - 9$ divided by $(x-3)$, remainder is

15. Find a .

$$P(x) = 15$$

$$P(3) = 15$$

把 $x=3$ 代入

$$(15 + 12 + 9) \div a = 4$$

③ Determine the polynomial $P(x) = 3x^3 + ax^2 + bx$ which have a remainder of -22 when divided by $(x+1)$ and is exactly divisible by $(x-1)$. Find a, b

$$P(-1) = -22$$

$$= a - b - 3$$

$$= a - b = -17$$

$$P(1) = a + b + 3 - 2 = 0$$

$$= a + b = -1$$

$$\text{①} + \text{③} : 2a = -18$$

$$a = -9, b = 8$$

$$P(x) = 3x^3 - 9x^2 + 8x - 2$$

④ Given $(x-4)$ is a factor of $x^3 - x^2 - 10x - 8$

$$\begin{array}{r} x^2 \\ x-4 \overline{) x^3 - x^2 - 10x - 8} \\ \underline{x^3 - 4x^2} \\ 0 \end{array}$$

$$(x^2 + 3x + 2)(x-4) = (x+2)(x+1)(x-4)$$

⑤ Factorise $P(x) = x^3 + 3x^2 - 13x - 15$

用 1, -1, 2, -2 ... 去代入 x 看什么时候 $= 0$

$$P(-1) = 0$$

$$\begin{array}{r} x^2 + 2x - 15 \\ x+1 \overline{) x^3 + 3x^2 - 13x - 15} \\ \underline{-(x^3 + x^2)} \\ 0 \end{array}$$

$$(x^2 + 2x - 15)(x+1) = (x+1)(x+5)(x-3)$$

⑥ USE CAS solve for x

$$x^3 - 13x^2 + 34x + 48$$

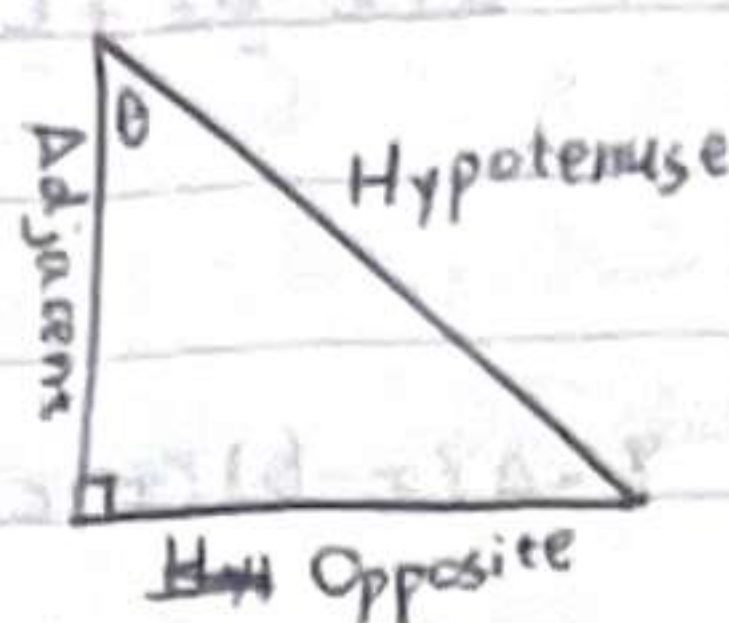
$$\text{Factor } (x^3 - 13x^2 + \dots) = (x+1)(x-6)(x-8)$$

$$x = -1, 6, 8$$

Trigonometry

Trigonometric Ratios

- SOH, sine, $\sin \theta = \frac{O}{H}$
- CAH, cosine, $\cos \theta = \frac{A}{H}$
- TOA, tangent, $\tan \theta = \frac{O}{A}$



Measurement

- 1 degree = 1 hour = 60 minutes
- 1 minute = 60 seconds
- $50^{\circ} 33' 48''$ means 50 degrees, 33 minute and 48 seconds
- 任一单位超30时, 四舍五入

Degrees to values on CAS

Example Questions: ① $\sin 30^{\circ}$ ② $\sin 40^{\circ} 30'$ ③ $\tan 30^{\circ} 40' 30''$

CAS Instructions: ① Set mode as 'Deg'

② Type Sin on Main screen

③ Interactive - Transformation - DMS - dms =

④ Input Degree, Minute, Second

⑤ 'OK'

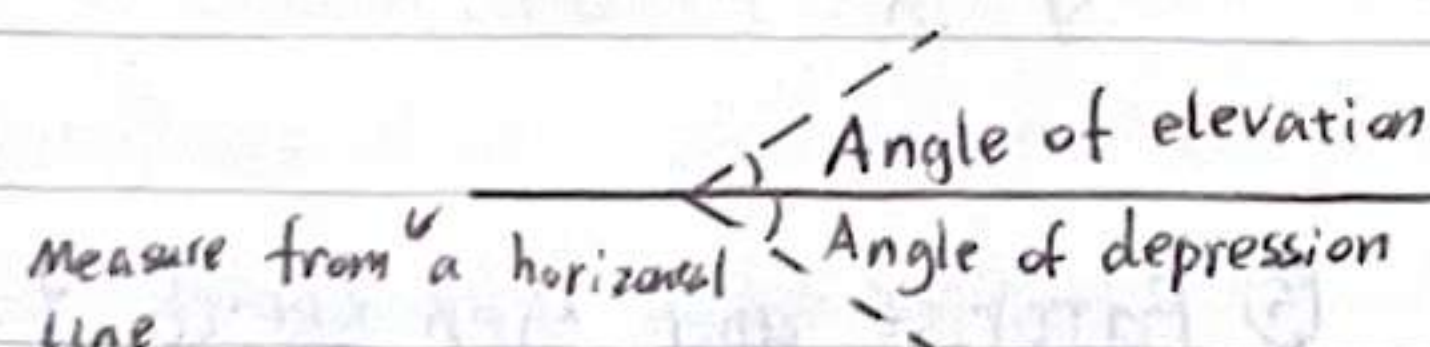
Value to Degree on CAS

CAS Instructions: ① Interactive - Transformation - DMS - toDMS =

② Type the value with ratio. Example: $\sin^{-1}(0.45)$

③ 'OK'

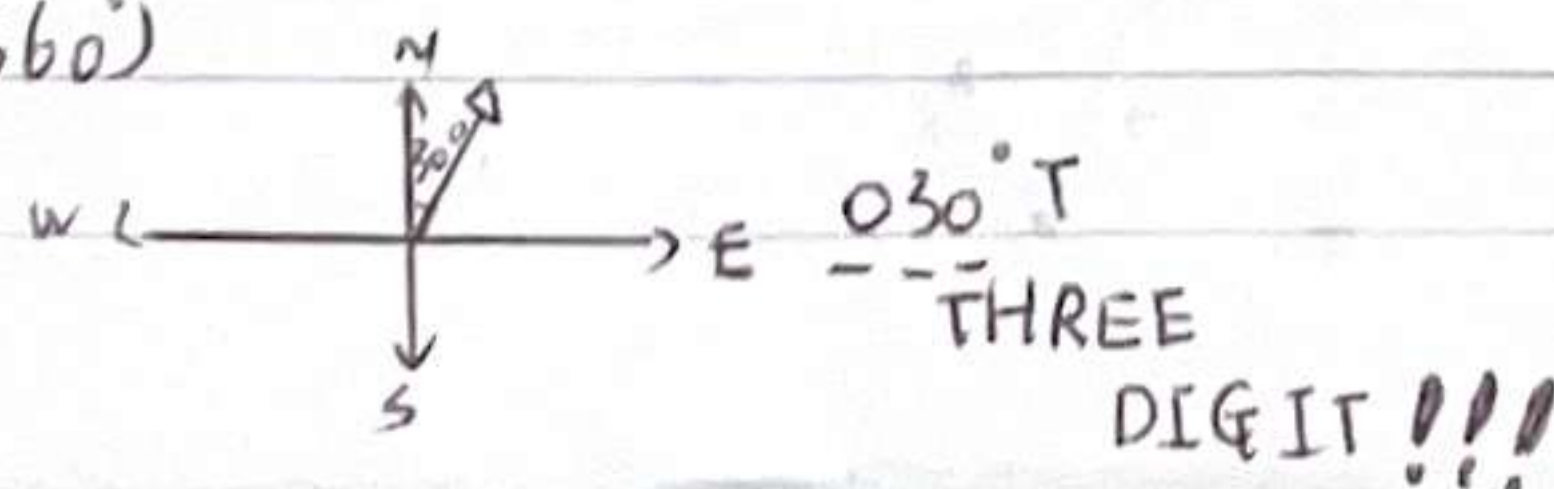
Angles of elevation and depression



Bearings

- Compass bearings. N 30 E *Always from N or S to W or E

- True bearings (0-360)



Cubic Graphs

* For basic form, power is odd like 3, 5 etc
求 x -in 时没有 $\pm$, 参考 (a)

Basic form: $y = a(x-b)^3 + c$

Positive (+)

Negative (-)

Factor form: $y = a(x-b)(x-c)(x-d)$

Positive (+)

Negative (-)

Sketch Question

① State the point of inflection

(a) $(x+5)^3 - 8 = y$

PI: $(-5, -8)$

(b) $y = -7x^3$

PI: $(0, 0)$

(c) $2 - (x-2)^3$

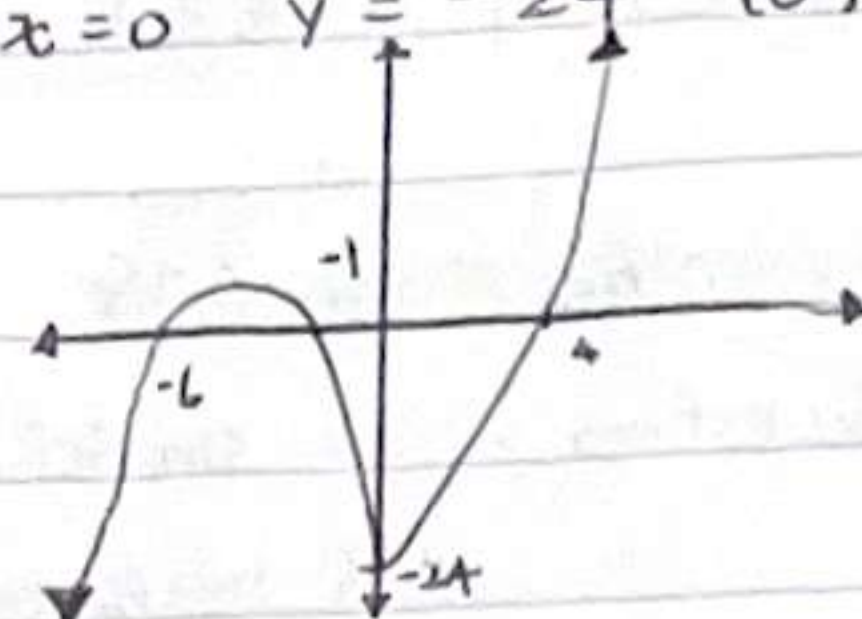
PI: $(2, 2)$

(c) $y = (x+1)(x+6)(x-4)$

Sketch without attempt to get PI.

x -in: $(-1, 0), (-6, 0), (4, 0)$

y -in: $x=0, y=-24, (0, -24)$



② Sketch

(a) $y = (x-1)^3 - 8$

PI: $(1, -8)$

x -in: $\frac{dy}{dx} = 0$

$(x-1)^3 - 8 = 0$

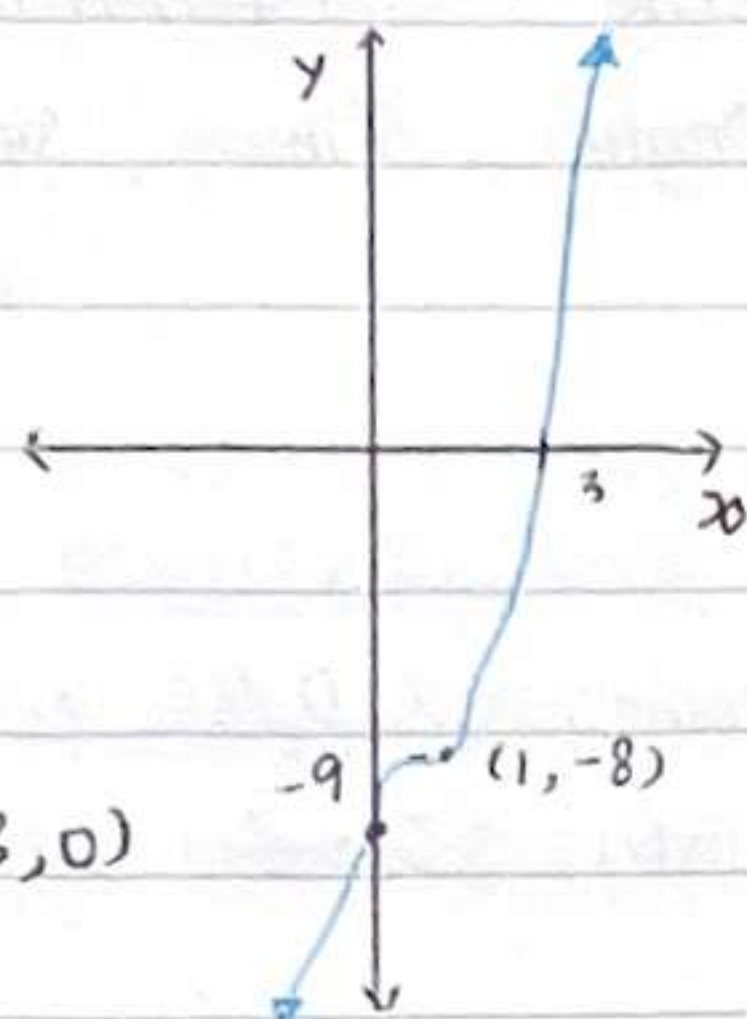
$(x-1)^3 = 8$

$x-1 = 2$

$x = 3, (3, 0)$

y -in: $\frac{dy}{dx} = 0$

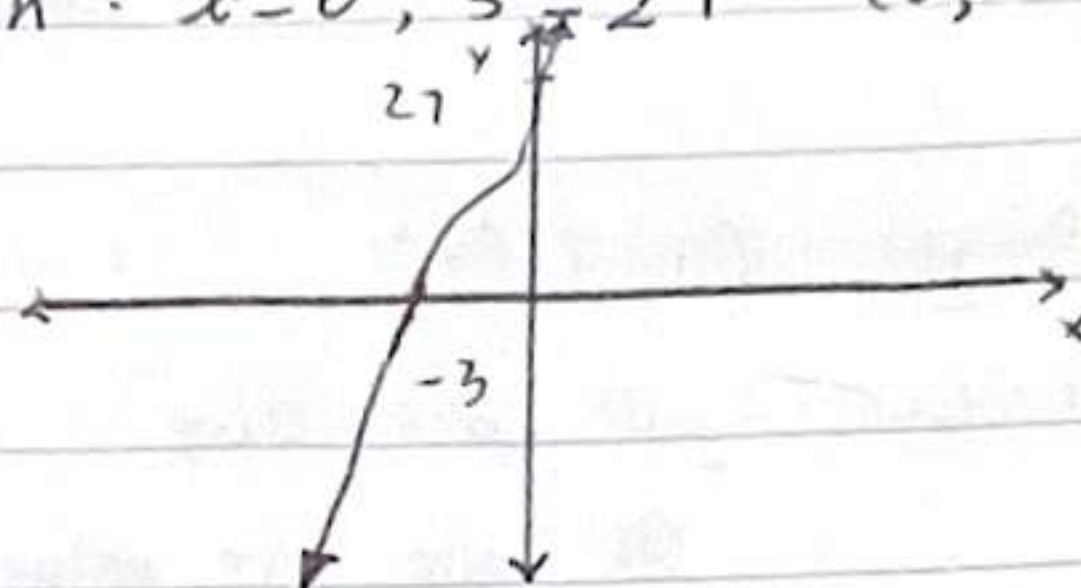
$-1 - 8 = -9, (0, -9)$



(d) $y = (x+3)^3$

x -in: $(-3, 0)$

y -in: $x=0, \frac{dy}{dx} = 27, (0, 27)$

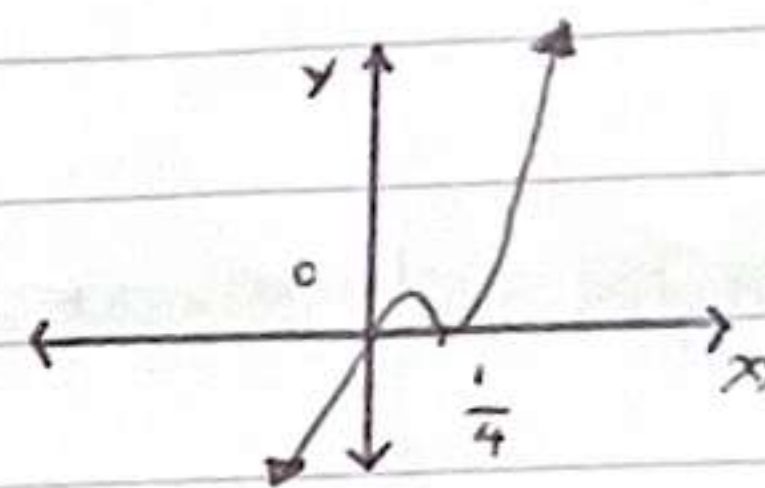


(e) $4y = x(4x-1)^2$

$y = \frac{x}{4}(4x-1)^2$

x -in: $(\frac{1}{4}, 0)$

y -in: $(0, 0)$



(b) $y = (3-x)^3 + 1$

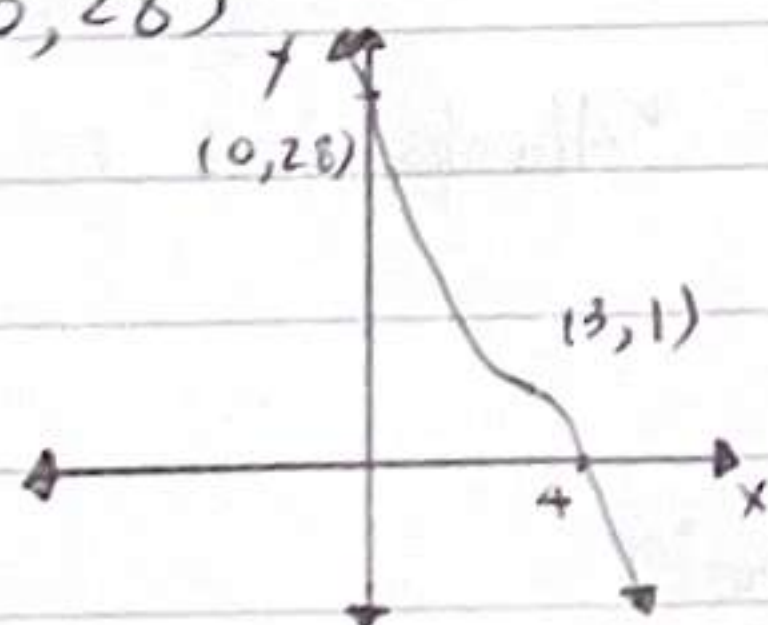
PI: $y = (-(x-3))^3 + 1, (3, 1)$

x -in: $y=0, (3-x)^3 = -1$

$3-x = \sqrt[3]{-1}$

$x = 4, (4, 0)$

y -in: $x=0, y=28, (0, 28)$

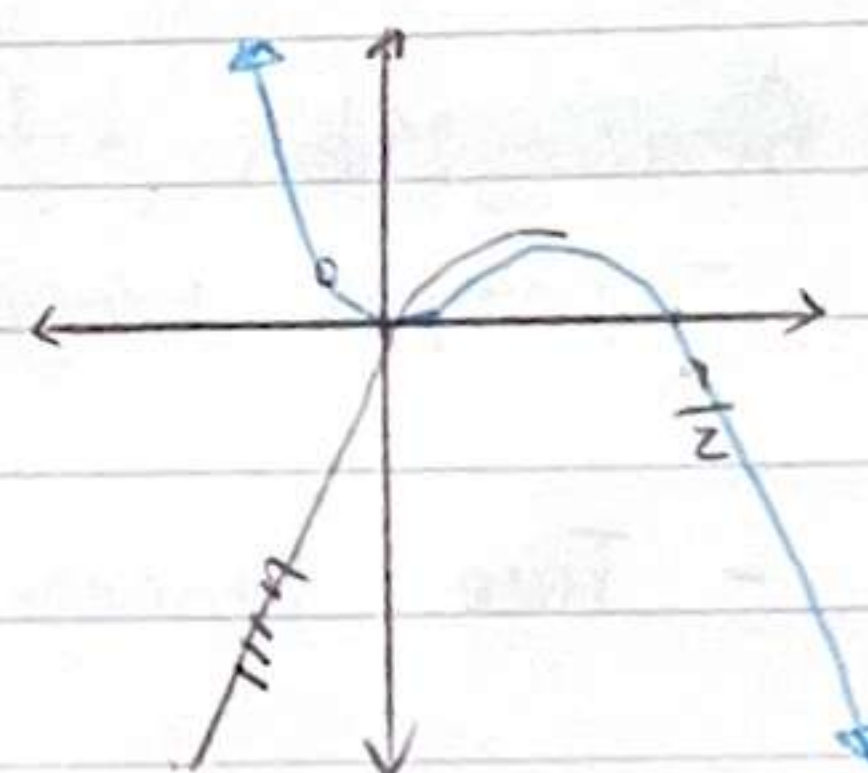


③ Factorise and then Sketch $y = 9x^2 - 2x^3$

$x^2(9-2x)$

x -in: $y=0, (\frac{9}{2}, 0)$

y -in: $x=0, (0, 0)$



Form equation and solve

① Determine the equation

(a) Point of inflection: $(3, -7)$

x -in: $(10, 0)$

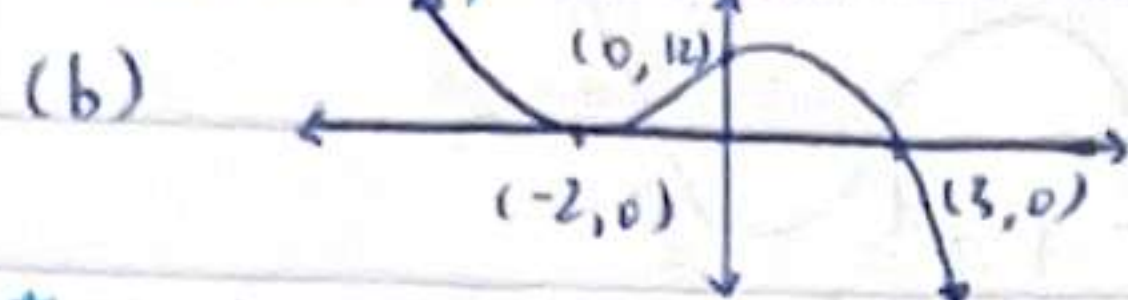
只有PT和另一个点时 $y = a(x-h)^3 + k$

套 (h, k) $y = a(x-3)^3 - 7$

套另一个点求 a $0 = a(10-3)^3 - 7$

$$a = \frac{1}{49}$$

In Summary: $y = \frac{1}{49}(x-3)^3 - 7$



有 touch x -in 的点
时

$$y = a(x-b)(x-c)(x-d)$$

$$y = a(x+2)^2(x-3)$$

套点求 a

$$12 = a(2)^2(-3)$$

$$a = -1$$

$$y = -(x+2)^2(x-3)$$

② Use sign diagram to solve the inequality

$$(a) (x-2)^2(6-x) > 0$$

x -in: $x = 2, 6$



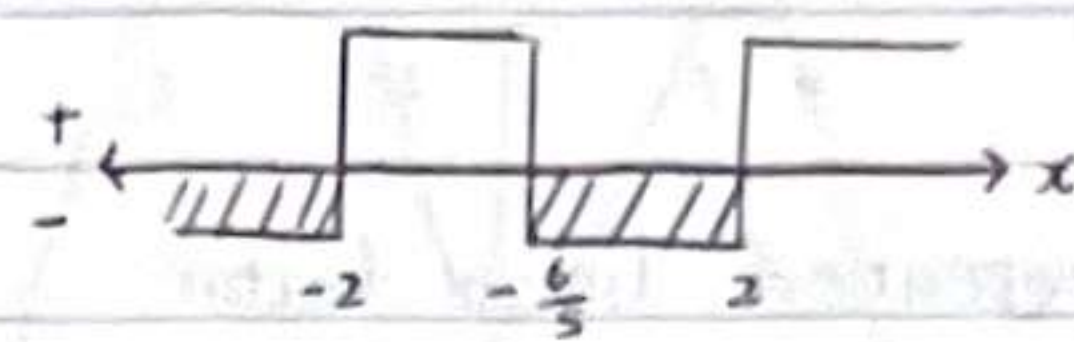
因为 $(x-b)^2$ 所以 just touch

Sign: $+$ $-$ $+$ $-$ $+$

(b) 写为: $x < 6, x \neq 2$

$$(b) 5x^3 + 6x^2 - 20x - 24 < 0, \text{ CAS Active}$$

Solve (equa, x) $x = -2, 2, -1, -2$

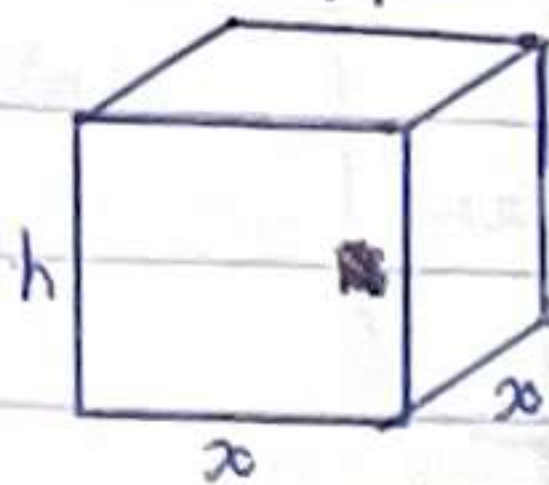


$$(b-x)(c-x)(d-x) = y$$

写为: $x < 2$ or $-\frac{6}{5} < x < 2$

Cubic Application

①



Total length of its 12 edges is 6 m.

(a) express h in terms of x

$$6 = 4h + 8x$$

$$h = \frac{3}{2} - 2x$$

(b) show $V_m^3 = 1.5x^2 - 2x^3$

$$(1.5 - 2x)x^2$$

$$= 1.5x^2 - 2x^3$$

(c) state restriction of x

x should not be negative so $x > 0$

Factorise $(1.5 - 2x)x^2$, $1.5 - 2x = 0$

套 x -in

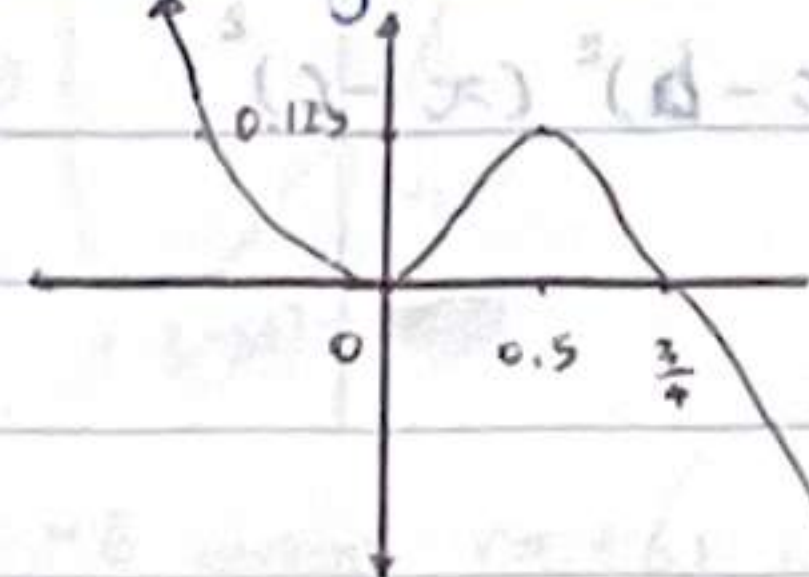
$$2x = 1.5$$

$$x = \frac{3}{4}, 0$$

$$x > 0, x < \frac{3}{4}$$

$$\text{Ans: } 0 < x < \frac{3}{4}$$

(d) Sketch given TP $(0.5, 0.125)$



$-2x^3$ is negative

(e) Find dimension with greatest volume.

As TP indicates, when $x=0.5$, we get greatest volume

Width: $x=0.5$ m

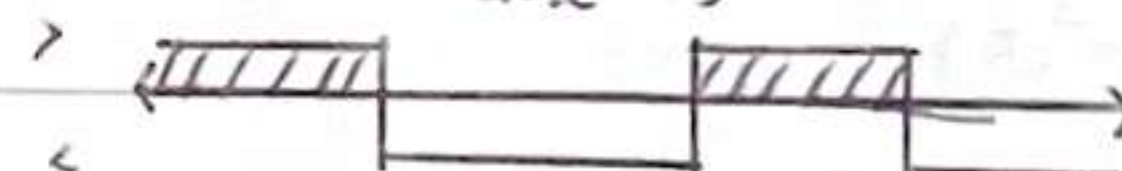
Length: $x=0.5$ m

Height: $1.5 - 1 = 0.5$ m

② $x(6-x)(2x-6)$, For what value of x is valid.

$\Rightarrow -2x^3$ is negative $\Rightarrow x = 0, 3, 6$

x 不能是 negative



refute

$$x < 0, x > 3, x < 6$$

$$3 < x < 6$$

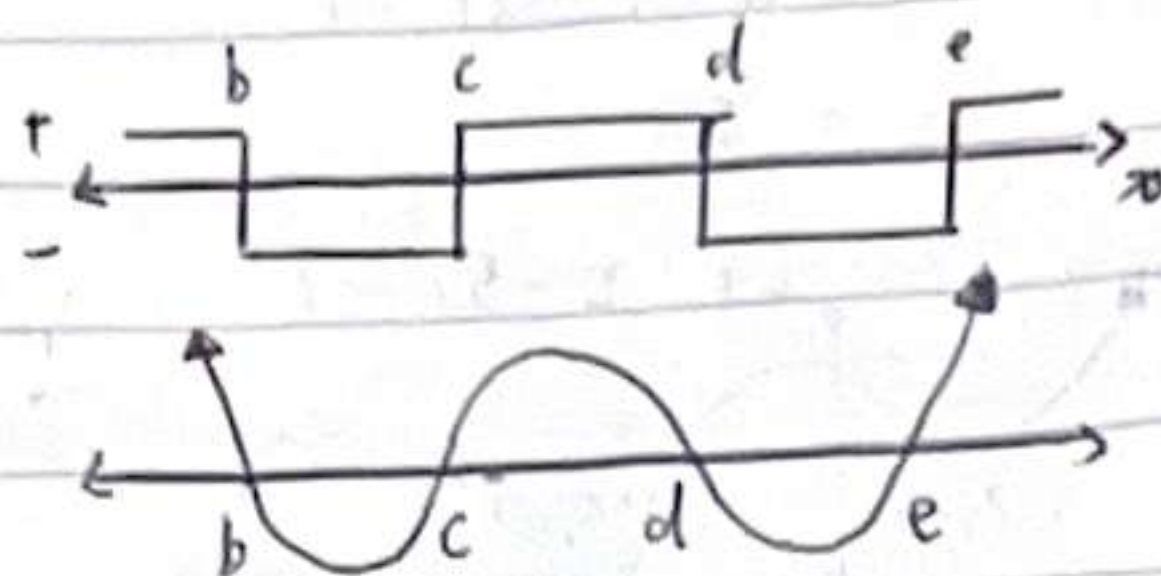
Quartic Polynomials

Definition

- A quartic polynomial may have up to 4 linear factors since it is of fourth degree.
The possible combinations of these linear factors are:

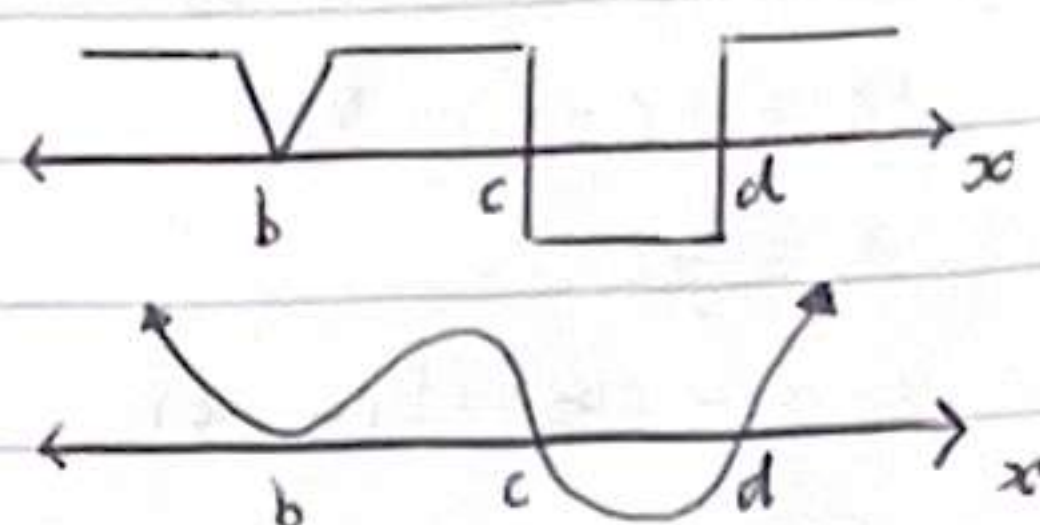
- Four distinct linear factors

$$y = a(x-b)(x-c)(x-d)(x-e)$$



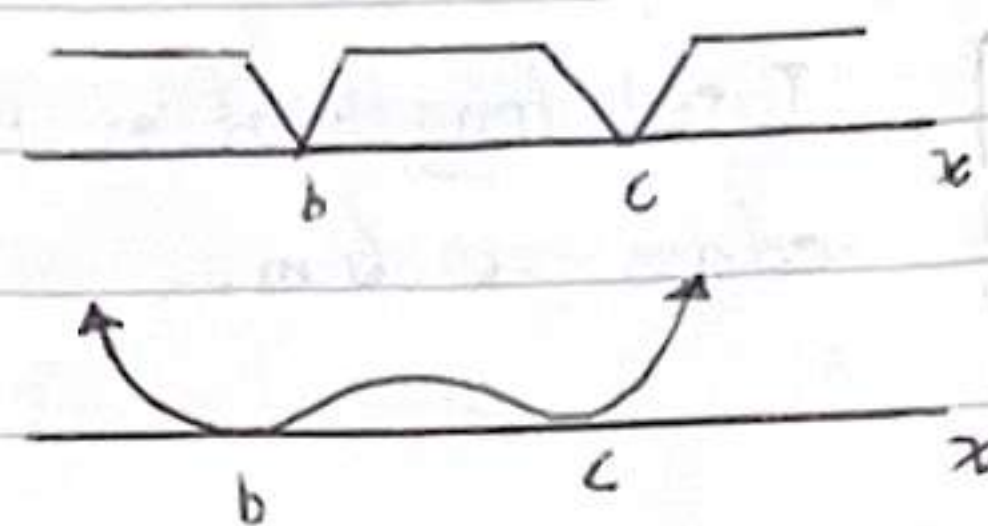
- One repeated linear factor

$$y = a(x-b)^2(x-c)(x-d)$$



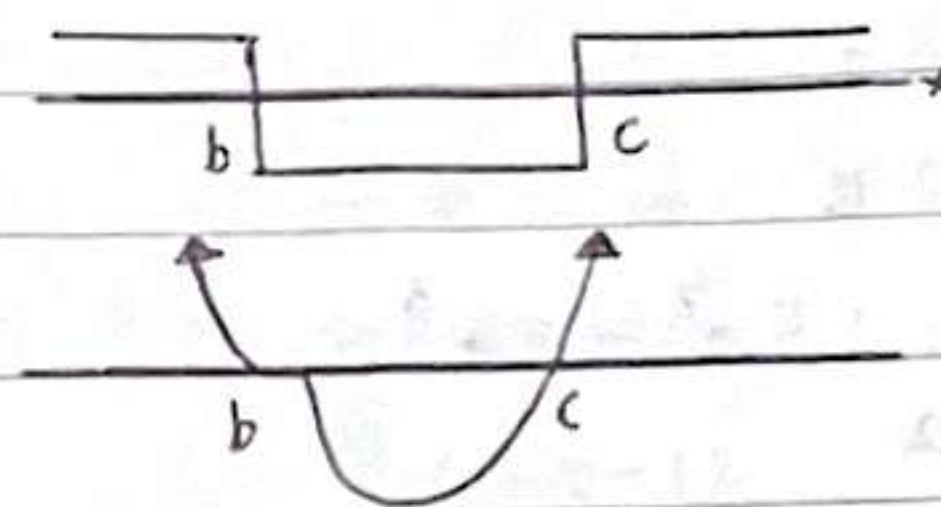
- Two repeated linear factors

$$y = a(x-b)^2(x-c)^2$$



- One factor of multiplicity 3

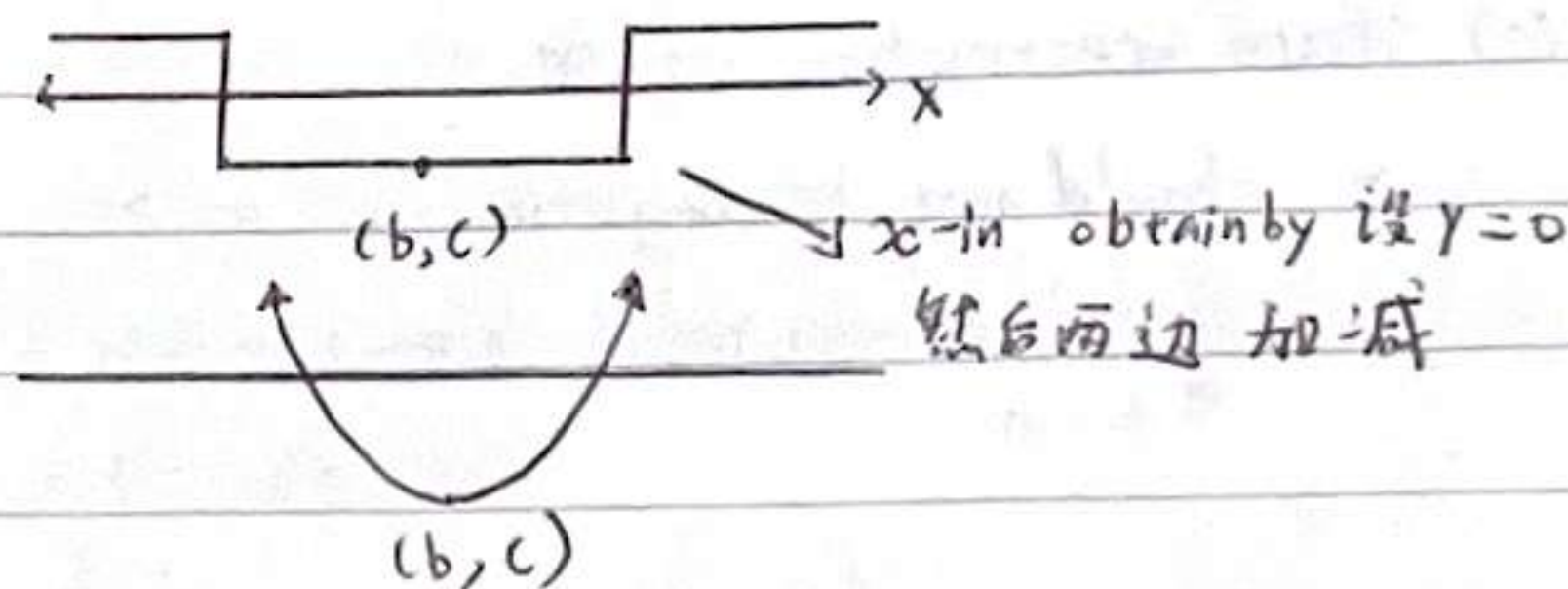
$$y = a(x-b)^3(x-c)$$



- One factor of multiplicity 4

$$y = a(x-b)^4 + c$$

This case in which the graph has a minimum TP at (b, c)



Sketch question

$$a(x-b)^4 + c$$

① State the nature of the TP, $y = \frac{1}{8}(x+2)^4 - 2$

This is a Minimum Turning Point since $a > 0$

② Sketch

(a) $y = (x-2)^4 - 1$

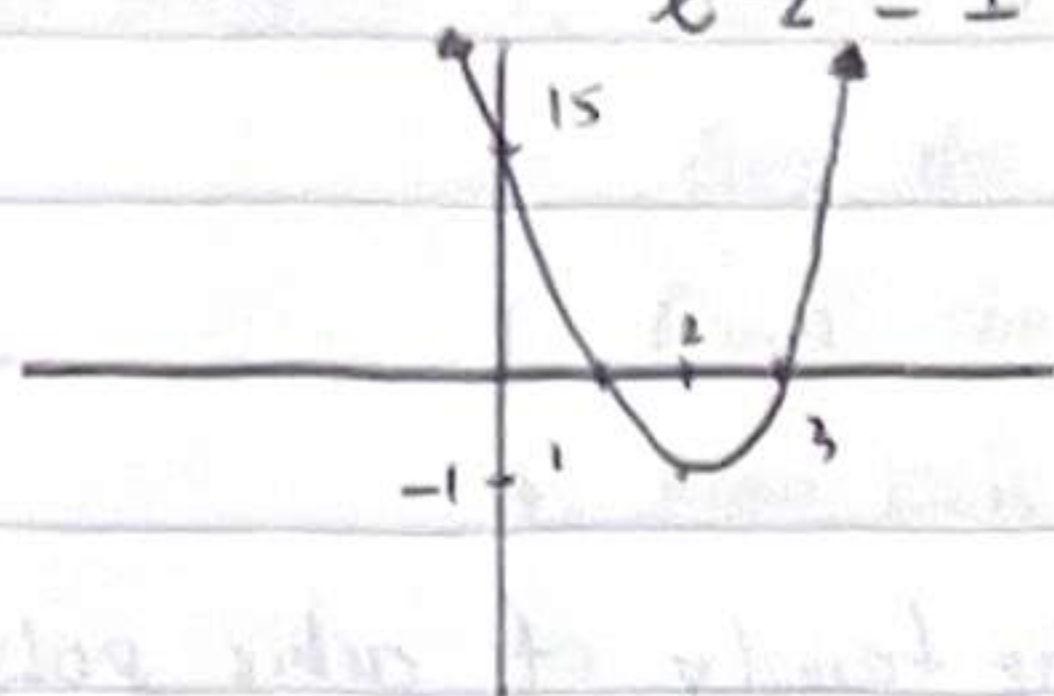
MinTP: $(2, -1)$

y-in: $x = 0, y = 15$ $(0, 15)$

x-in: $y = 0$

$$(x-2)^4 - 1 = 0$$

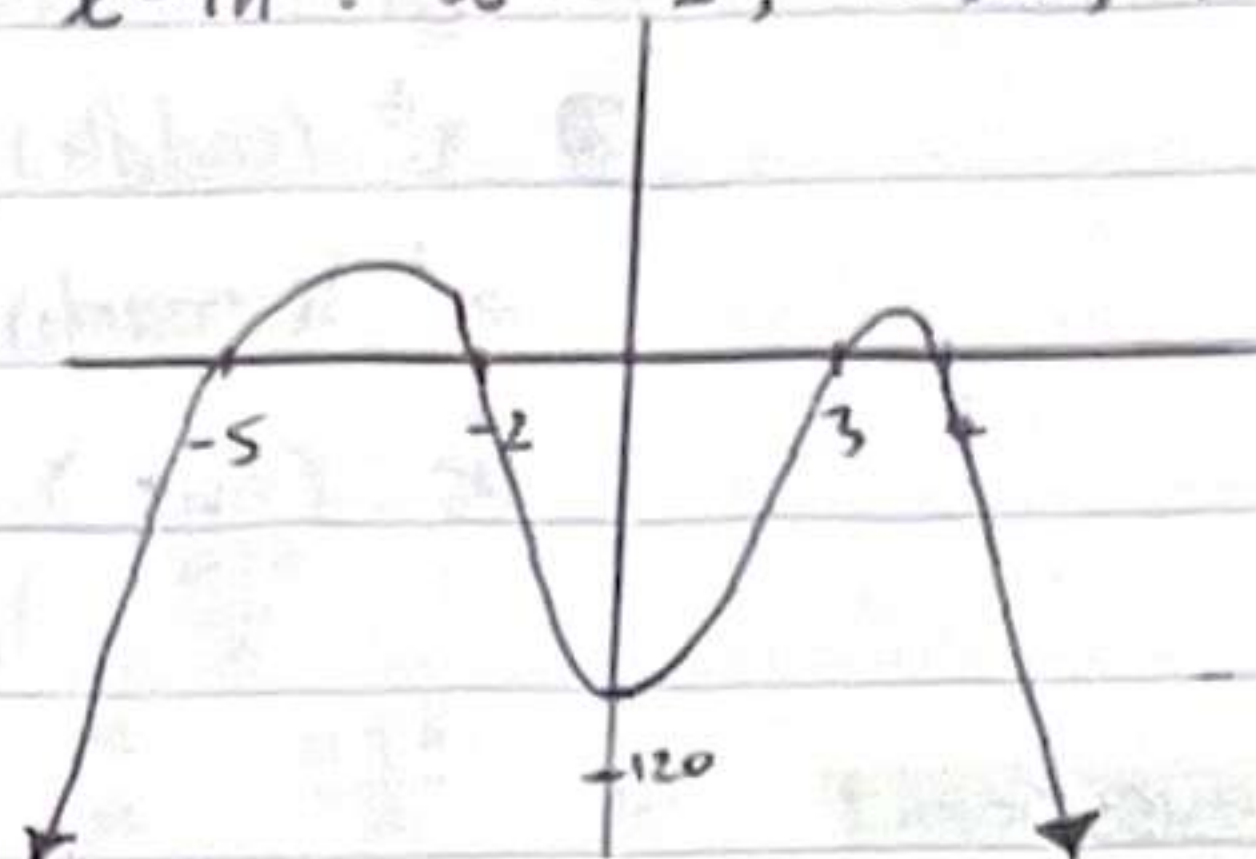
$$x-2 = \pm 1 \quad x = 1, 3$$



(b) $y = -(x+2)(x-3)(x-4)(x+5)$
 $-x^4$ is negative

y-in: $x = 0, y = -120$ $(0, -120)$

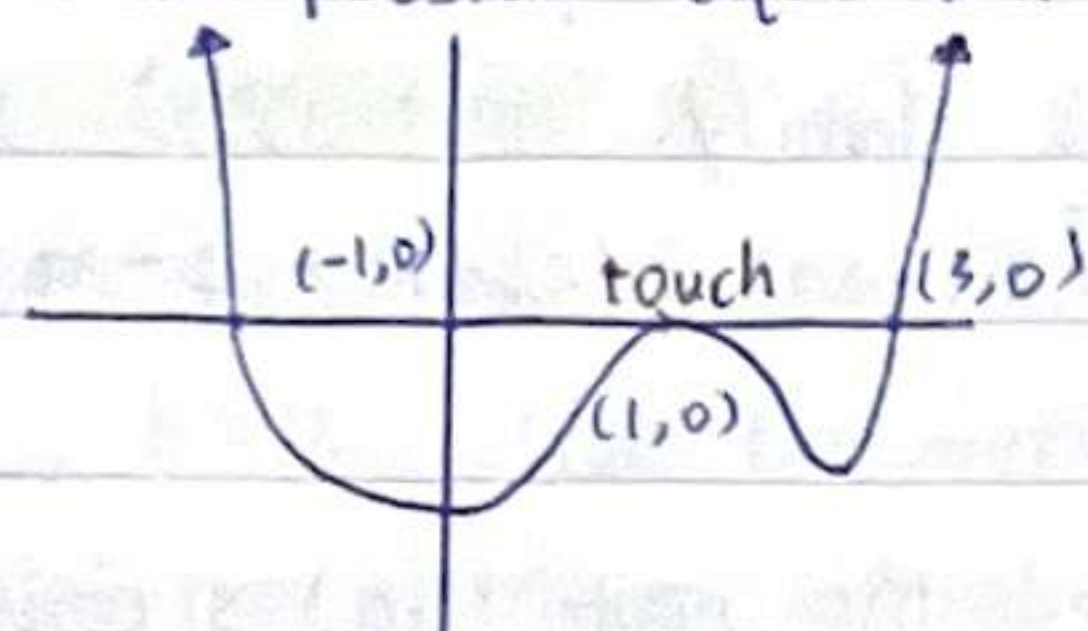
x-in: $x = -2, -5, 3, 4$



Form equation Question

① Form a possible equation

(a)



$$x = -1, 1, 3$$

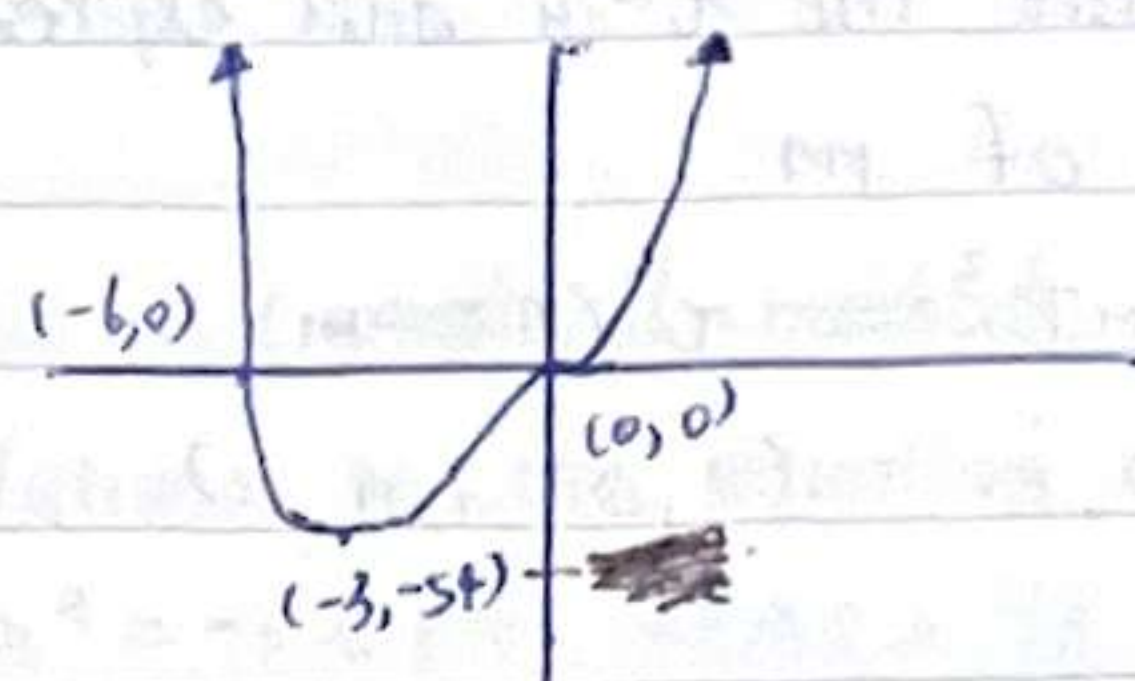
$x = 1$ (touch) means $(x-1)^2$ is a factor

$x = 3$ means $(x-3)$ is a factor

$x = -1$ means $(x+1)$ is a factor

$$y = a(x+1)(x-3)(x-1)^2$$

(b)



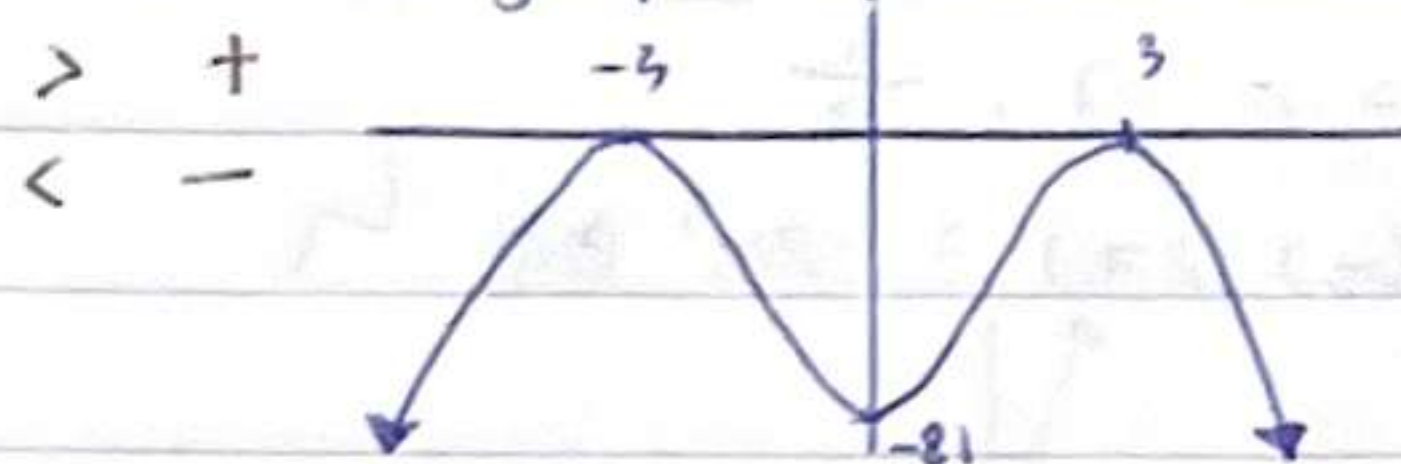
$x = -6$ means $(x+6)$ is a factor

$x = 0$ (saddle cut) means x^3 is a factor

$y = a(x+6)x^3$ 套进去找 a

Solve Inequation and other Question

① Use the graph to solve $\{x: -x^4 + 18x^2 - 81 > 0\}$



$$\{x: -x^4 + 18x^2 - 81 > 0\} = \emptyset$$

② Factorise $-x^4 + 18x^2 - 81$

$$\text{Let } x^2 = a, \quad = -(a^2 - 18a + 81)$$

$$= -(a-9)^2$$

$$= -(x^2-3)^2$$

$$= -(x+3)^2(x-3)^2$$

因式分解

Re 5.3 Families of polynomials

Sketch Tips: ① 画图前先看次方

负数单次方 [负单]	从上往下走	如: $-x^3$
负数双次方 [负双]	从下往上走	如: $-x^4$
正数单次方 [正单]	从下往上走	如: x^3
正数双次方 [正双]	从上往下走	如: x^4

② 如果次方是双数且 x 只出现了一次, 如: $y = (2x+1)^6 - 6$

那么图长得应该是 parabola $\cup$ 那样 或 $\cap$

求 TP: $x = -\frac{1}{2}$ 套进去找 y

求 y -in: 设 x 为 0

③ x^3 (saddle): touch and turn $\nmid$

x^2 (touch): only touch $\nmid$

x (cut): $\nmid$

5.3 example questions

① Consider the family of quartic polynomials defined by the equation $y = x^4 - mx^3$ where m is a real non-zero constant.

(a) calculate the x -in and express in terms of m

$$x^4 - mx^3 = x^3(x - m)$$

$$x = 0, m \quad (0, 0) \quad (m, 0)$$

② Consider the family of cubic polynomials for which $y = ax^3 + (3-2a)x^2 + (3a+1)x - 4-2a$ where $a \in \mathbb{R} \setminus \{0\}$

(a) show that the point $(1, 0)$ is common to all this family.

$$0 = a + (3-2a) + (3a+1) - 4-2a = 0$$

Therefore, the point $(1, 0)$ is common to all this family.

(b) For the member of the family which pass through the origin, form equation and sketch

Sub $(0, 0)$ to the equation for a :

$$0 = 0 + 0 + 0 - 4 - 2a$$

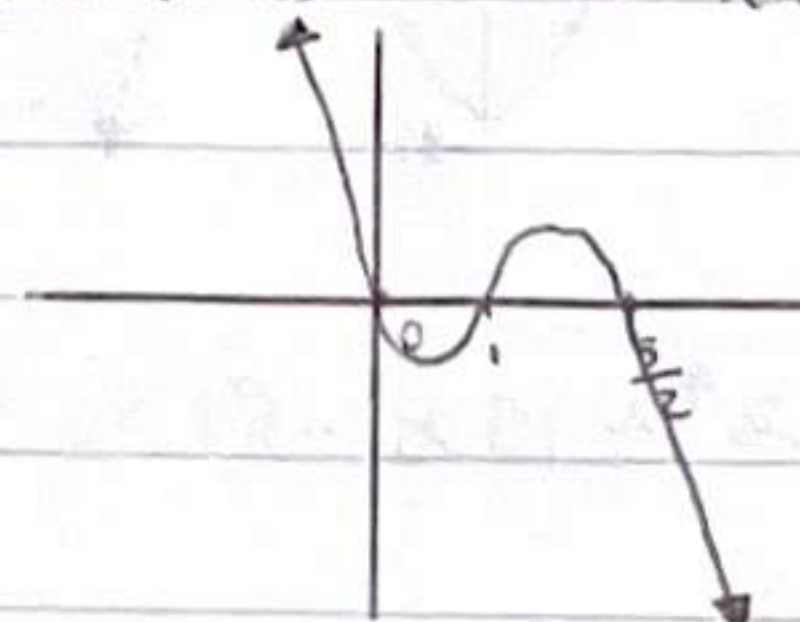
$$a = -2$$

∴ The equation now become: $y = -2x^3 + 7x^2 - 5x$

$$\text{Factorise: } y = -x(x-1)(2x-5)$$

$$(\text{cut}) x = 0, 1, \frac{5}{2}$$

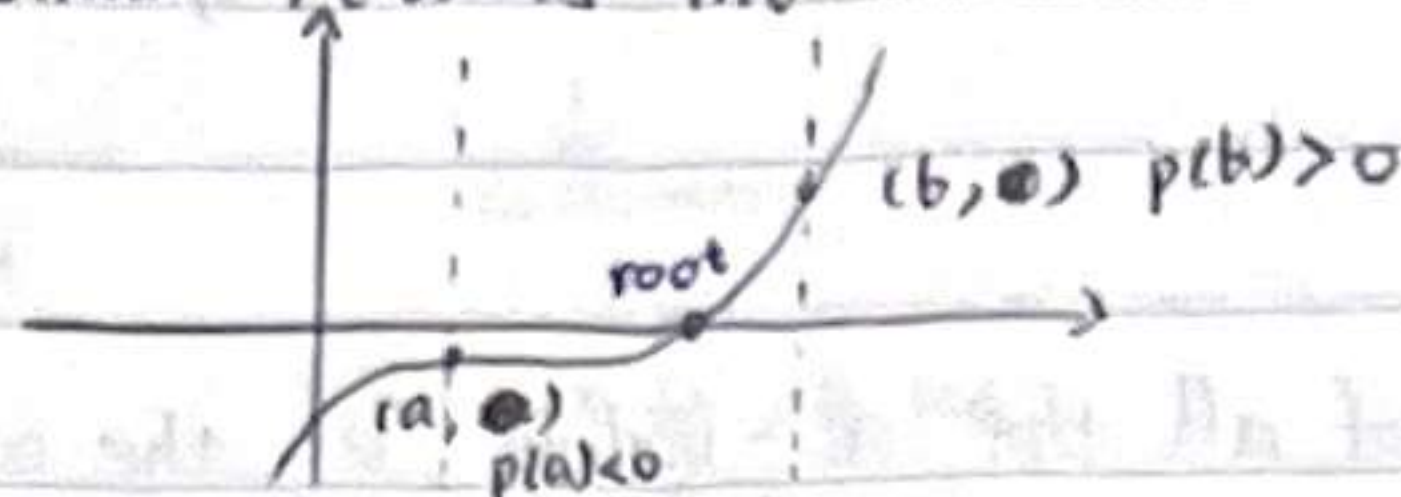
$$(-x)(x)(2x) = -2x^3 \text{ 负单 } \nmid$$



5.4 Numerical Approximation

0 Root of a polynomial

- The root of a polynomial $P(x)$ is the value of x that makes $P(x)$ equal zero.



root is in between $[a, b]$

0 Method of bisection

- a procedure for improving the estimate of root by halving the interval in which the root is known to lie.

- Six steps: 1. Check if a root lies within an interval

2. Get the interval $[a, b]$

3. Find the ^{midpoint} mid-value of the interval $\frac{a+b}{2}$

4. Draw table and get y-value

x a $\frac{a+b}{2}$ b

y $\square$ $\square$ $\square$

y-值

5. pick the two value from $a, \frac{a+b}{2}, b$ that has less value and $-1 > 0, -1 < 0$

6. A new interval generated.

0 Example Question ☆:

① For the polynomial equation $x^2 - 12x + 1 = 0$, $a = 10, b = 12$. Use the method of bisection to obtain two narrower intervals in which the root lies and hence give an estimate of the root which lies in the interval $[a, b]$

① x $10, \frac{10+12}{2}, 12$

y $-19, -10, 1$ $[11, 12]$

② x $11, 11.5, 12$

y $-10, -4.75, 1$ $[11.5, 12]$

Estimate of the root: $11.75, x \approx 11.75$

② Use a graphical method to estimate any solutions to the equation $x^4 + 3x - 4 = 0$

$$x^4 = -3x + 4$$

把两个x分开

The solu of the equa can be obtained from the intersection of the graphs of $y = x^4$ and the line $y = -3x + 4$

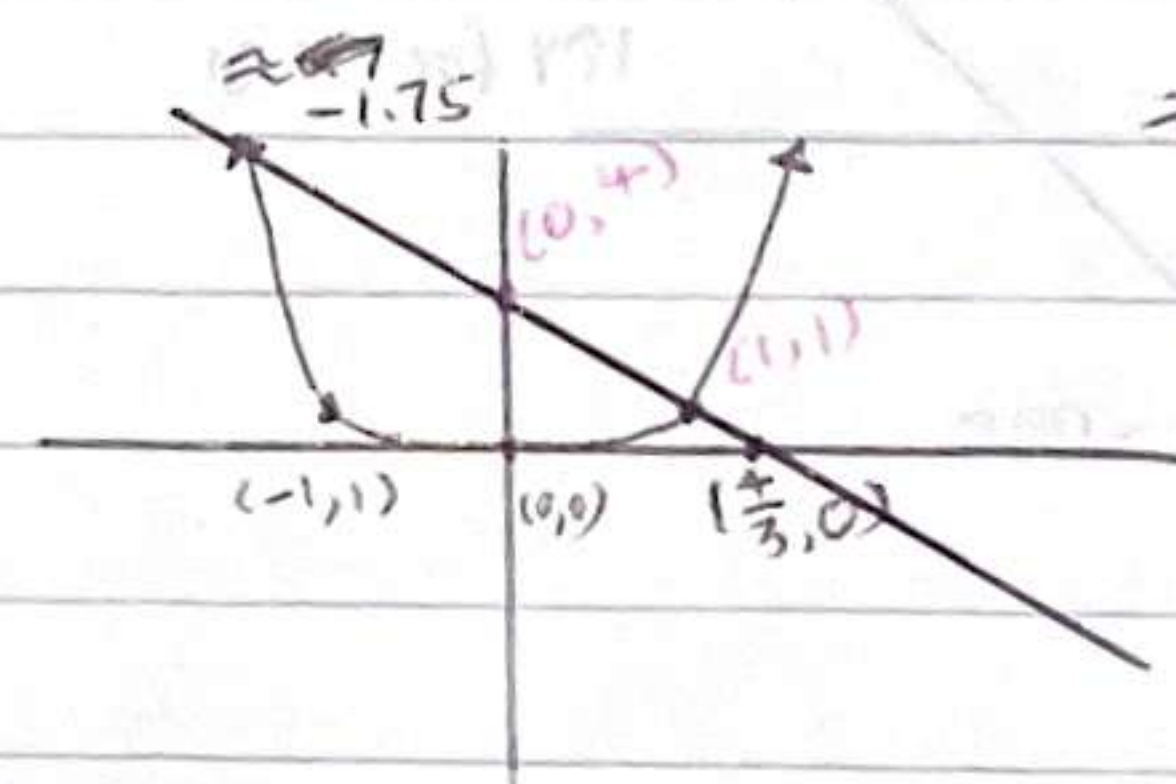
把他们当两个 x^4 : a quartic graph U positive

图看:

pass through $(-1, 1)$ $(1, 1)$
随意找两个数x

$-3x + 4$: a line y -in: $(0, 4)$

x -in: $(\frac{4}{3}, 0)$



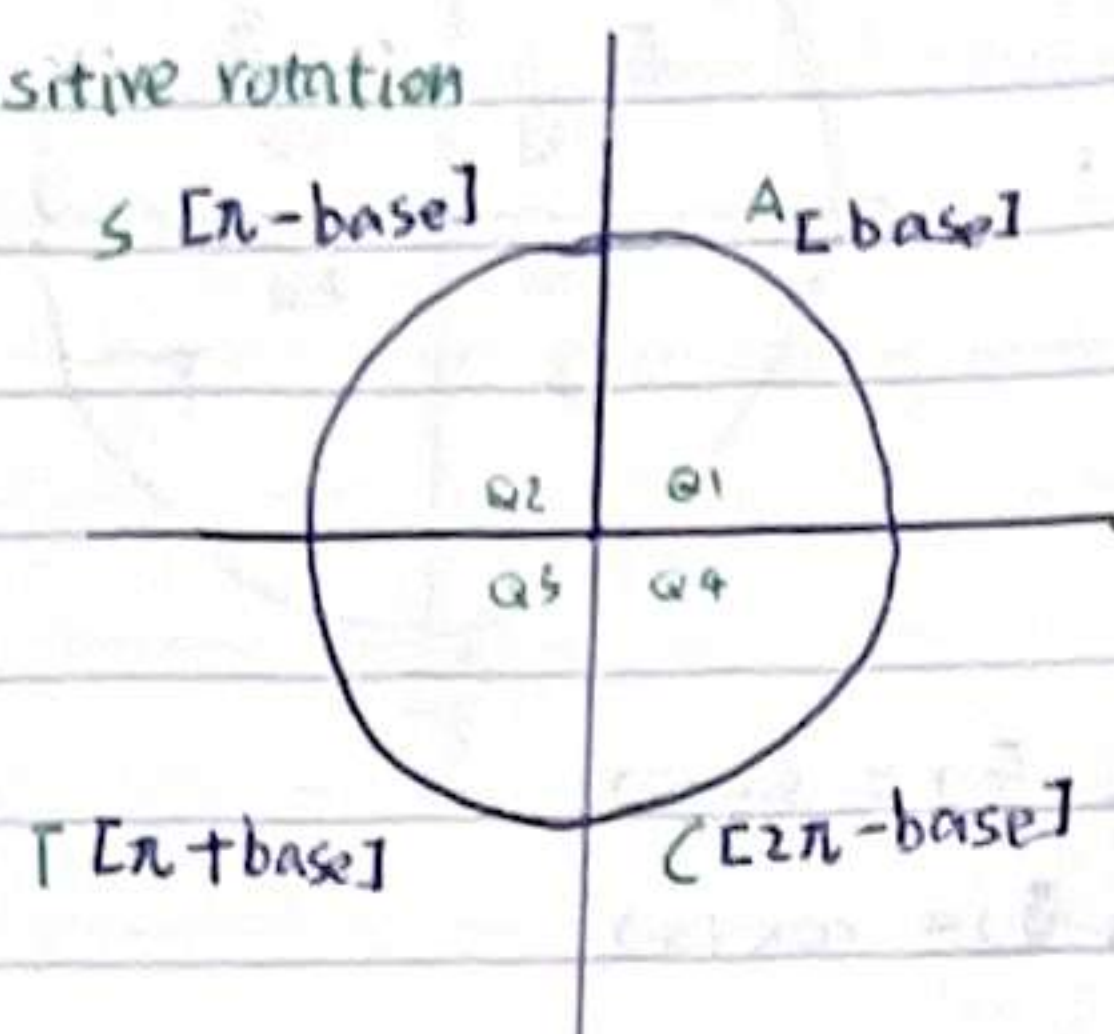
可能需要精美图表
按格走才可以

↑ 用CAS solve (equa, x)

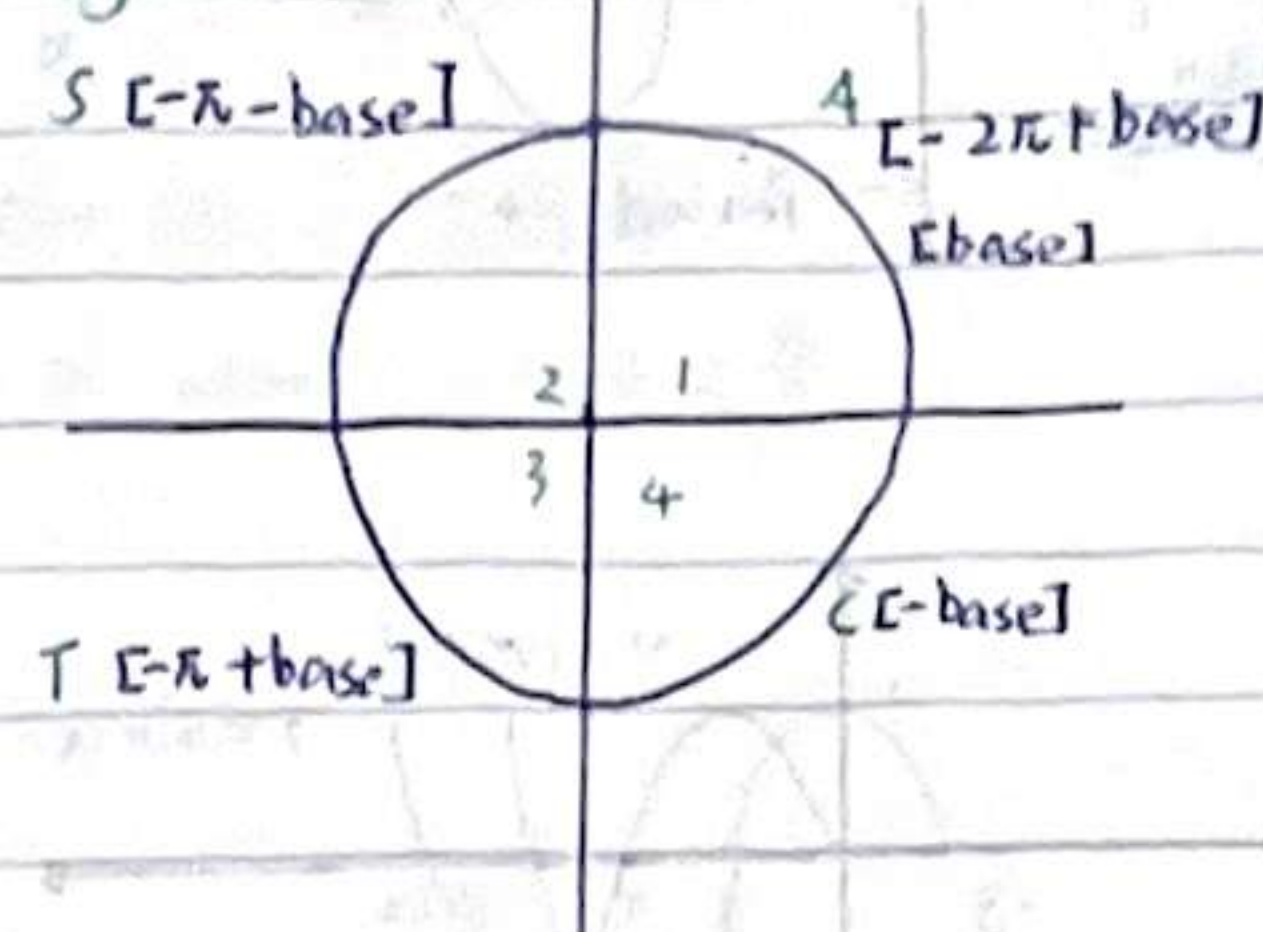
得 all intersection.

Symmetric forms

Positive rotation



Negative rotation

找 x value 步骤:

- Find exact value in the table
- 根据其正/反数 determine its quadrant (negative 在自己象限, positive 在自己象限)
- Find value of x by applying the positive/negative rotation graph
 根据 domain 值, $x \in [0, 2\pi]$ positive
 $x \in [-2\pi, 0]$ negative
 $x \in [-2\pi, 2\pi]$ 正负都看

例题:

Q1 solve for x given $0 \leq x \leq 2\pi$

(a) $2\sqrt{3} \cos(x) + 3 = 0$

$$\cos(x) = -\frac{\sqrt{3}}{2}$$

① $\cos(x) = \frac{\sqrt{3}}{2} = \frac{\pi}{6}$ $\leftarrow$ base

② $-\frac{\sqrt{3}}{2}$ 是负数, $\cos(x)$ 的负象限在 Q2, Q3③ Positive rotation since $x \in [0, 2\pi]$

in Q2: $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$

Q3: $\pi + \frac{\pi}{6} = \frac{7\pi}{6}$

$$x = \frac{5\pi}{6}, \frac{7\pi}{6}$$

(b) $\sin(2x) + \cos(2x) = 0$

$$\sin(2x) = -\cos(2x)$$

$$\frac{\sin(2x)}{\cos(2x)} = -1$$

$$\star \tan(2x) = -1$$

$$0 \leq x \leq 2\pi$$

$$= 0 \leq 2x \leq 4\pi$$

$$\tan(2x) = -1 \rightarrow \frac{\pi}{4} \text{ 负数 } Q2 + Q4$$

$$2x = \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}, 2\pi + \frac{7\pi}{4} = \frac{15\pi}{4}$$

$$x = \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{11\pi}{8}, \frac{15\pi}{8}$$

(c) $6 \tan(\theta) = -6$ solve in $[0^\circ, 360^\circ]$

CAS

根据限制设 mode
如 $0^\circ-360^\circ$ 用 degree

solve $(6 \tan(\theta) = -6 \mid 0^\circ \leq \theta \leq 360^\circ, \theta)$

math 3

不能省略!!

得 $\theta = 135^\circ, 315^\circ$

The graphs of $y = a \sin(n\pi x) + k$ and $y = a \cos(n\pi x) + k$ have

• amplitude a for $a > 0$; the graphs are reflected in the x -axis (inverted) if $a < 0$.

• period $\frac{2\pi}{n}$ (for $n > 0$)

• equilibrium or mean position $y = k$

• Range $[k-a, k+a]$

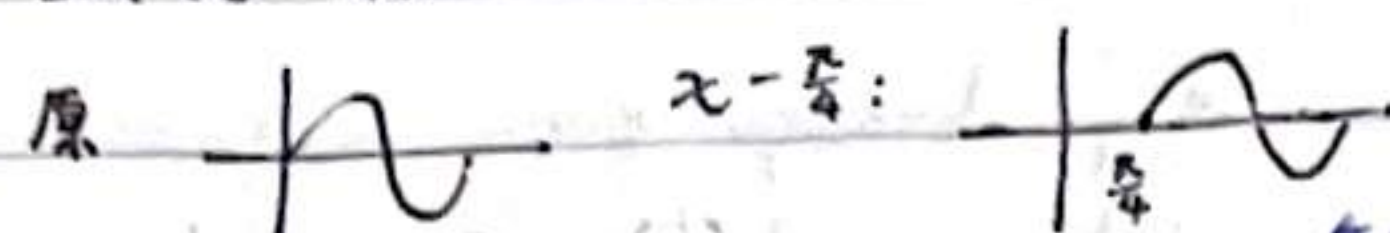
$$y = a \sin(n(x-h)) + k$$

$$y = a \cos(n(x-h)) + k$$

• phase shift of h from graph of $y = a \sin(n\pi x)$ or $y = a \cos(n\pi x)$

平行移动 h units

$$\text{如 } \sin(x - \frac{\pi}{4})$$



每个 x -value $+ \frac{\pi}{4}$ 后再算 y -值

例题: ① Sketch $y = 2 \cos(x) - 1$, $0 \leq x \leq 2\pi$

amplitude $a = 2$, $n = 1$, $k = -1$ $\cos$

Amplitude = 2, period = $\frac{2\pi}{1} = 2\pi$

mean position $y = -1$

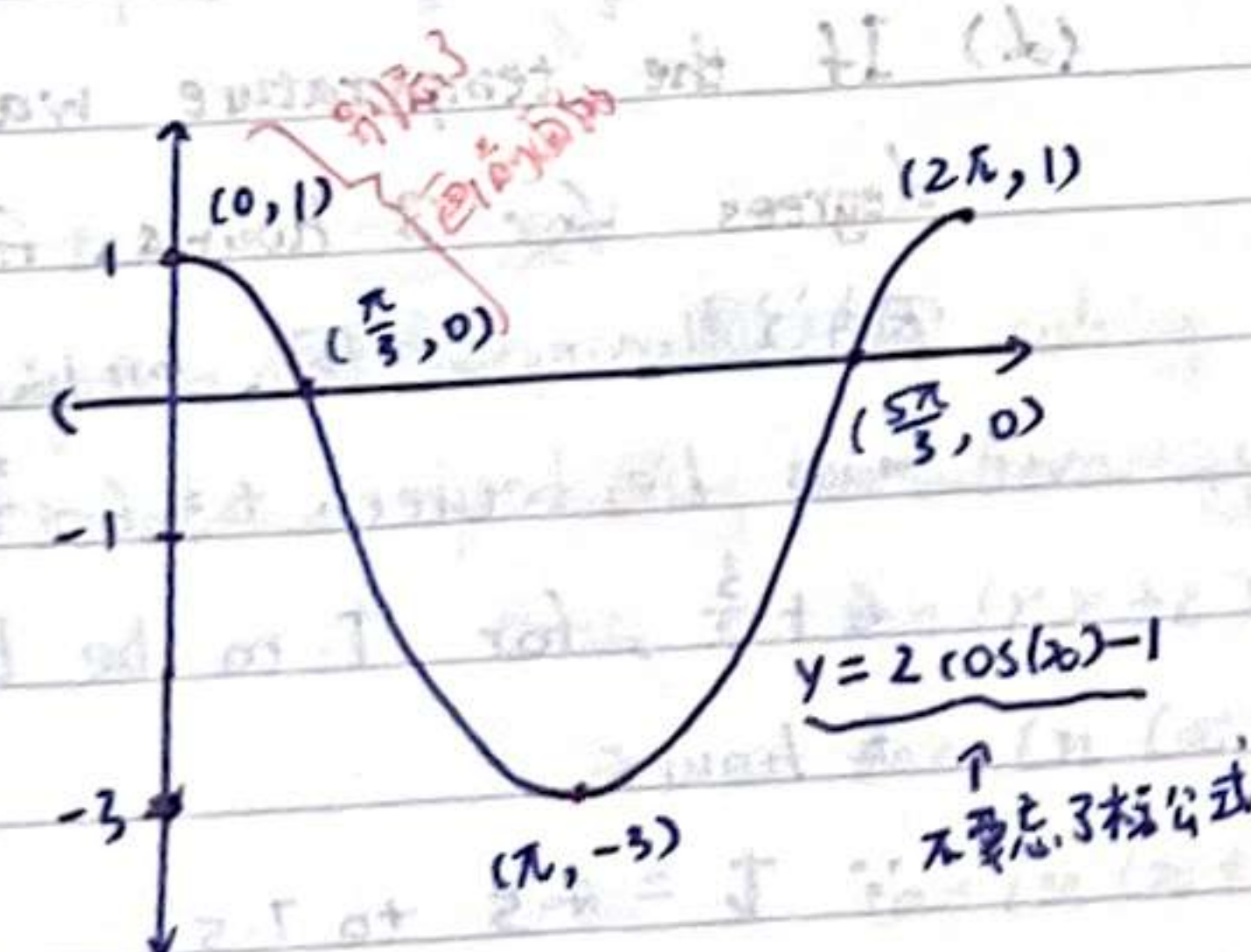
Range $[-3, 1]$

x -intercepts: $y = 0$

$$2 \cos(x) - 1 = 0$$

$$\cos(x) = \frac{1}{2} \text{ 在 } [0, 2\pi] \text{ 上}$$

$$\text{Base } \frac{\pi}{3}, x = \frac{\pi}{3}, 2\pi - \frac{\pi}{3} = \frac{5\pi}{3} \quad (\frac{\pi}{3}, 0), (\frac{5\pi}{3}, 0)$$



② Sketch $y = \sin(2x - \frac{\pi}{6})$, $0 \leq x \leq \pi$

$$a = 1, y = \sin(2(x - \frac{\pi}{12}))$$

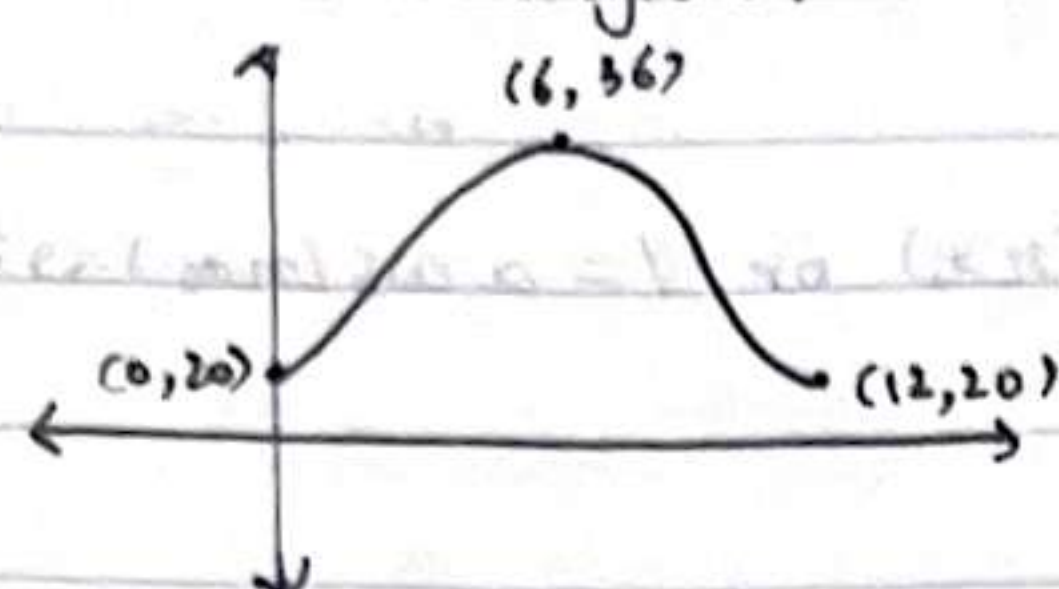
$$\text{period} = \frac{2\pi}{2} = \pi$$

$$\text{phase, } h = \frac{\pi}{12} \quad \sin$$

Applications of sine and cosine functions

Example Question

① Temperature from 8am to 10pm

on a day is shown. $T^{\circ}\text{C}$, t
represent t hours after 8am.Form the equation and use it
to calculate the times during
the day when the temperature
exceeded 30° degrees.Period is 12 hours, Range $[20, 36]$

$$\text{Amplitude} = \frac{36-20}{2} = 8$$

$$\text{Equilibrium} = 20 + 8 = 28, k = 28$$

$$T = a \cos(\pi t) + k$$

$$T = -8 \cos(\pi t) + 28$$

$$\frac{2\pi}{\pi} = 12, \pi = \frac{\pi}{6}$$

$$T = -8 \cos\left(\frac{\pi}{6}t\right) + 28$$

When $T = 30$:

$$30 = 28 - 8 \cos\left(\frac{\pi}{6}t\right)$$

$$\cos\left(\frac{\pi}{6}t\right) = -\frac{1}{4}$$

$$\frac{\pi}{6}t = \cos^{-1}\left(-\frac{1}{4}\right)$$

负cos在Q2+Q3

用CAS找 Exact value $\cos^{-1}\left(-\frac{1}{4}\right) = 1.32$

$$\text{在Q2: } \pi - 1.32 \quad \text{Q3: } \pi + 1.32$$

$$\therefore \frac{\pi}{6}t = \pi \pm 1.32$$

$$\text{用CAS算 } t = \frac{(\pi + 1.32)}{(\pi/6)}, \frac{(\pi - 1.32)}{(\pi/6)}$$

$$\approx 3.48, 8.52 \rightarrow 4.31 \text{ pm}$$

$$60 \times 0.48 = 28.8 \text{ 分钟}$$

3.1小时29分钟 after 8am

$$= 11:29 \text{ am}$$

Ans: The temperature exceeds 30°
between 11:29 am and 4:31 pm.② Temperature = $19 - 3 \sin\left(\frac{\pi}{12}t\right) + \text{hours}$
after midnight:

(a) state the Max T and when it occur.

从equa中能看出当 $\sin\left(\frac{\pi}{12}t\right) = -1$ 时 $19 - 3(-1)$ 得最大数。所以The max T occur when $\sin\left(\frac{\pi}{12}t\right) = -1$

$$19 - 3(-1) = 22^{\circ}$$

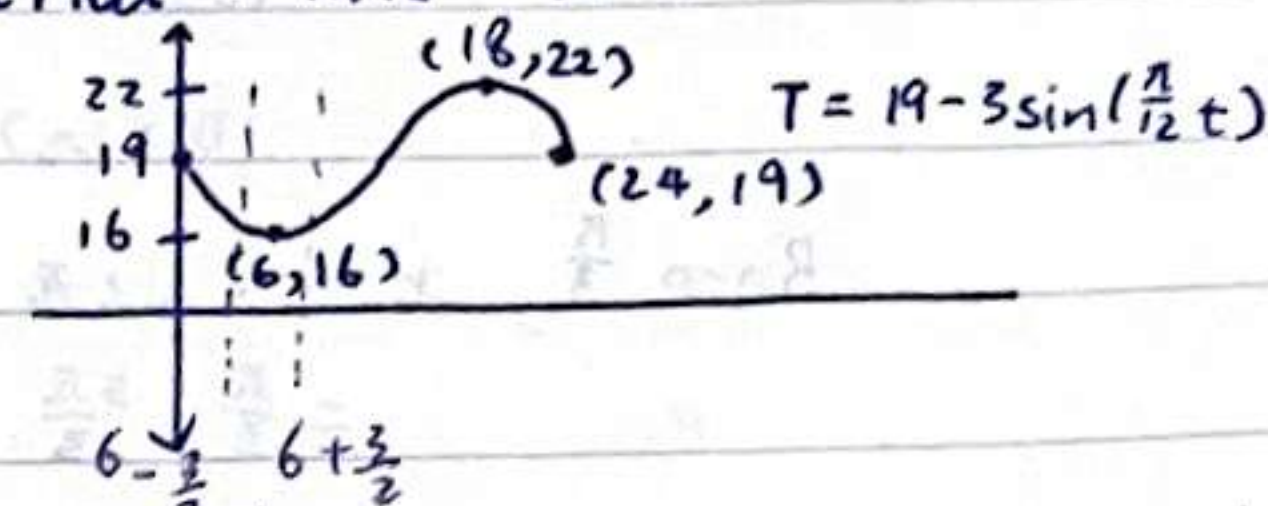
Find t (time): $\sin\left(\frac{\pi}{12}t\right) = -1$

$$\frac{\pi}{12}t = \sin^{-1}(-1)$$

$$\frac{\pi}{12}t = -\frac{\pi}{2}$$

$$\star \frac{\pi}{12}t = \frac{3\pi}{2}$$

$$t = 18$$

(b) Over what interval did the
Temperature vary that day?求 Range $[\text{Min } T, \text{Max } T]$ Min T: when $\sin\left(\frac{\pi}{12}t\right) = 1, 19 - 3 = 16$ Max T: when $\sin\left(\frac{\pi}{12}t\right) = -1$ Range $[16, 22]$ (c) State the period and sketch the
 $[0, 24]$ Period: $\frac{2\pi}{\pi/12} = 24$ (d) If the temperature was below k
degrees for 3 hours, find k 一位小数
因为这是many-one图, 所以 the interval
must lie between $t = 6 - \frac{3}{2}$ and
 $6 + \frac{3}{2}$, for T to be below k for
3 hours.

$$\therefore t = 4.5 \text{ to } 7.5$$

然后用 4.5 或 7.5 求 T 都可, 结果是一样的.

$$T = 19 - 3 \sin\left(\frac{\pi}{12} \times \frac{9}{2}\right)$$

$$\approx 16.2^{\circ}\text{C}$$

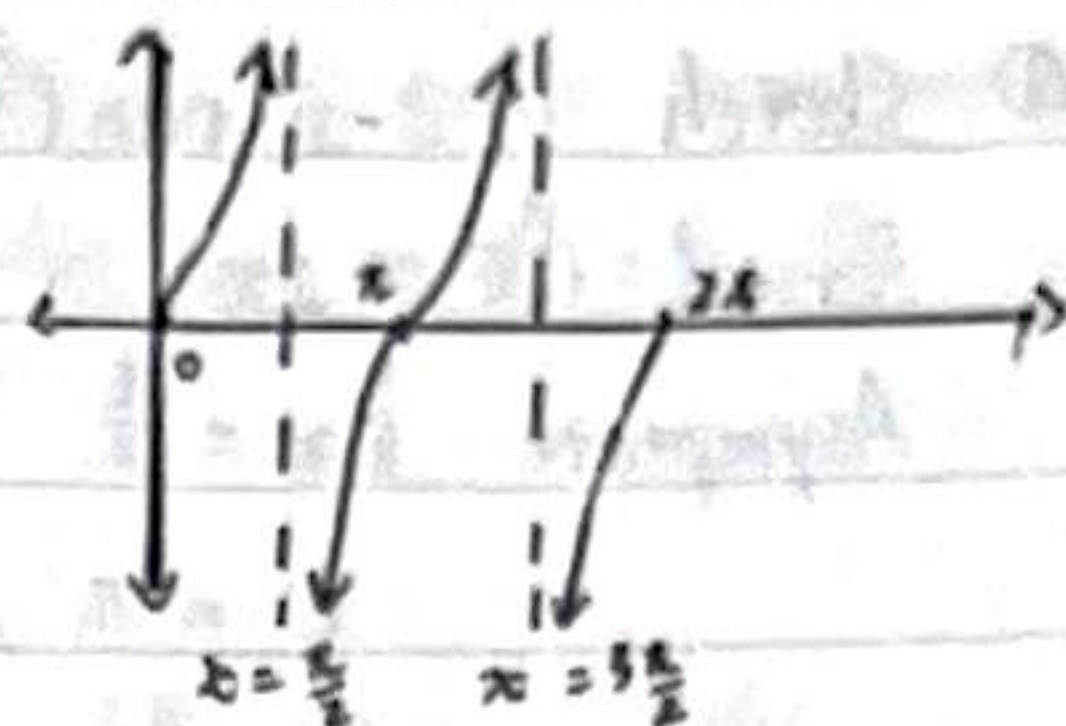
The Tangent Function

The relationship $\tan(x) = \frac{\sin(x)}{\cos(x)}$ shows that:

- $\tan(x)$ will be undefined whenever $\cos(x) = 0$
- $\tan(x) = 0$ whenever $\sin(x) = 0$

$\tan(x)$ 定义域

$x \in [0, 2\pi]$



Transformation

① $y = a \tan(x)$

- if $a > 1$, the graph narrower than $y = \tan(x)$
- if $0 < a < 1$, the graph wider than $y = \tan(x)$
- if $a < 0$, the graph is reflected in the x -axis

* The dilation does not affect vertical asymptote and period.

Period: π

* Asymptotes are one period apart

* x -intercepts are one period apart

* 凡是无 vertical translation ($\tan(x)$ 状)

的, 其 x -intercepts 都在两个 asymptote 中间

② $y = \tan(x) + k$ ← mean position $y = k$

- if $k > 0$, $y = \tan(x)$ move upward by k
- if $k < 0$, $y = \tan(x)$ move downward by k

* The vertical translation does not affect asymptote and period

however, the x -intercept changed as the mean position changed.

can be find by set $y = 0$, 然后根据其 domain, 用得出的 x 任意加减 period

③ $y = \tan(nx)$

↳ Dilation factor of n from the y -axis

- Period = $\frac{\pi}{n}$

* Asymptote 会有变化, Asymptote's equation can be obtain by $nx = \frac{\pi}{2}$, 然后 solve for x .

求得 Asymptote 后, 剩余的 asymptotes can obtain by adding or subtracting multiples of the period

④ $y = \tan(x - h)$ 相位!

* Asymptote 会有变化, obtain by solving $x - h = \frac{\pi}{2}$ and other asymptotes can be obtain by adding or subtracting multiples of the period.

if $y = a \tan(nx + c)$

$= a \tan(n(x + \frac{c}{n}))$

$= a \tan(n(x + h))$

$-\frac{c}{n} = -h$, h 是 horizontal translation

- 负 h 往左 move h units, 正 h 往右 move h units.

10.5 tangent Function

Example Question

① Sketch $y = -2 \tan\left(\frac{x}{2}\right)$, $0 \leq x \leq 2\pi$

Period $\frac{\pi}{1/2} = 2\pi$

Asymptote $\frac{1}{2}x = \frac{\pi}{2}$

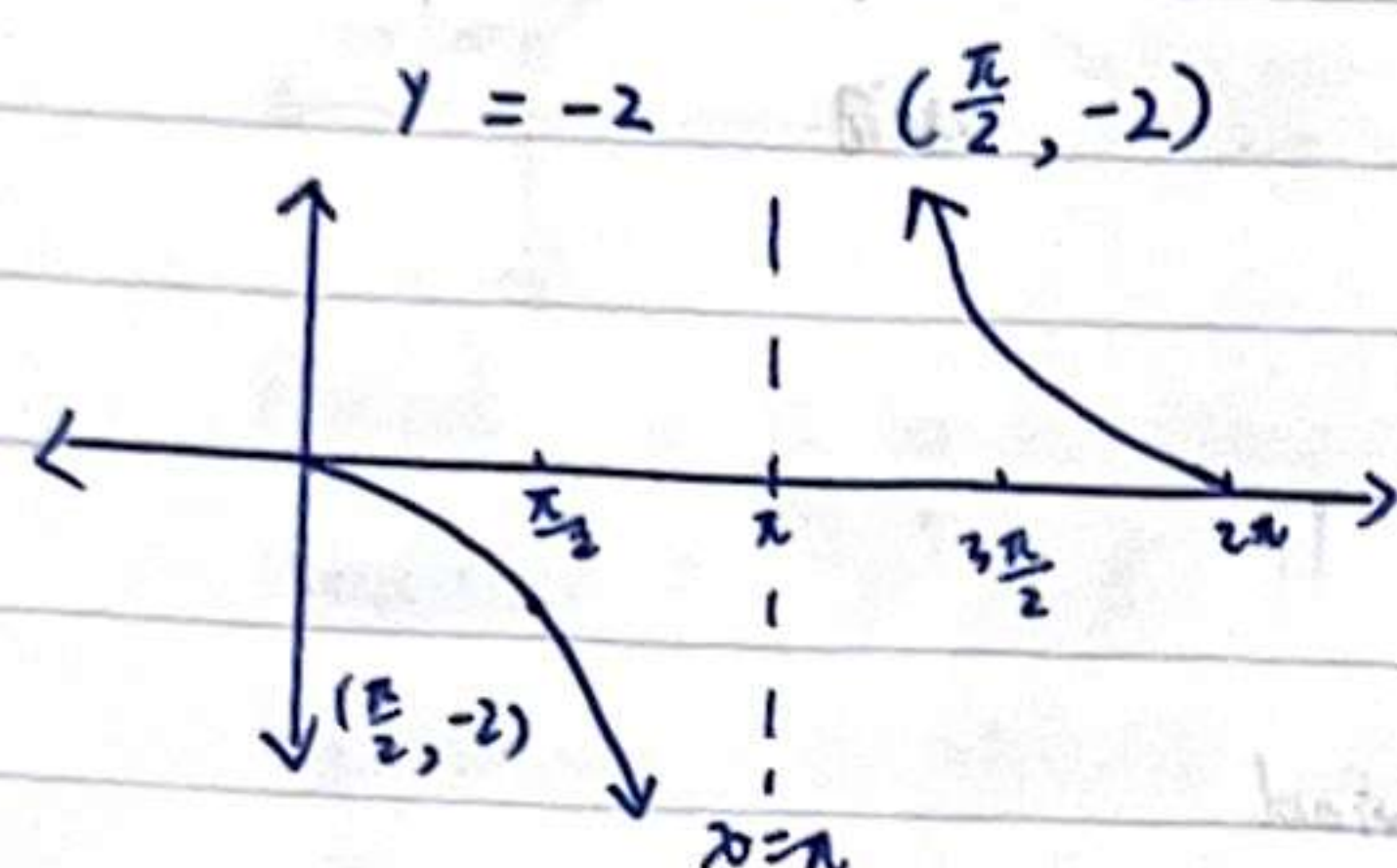
$x = \pi$, $\frac{3\pi}{2}$
 $\rightarrow$ rejected as it $> 2\pi$
 $\rightarrow$ get from $\pi + 2\pi$

x-intercepts: $(3\pi + \pi) \div 2 = 2\pi$

$x = 2\pi$, $0 \leftarrow 2\pi - 2\pi = 0$

Dilation from the x-axis by a factor of 2 and reflected

When $x = \frac{\pi}{2}$



② Sketch $y = \tan\left(2x + \frac{\pi}{4}\right)$, $0 \leq x \leq \pi$

$= \tan\left(2\left(x + \frac{\pi}{8}\right)\right)$

Period: $\frac{\pi}{2}$ horizontal translation $\frac{\pi}{8}$ to the left

Endpoints: $x = 0$ $y = 1$ $(0, 1)$

$x = \pi$ $y = 1$ $(\pi, 1)$

Asymptotes at equation: $2x + \frac{\pi}{4} = \frac{\pi}{2}$

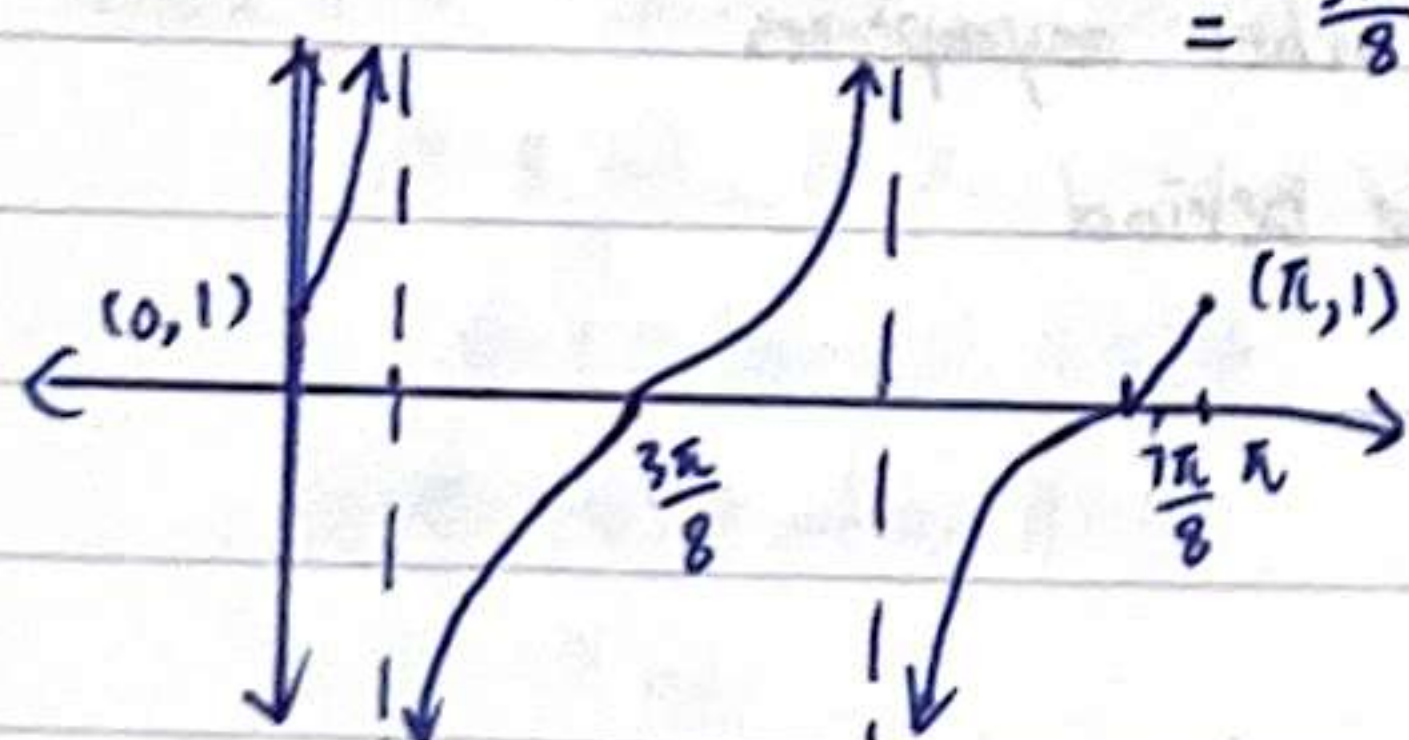
$2x = \frac{\pi}{4}$

$x = \frac{\pi}{8}, \frac{\pi}{8} + \frac{\pi}{2}$

$= \frac{\pi}{8}, \frac{5\pi}{8}$

x-intercepts: $\left(\frac{5\pi}{8} + \frac{\pi}{8}\right) \div 2 = \frac{3\pi}{8}, \frac{3\pi}{8} + \frac{\pi}{2}$

$= \frac{3\pi}{8}, \frac{7\pi}{8}$



Trigonometric relationships

Pythagorean identity:

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

It is a true statement

$$\sin^2(\theta) = 1 - \cos^2(\theta)$$

for any value of θ

$$\cos^2(\theta) = 1 - \sin^2(\theta)$$

Complementary properties: $\sin(\frac{\pi}{2} - \theta) = \cos(\theta)$ and $\cos(\frac{\pi}{2} - \theta) = \sin(\theta)$

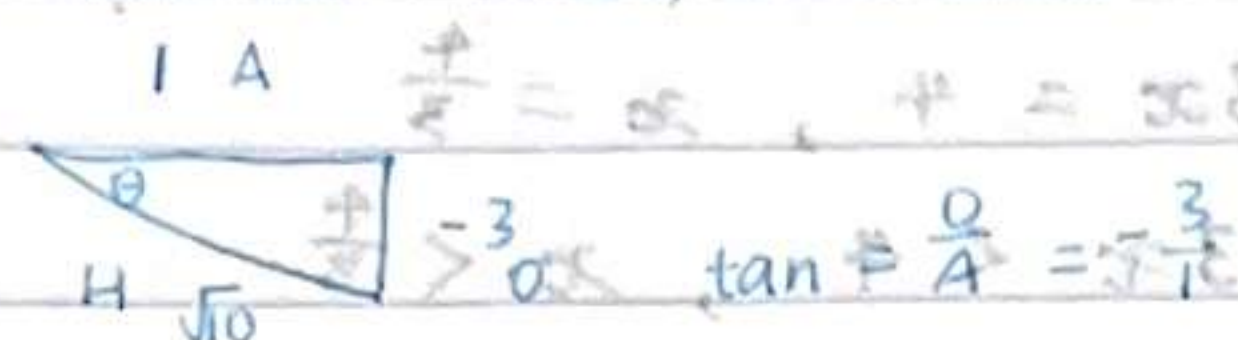
Example Question:

① Simplify $4 - 4\cos^2(\theta)$

$$4(1 - \cos^2(\theta)) = 4\sin^2(\theta)$$

② Given $\tan(x) = -3$, $\pi \leq x \leq 2\pi$ Find $\sin(x)$ and $\cos(x)$

$$\pi \leq x \leq 2\pi \quad \begin{array}{c} \text{S} \\ \text{Q3} \quad \text{Q4} \\ \text{T} \quad \text{C} \end{array}$$

Since $\tan(x)$ is negative, it is in Q4 where the reference angle is θ .

$$H = \sqrt{9+16} = 5 \quad \text{Thus } \sin(\frac{\pi}{2} + \theta) + \sin(\pi + \theta)$$

$$\sin(x) = \frac{y}{H} = \frac{-3}{5} \quad \cos(x) = \frac{x}{H} = \frac{4}{5}$$

$$\cos(x) = \frac{4}{5} = \frac{4}{5}$$

$$x < \pi$$

③ Simplify $\cos(\frac{3\pi}{2} - \theta)$

$$\cos(\frac{3\pi}{2} - \theta)$$

$$0 = (\pi + \frac{\pi}{2}) - \theta = \frac{3\pi}{2} - \theta$$

因为在 Q3, $\cos$ is negative. Thus $\sin(\frac{\pi}{2} + \theta) = \cos(\theta)$ 求 Base, Base 永远只以 $\sqrt{a^2+b^2}$ 的形式出现现有角度: $\frac{3\pi}{2} - \theta$

$$\text{求 } \frac{3\pi}{2} - \theta$$

$$\text{Base} = \frac{3\pi}{2} - \theta = \frac{3\pi}{2} - \theta$$

$$\text{Therefore } \cos(\frac{3\pi}{2} - \theta) = -\cos(\theta)$$

$$\cos(\frac{3\pi}{2} - \theta) = -\cos(\theta)$$

$$\cos(\frac{3\pi}{2} - \theta) = -\cos(\theta)$$

余弦函数

④ Simplify $\sin(\frac{\pi}{2} + \theta) + \sin(\pi + \theta)$ $\sin(\frac{\pi}{2} + \theta)$ in Q2, $\sin$ is positive

$$\text{Base} = \frac{\pi}{2} - (\frac{\pi}{2} + \theta) = -\theta$$

$$\sin(\frac{\pi}{2} + \theta) = \sin(\frac{\pi}{2} - \theta)$$

$$\sin(\frac{\pi}{2} - \theta) = \cos(\theta)$$

$$\sin(\pi + \theta) = -\sin(\theta)$$

$$\text{Base} = \pi - \theta$$

⑤ USE CAS simplify $\sin(\frac{\pi}{2} + \theta)$

Alg Standard Real Rad

Type $\sin(\frac{\pi}{2} + \theta)$

$$\sin(\frac{\pi}{2} + \theta) = \cos(\theta)$$

$$\sin(\frac{\pi}{2} + \theta) = \cos(\theta)$$

Indices as exponents

8 can be written as 2^3

2 ← known as 'Base'
 3 ← known as 'power', 'index', 'exponent'
 指数

Index Law

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n} \text{ and } \frac{1}{a^{-n}} = a^n$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m \text{ or } = \sqrt[n]{a^m}$$

Indices equation 指数方程

• defin: An indices equation has the unknown variable as an exponent.

• In the case of $2^{3x} = 2^4$, $3x = 4$, $x = \frac{4}{3}$
 or $2^{3x} < 2^4$, $3x < 4$, $x < \frac{4}{3}$

• Note that, you must ensure the base is greater than 1.

For example: in the case of $\left(\frac{1}{2}\right)^x < \left(\frac{1}{2}\right)^3$

you first need to convert it to $2^{-x} < 2^{-3}$

$$\text{then: } -x < -3$$

$$x > 3$$

To solve $p \times a^{2x} + q \times a^x + r = 0$

Hints: $\nabla a^{2x} = (a^x)^2$, 设 a^x 为 k , 得如: $(k-2)(k+3) = 0$

• Since a^x must always be positive, after we solve for a^x , we must reject any negative or zero value for a^x .

• Remember, we aim to solve for x , not a^x .

* Significant figures 有效数字

When a number is expressed in scientific notation as either $a \times 10^b$ or $a \times 10^{-b}$, the number of digits in a

determines the number of significant figures in the basic numeral.

$$\text{如: } 4.54 \times 10^9 = \underline{454}0000000$$

3位有效数字

Example Question

① $3^{1-5x} = 1$, solve for x .

$$3^{1-5x} = 3^0$$

$$1-5x = 0$$

$$x = \frac{1}{5}$$

② Solve $2^x - 48 \times 2^{-x} = 13$ for x

$$2^x - \frac{48}{2^x} = 13$$

$$\text{let } a = 2^x$$

$$a - \frac{48}{a} = 13$$

$$a^2 - 48 = 13a$$

$$(a-16)(a+3) = 0$$

$$a = 16, -3 \text{ rejected}$$

$$2^x = 16$$

$$2^x = 2^4$$

$$x = 4$$

CAS Active:

用 solve ()

· 用 Math 1 最左列的 $\square$ 输入 index, 这样比较明了的看到 input.

Example Question



① Sketch $y = 3 \times 2^x$

② Sketch $y = 3 \times 2^{-x}$

③ Sketch $y = 3 \times 2^{x-1}$

④ Sketch $y = 3 \times 2^{x+1}$

⑤ Sketch $y = 3 \times 2^{x-2}$

⑥ Sketch $y = 3 \times 2^{x+2}$

⑦ Sketch $y = 3 \times 2^{x-3}$

⑧ Sketch $y = 3 \times 2^{x+3}$

⑨ Sketch $y = 3 \times 2^{x-4}$

⑩ Sketch $y = 3 \times 2^{x+4}$

⑪ Sketch $y = 3 \times 2^{x-5}$

⑫ Sketch $y = 3 \times 2^{x+5}$

⑬ Sketch $y = 3 \times 2^{x-6}$

⑭ Sketch $y = 3 \times 2^{x+6}$

⑮ Sketch $y = 3 \times 2^{x-7}$

⑯ Sketch $y = 3 \times 2^{x+7}$

⑰ Sketch $y = 3 \times 2^{x-8}$

⑱ Sketch $y = 3 \times 2^{x+8}$

⑲ Sketch $y = 3 \times 2^{x-9}$

⑳ Sketch $y = 3 \times 2^{x+9}$

㉑ Sketch $y = 3 \times 2^{x-10}$

㉒ Sketch $y = 3 \times 2^{x+10}$

㉓ Sketch $y = 3 \times 2^{x-11}$

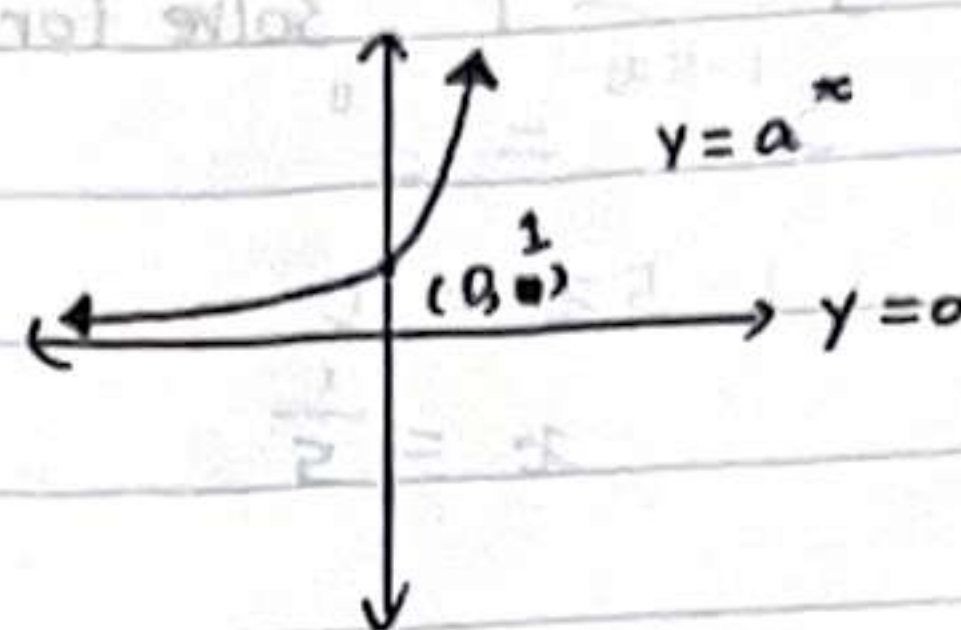
㉔ Sketch $y = 3 \times 2^{x+11}$

㉕ Sketch $y = 3 \times 2^{x-12}$

11.4

Graphs of exponential functions

$$y = a^x \text{ where } a > 1$$

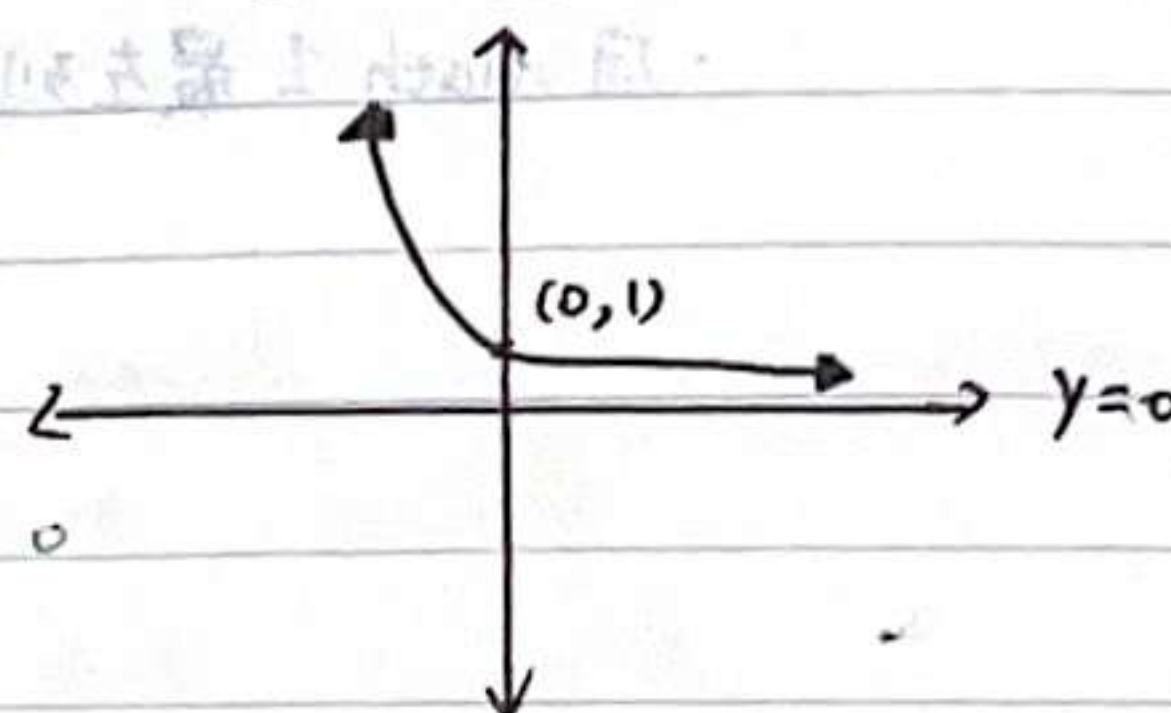


For any function $y = a^x$ where $a > 1$

- horizontal asymptote with equation $y = 0$
- y-intercept is $(0, 1)$
- shape is of 'exponential growth'
- Domain $\mathbb{R}$
- Range $\mathbb{R}^+$
- one-to-one correspondence

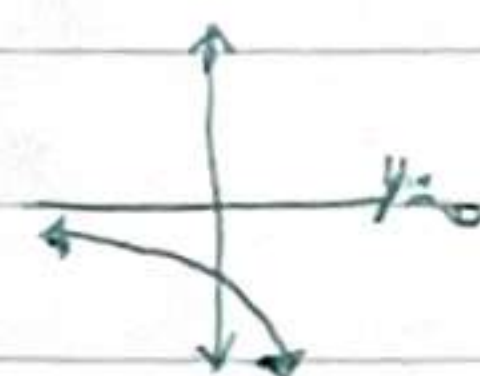
For any function $y = a^x$ where $0 < a < 1$ and $y = a^{-x}$ where $a > 1$

$$y = \left(\frac{1}{2}\right)^x \text{ equal to } y = 2^{-x}$$



- horizontal asymptote with equation $y = 0$
- y-intercept is $(0, 1)$
- shape is of 'exponential decay'
- Domain $\mathbb{R}$
- one-to-one correspondence
- * Reflection in the y-axis

As with other functions, $y = -a^x$ will be reflected in the x-axis



Translations of exponential graphs

$$y = a^x + k$$

- vertical translation of asymptote $y = k$
- x-intercepts can be found by solve $a^x + k = 0$

$$y = a^{x-h}$$

- no affect to asymptote
- y-intercept can be found by set $x = 0$

Dilation

$y = b \times a^x$ have been dilated by a factor b ($b > 0$) from the x -axis

• affect y -intercept

• no affect to asymptote.

$y = a^{nx}$ have been dilated by a factor $\frac{1}{n}$ ($n > 0$) from the y -axis

• affect steepness of the graph

• no affect to y -in or asymptote.

Combinations

$$y = b \times a^{n(x-h)} + k$$

* Another point obtained could provide assurance of the shape.

Always aim to show at least two points on the graph.

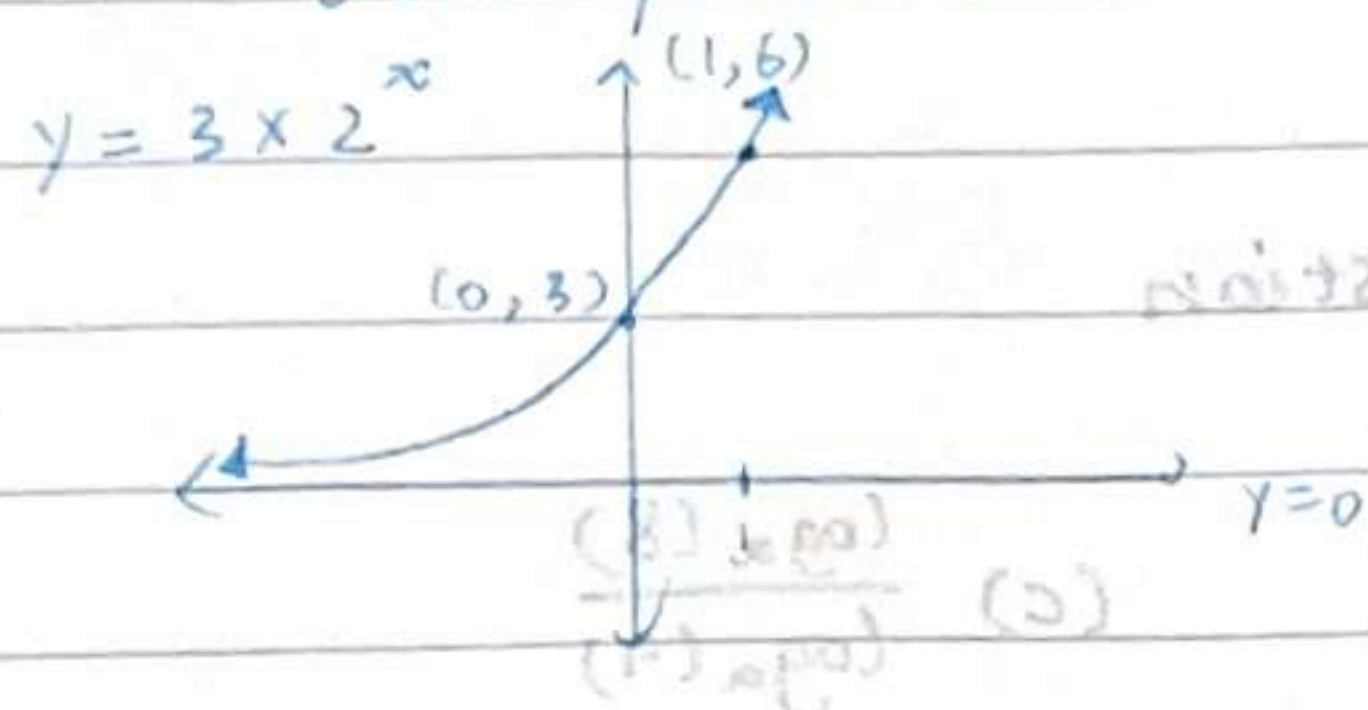
Example Question

① Sketch $y = 3 \times 2^x$

Asymptote: $y = 0$

y -in $(0, 3)$

Other point $(1, 6)$



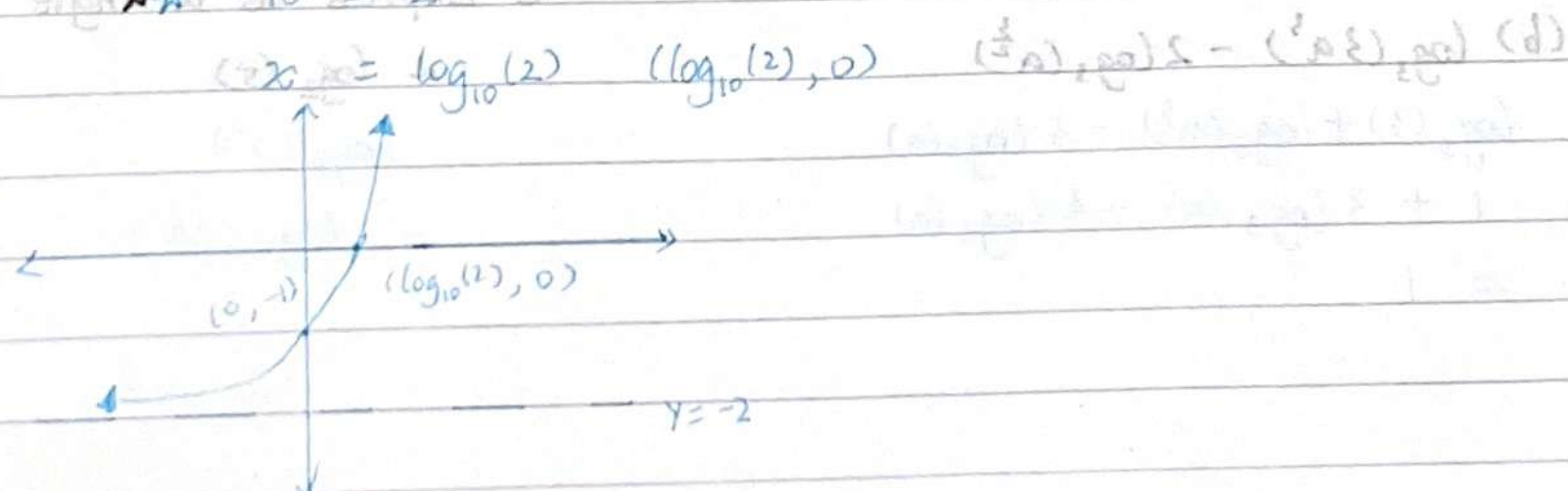
② $y = 10^x - 2$

Asymptote: $y = -2$

y -in: $(0, -1)$

x -in: $0 = 10^x - 2$

★ $10^x = 2$



Indices as logarithms

Index form

$$\underset{\substack{\uparrow \\ \text{Base number}}}{e}^{\overset{\substack{\uparrow \\ \text{index}}}{x}} = y$$

$$10^2 = 100$$

Logarithm form

$$\underset{\substack{\uparrow \\ \text{Base number}}}{\log_e}(y) = x \leftarrow \text{index}$$

$$\log_{10}(100) = 2$$

Logarithm laws:

For any $a, m, n > 0, a \neq 1$, the laws are:

$$\log_a(1) = 0$$

$$\log_a(a) = 1$$

$$\log_a(m) + \log_a(n) = \log_a(mn)$$

$$\log_a(m) - \log_a(n) = \log_a\left(\frac{m}{n}\right)$$

$$\log_a(m^p) = p \log_a(m)$$

Change of base law:

$$\log_a(P) = \frac{\log_{10}(P)}{\log_{10}(a)}$$

★ Example Question

① Simplify

$$(a) \frac{\log_a\left(\frac{1}{4}\right)}{\log_a(2)}$$

$$= \frac{\log_a(2^{-2})}{\log_a(2)}$$

$$= \frac{-2 \log_a(2)}{\log_a(2)}$$

$$= -2$$

$$(c) \frac{\log_a(8)}{\log_a(4)}$$

$$= \frac{\log_a(2^3)}{\log_a(2^2)}$$

$$= \frac{3}{2}$$

$$(b) \log_3(3a^3) - 2\log_3(a^{\frac{3}{2}})$$

$$\log_3(3) + \log_3(a^3) - 3\log_3(a)$$

$$1 + 3\log_3(a) - 3\log_3(a)$$

$$= 1$$

② Express as a single logarithm

$$2 \log_2(5)$$

$$\log_2(5^2)$$

$$= \log_2(25)$$

③ Solve for x : $\log_2(2x+1) + \log_2(2x-1) = 3 \log_2(3)$

$$(a+b)(a-b) = a^2 - b^2$$

$$\log_2(2x+1) + \log_2(2x-1) = \log_2(27)$$

$$2x+1 > 0 \quad 2x-1 > 0 \quad 2x > 1 \quad x > \frac{1}{2}$$

$$2x+1 = 27 \quad 2x = 26 \quad x = 13$$

$$x = 13$$

Example Question

① The value of a car after 1 year is given by $V = V_0 \cdot 2^{-t}$

(a) If 50% of the purchase value is lost in 2 years, find K .

② The value of a car after 1 year is given by $V = V_0 \cdot 2^{-t}$

and after 2 years the value is $V_0 \cdot 2^{-2}$

$$\log_2(V) = \log_2(V_0 \cdot 2^{-t})$$

$$\log_2(V) = \log_2(V_0) - t$$

$$t = \log_2(V_0) - \log_2(V)$$

$$f: \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \log_2(x)$$

③ The inverse of any exponential function is a logarithmic function

The inverse of $y = b \cdot a^{x-h} + k$ can be found by interchanging x & y

$$x = b \cdot a^{y-h} + k$$

$$\frac{x-k}{b} = a^{y-h}$$

$$y = \frac{1}{h} \log_a\left(\frac{x-k}{b}\right) + h$$

④ $\log(a^x) = x$ and $a^{\log(a)x} = x$

$$\log(a^x) = x \log(a)$$

$$x \log(a) = x$$

$$\log(a) = 1$$

$$a = 10$$

$$a^x = x$$

$$10^{\log(x)} = x$$

Applications of exponential function

- Models of exponential growth and exponential decay are often able to be expressed in the form $N = N_0 \times a^{kt}$ or $Q = Q_0 \times a^{-kt}$ where N_0 and Q_0 are symbols for the initial values of the quantities N and Q respectively, and k is a positive constant.

Example Question:

① The value of a car after t years is given by $V = V_0 \times 2^{-kt}$

(a) if 50% of the purchase value is lost in 5 years, find k .

$$\begin{aligned} \text{Purchase value} &= V_0 \times 2^{-k \times 0} \\ &= V_0 \end{aligned}$$

$$50\% \text{ of purchase value } V = \frac{1}{2} V_0$$

$$\text{when } t = 5, V = V_0 \times 2^{-k \times 5}$$

$$\frac{1}{2} V_0 = V_0 \cdot 2^{-k \times 5}$$

$$\frac{1}{2^{k \times 5}} = \frac{1}{2}$$

$$k \times 5 = 1, \quad k = \frac{1}{5}$$

Inverses of exponential functions

Exponential function:

$$y = a^x \quad (x, y) = Y \text{ to } x = a^y$$

Domain $\mathbb{R}$, range $\mathbb{R}^+$ horizontal asymptote $y=0$ y-intercept $(0, 1)$ point $(1, a)$

one-to-one correspondence

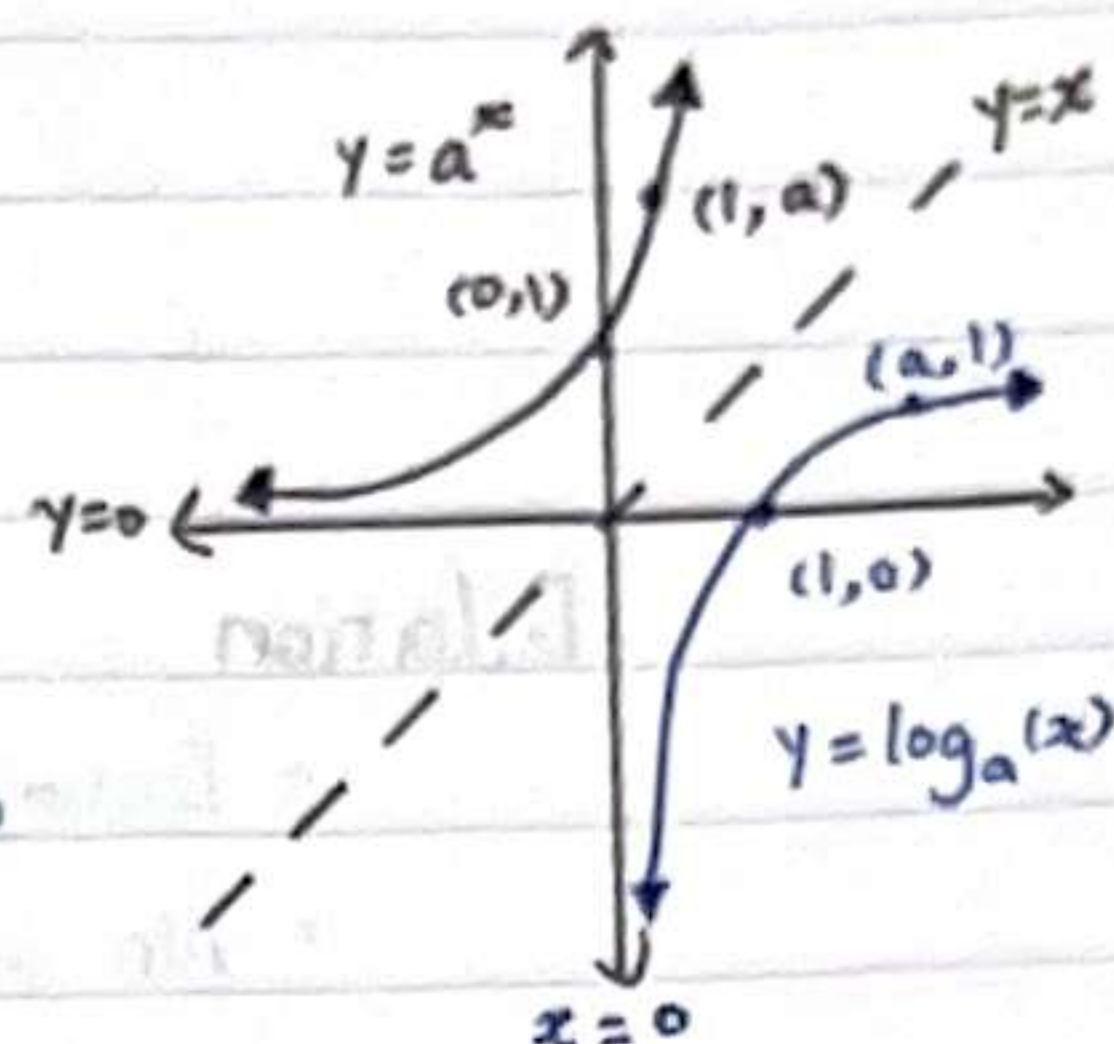
Inverse / Logarithmic function

$$x = a^y$$

$$y = \log_a(x)$$

Domain $\mathbb{R}^+$, Range $\mathbb{R}$ vertical asymptote $x=0$ x-intercept $(1, 0)$ point $(a, 1)$

one-to-one correspondence



- ⊕ Note that the logarithm growth is much slower than exponential growth and unlike a^x which is always positive

$$\log_a(x) \begin{cases} > 0, & \text{if } x > 1 \\ = 0, & \text{if } x = 1 \\ < 0, & \text{if } 0 < x < 1 \end{cases}$$

$$f: \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \log_a(x)$$

- ⊙ The inverse of any exponential function is a logarithmic function
The inverse of $y = b \times a^{n(x-h)} + k$ can be found by interchange x & y

$$x = b \times a^{n(y-h)} + k$$

$$\frac{x-k}{b} = a^{n(y-h)}$$

$$y = \frac{1}{n} \log_a\left(\frac{x-k}{b}\right) + h$$

$$\odot \log_a(a^x) = x \text{ and } a^{\log_a(x)} = x$$

$$\log_a(a^x) = x \log_a(a)$$

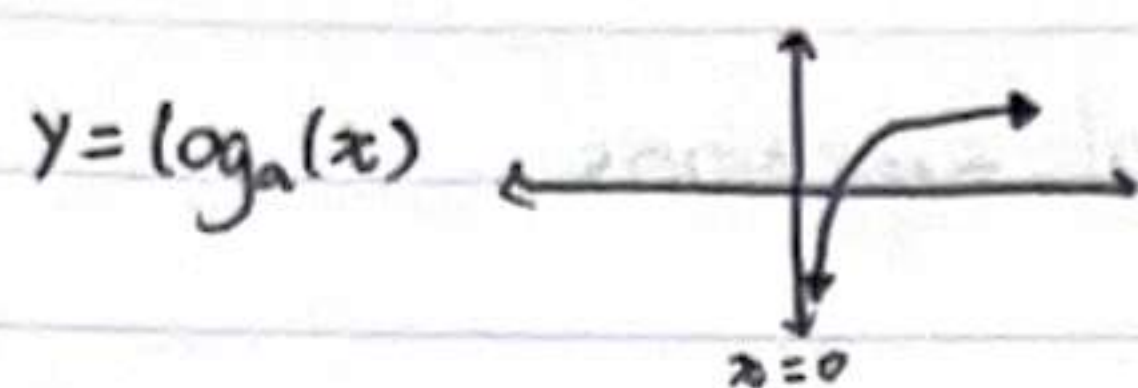
$$= x \times 1$$

$$= x$$

$$a^n = x$$

$$a^{\log_a(x)} = x$$





Transformation of $y = \log_a(x)$

Dilation

- Example: $y = 2 \log_a(x)$, $y = \log_a\left(\frac{x}{2}\right)$
- No impact on vertical asymptote $x = 0$
- x -intercept would **change**

Horizontal translation $y = \log_a(x-h)$

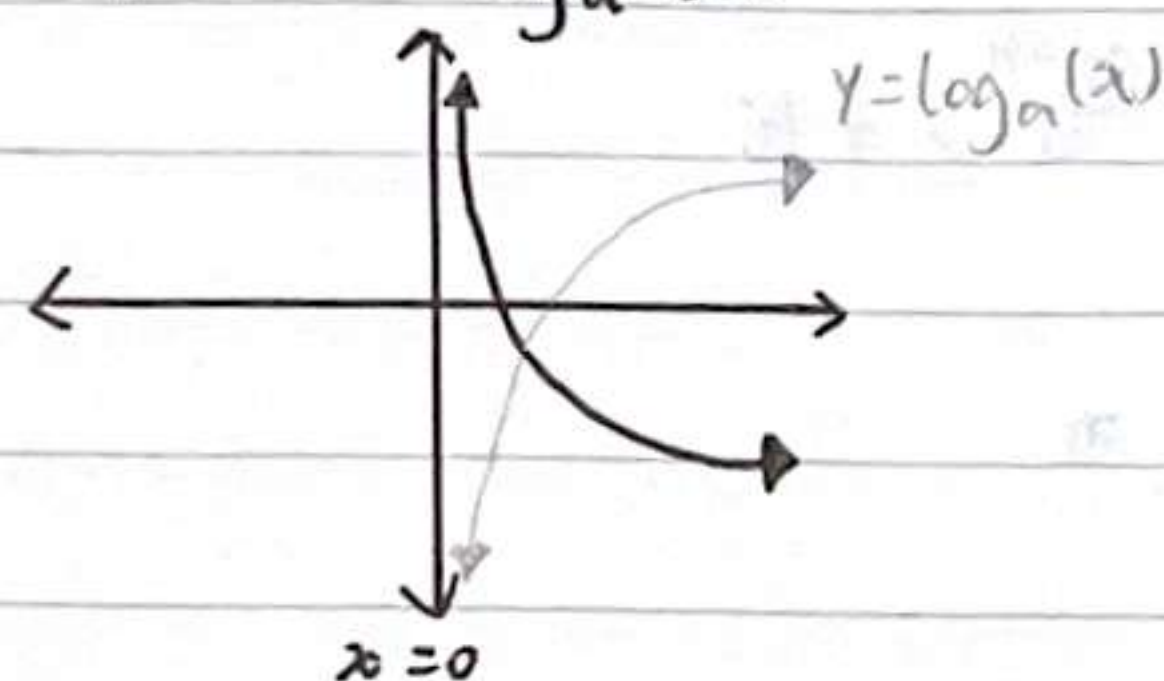
- Vertical asymptote $x = 0$ and thus domain would **change**
- Equation of asymptote $x = 0 \rightarrow x = h$
- domain $\{x : x > 0\} \rightarrow \{x : x > h\}$
- x -intercept $(1, 0) \rightarrow (1+h, 0)$

Vertical translation $y = \log_a(x) + k$

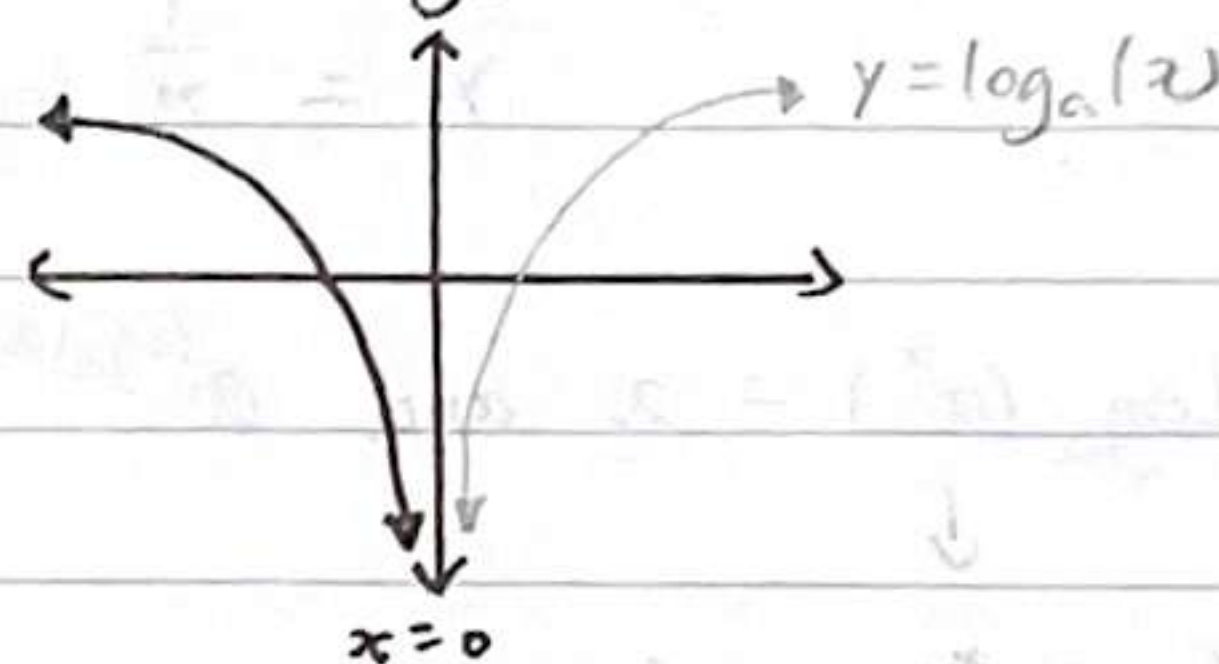
- No impact on asymptote $x = 0$ or domain
- x -intercept would **change**

Reflection $y = -\log_a(x)$ and $y = \log_a(-x)$

- No impact on asymptote
- $y = -\log_a(x)$



$$y = \log_a(-x)$$



Example Question:

① Sketch $y = \log_5(x-2)$

Domain: $x-2 > 0$

$$x > 2$$

$$(2, \infty)$$

Range: $\mathbb{R}$

Asymptote: $x = 2$

y-intercept:

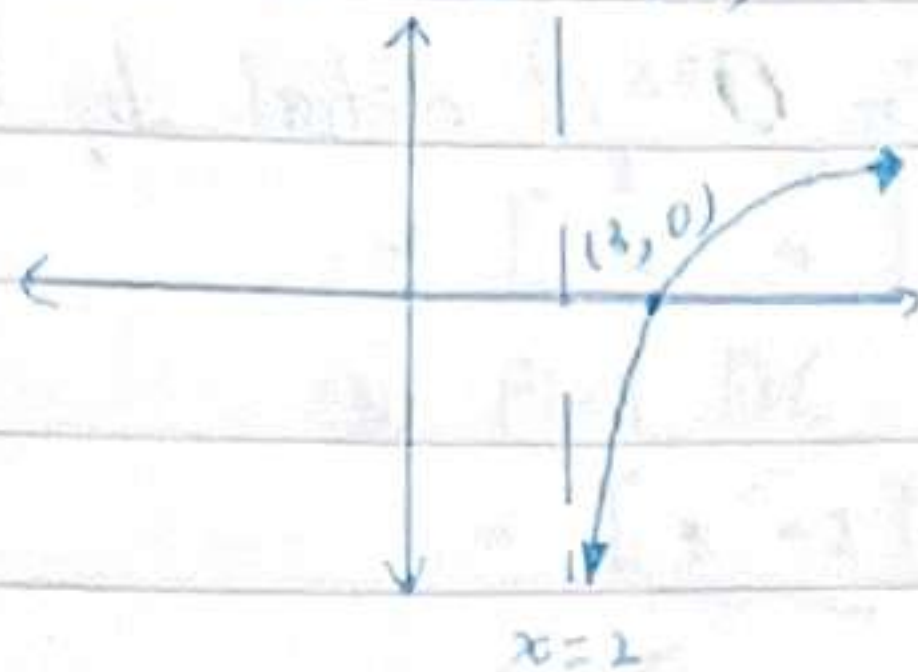
$$y = \log_5(0-2), \text{undefined}$$

x-intercept:

$$0 = \log_5(x-2)$$

$$5^0 = x-2$$

$$x = 3 \quad (3, 0)$$



Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 3 & 3 \\ 7 & 7 \end{bmatrix}$$

to multiply two matrices, the number of columns of the first matrix must equal the number of rows of the second matrix. Thus, $A+B$ does not exist.

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$E = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

(a) Find D^E

A important: D is 2×3

type D : 2×3 matrix

transpose D^T is 3×2

type E : 3×2 matrix

Define D : 2×3 matrix

transpose D^T is 3×2

type E : 3×2 matrix

since the number of columns of D is not equal to the number of rows in E , the matrices cannot be multiplied. Thus, D^E does not exist.

7.2

Matrices 矩阵的加减

$$C = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

C is a 3×1 matrix as it has three rows and one column. If a column has only one column, it is also known as 'column' or 'vector matrix'.

★ only two matrices with the same size or order can be added or subtracted.

Identity matrix

The 2×2 identity matrix I is defined by $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Zero matrix

The 2×2 null matrix or zero matrix O is defined by $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Example:

$$\textcircled{1} A = \begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & -3 \\ 4 & 7 \end{bmatrix}$$

$$A + B = ?$$

$$= \begin{bmatrix} 7 & 1 \\ 2 & 12 \end{bmatrix}$$

For 2×2 matrices if $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$

$$A \cdot B = \begin{bmatrix} a_1 b_1 + a_2 b_3 & a_1 b_2 + a_2 b_4 \\ a_3 b_1 + a_4 b_3 & a_3 b_2 + a_4 b_4 \end{bmatrix}$$

Thus, $AB \neq BA$

Example Question [include (AS)]:

① $C = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ $D = \begin{bmatrix} 3 & -2 \end{bmatrix}$

(a) Find CD

$$\begin{aligned} & \begin{bmatrix} 1 \\ -2 \end{bmatrix} \times \begin{bmatrix} 3 & -2 \end{bmatrix} \quad \text{行对-列} \\ &= \begin{bmatrix} 1 \times 3 & 1 \times -2 \\ -2 \times 3 & -2 \times -2 \end{bmatrix} \\ &= \begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix} \end{aligned}$$

(b) Find DC

$$\begin{aligned} & \begin{bmatrix} 3 & -2 \end{bmatrix} \times \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \text{行对-列} \\ &= \begin{bmatrix} 3 \times 1 + -2 \times -2 \end{bmatrix} \\ &= \begin{bmatrix} 7 \end{bmatrix} \end{aligned}$$

② CAS, $D = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -4 \end{bmatrix}$

$$E = \begin{bmatrix} 6 & 2 \\ 3 & -1 \\ 3 & -3 \end{bmatrix}$$

(a) Find DE

在 keyboard 点 $\begin{bmatrix} \square & \square & \square \end{bmatrix}$

type D 再按一下 $\begin{bmatrix} \square & \square \end{bmatrix}$ (行横列)

就能扩成 $\begin{bmatrix} \square & \square & \square \end{bmatrix}$

填 $\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -4 \end{bmatrix}$

Define D : 然后点 $\Rightarrow$ Math51

然后去 abc 选 d

得 : $\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -4 \end{bmatrix} \Rightarrow d$

E 的输入同上, $D \times E = \begin{bmatrix} \circ & \circ \\ \circ & \circ \end{bmatrix}$

② $D = \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 5 \end{bmatrix}$ $E = \begin{bmatrix} 2 & -2 & 4 \\ 1 & 4 & -3 \end{bmatrix}$

(a) find DE

$$\begin{aligned} & \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & 4 \\ 1 & 4 & -3 \end{bmatrix} \quad \text{行对-列} \\ &= \begin{bmatrix} 1 \times 2 + 4 \times 1 & 1 \times -2 + 4 \times 4 & 1 \times 4 + 4 \times -3 \\ -3 \times 2 + 2 \times 1 & -3 \times -2 + 2 \times 4 & -3 \times 4 + 2 \times -3 \\ 2 \times 2 + 5 \times 1 & 2 \times -2 + 5 \times 4 & 2 \times 4 + 5 \times -3 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 6 & 14 & -8 \\ -4 & 14 & -18 \\ 9 & 16 & -7 \end{bmatrix}$$

(b) find E + D

★ we can only add matrices of same size. Thus, $E + D$ does not exist

(c) find D^2

$$\begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 2 & 5 \end{bmatrix} \quad \begin{matrix} \text{行对-列} \\ \text{对不上} \end{matrix}$$

since the number of columns of D is not equal to the number of rows in D, the matrices cannot be multiplied. Thus, D^2 does not exist.

Translations using Matrices

The transformation T , is written as:

$$T: \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix}$$

this means T maps the points of the original or the pre-image (x, y) onto a new position known as the image point (x', y')

- * An invariant point or fixed point is a point which is unchanged by the transformation.
- * General order of translation: dilation, reflection and translation (DRT).

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

Example Questions:

- ① Find the image point for the pre-image point $(1, 2)$ using translation equa $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$$\begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Image point $(4, 0)$

- ② Find the translation matrix

if $\triangle ABC$ $A(0, 0)$ $B(2, 3)$

$C(-3, 4)$ is translated to

$\triangle A'B'C'$ with $A'(1, -2)$ $B'(3, 1)$

$C'(-2, 2)$

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\text{Verify } \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} \checkmark$$

Translation matrix =

$$\begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -2 \end{bmatrix}$$

- ③ Find image equation when

$y = x - 1$ is transformed by $T = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$x' = x + 3 \quad y' = y - 2$$

$$x = x' - 3 \quad y = y' + 2$$

$$\text{Sub } y = x - 1$$

$$y' + 2 = x' - 3 - 1$$

$$y' = x' - 6$$

Image equa: $y = x - 6$

- ④ Find $T = \begin{bmatrix} a \\ b \end{bmatrix}$ if $y = x^2$ changed

with $y = (x-7)^2 + 3$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

$$x = x' - a \quad y = y' - b$$

$$\text{Sub } y = x^2, \quad y' - b = (x' - a)^2$$

$$\text{Image equa: } y = x^2 - 2ax + a^2 + b$$

$$\text{Image equa: } y = (x-7)^2 + 3$$

$$= x^2 - 14x + 52$$

$$-2a = -14, \quad a = 7$$

$$a^2 + b = 52, \quad b = 3 \quad T = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

Reflection

① Reflection in the x -axis

- Reflection point in the x -axis of the image point

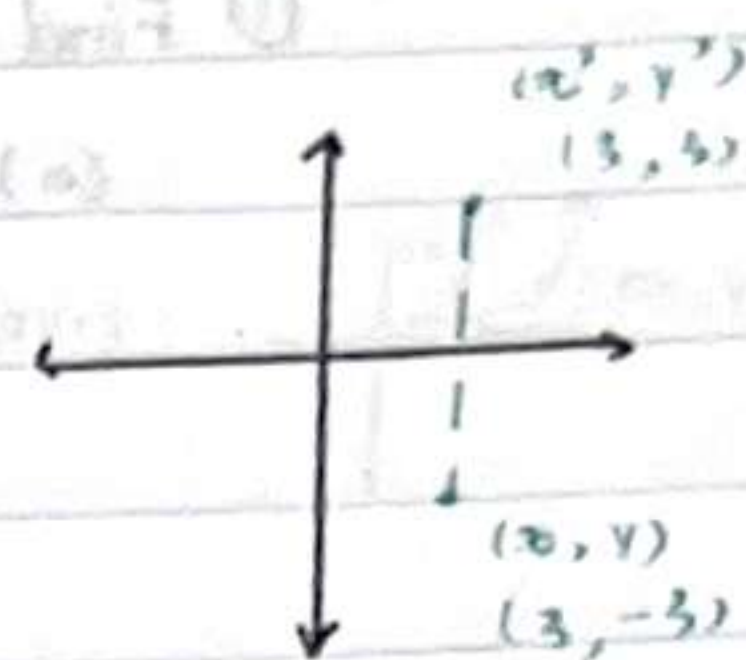
$$(x', y') = (x, -y)$$

- The matrix for reflection mapping in the x -axis:

$$M_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- it $\hat{=}$ reflection for any point in the x -axis

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{eg.} \quad \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

② Reflection in the y -axis

- The reflection point in y -axis of $(x', y') = (-x, y)$

- it $\hat{=}$ reflection for any point in the y -axis

$$M_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

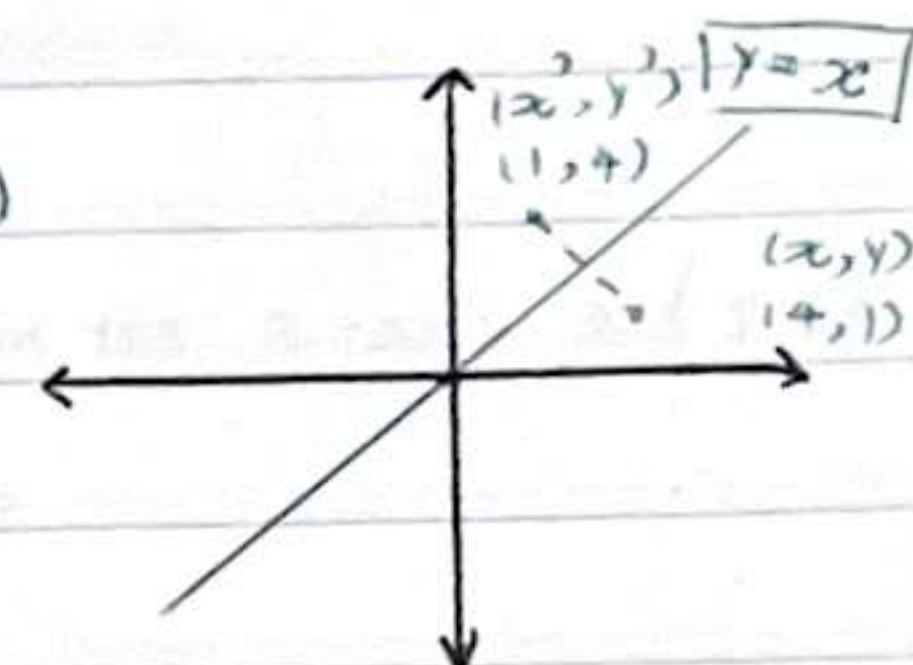
③ Reflection in the line with equation $y=x$

- The reflection point in the line $y=x$ of $(x', y') = (y, x)$

- it $\hat{=}$ any reflection point in the line $y=x$:

$$M_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$



④ Reflection in a line parallel to either axis

eg. $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ for $x=1$ and $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ for $y=-1$

(a) Reflection in $x=a$

$$(x', y') = (2a - x, y)$$

- it $\hat{=}$ any reflection point in the line $x=a$:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2a \\ 0 \end{bmatrix}$$

(b) Reflection in $y=b$

$$(x', y') = (x, 2b - y)$$

- it $\hat{=}$ any reflection point in the line $y=b$:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 2b \end{bmatrix}$$

1.1 Example Question:

① Find image of $(8, -4)$ after reflected

(a) M_x

$$\begin{aligned} M_x: \begin{bmatrix} x' \\ y' \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 8 \\ -4 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 8 + 0 \times -4 \\ 0 \times 8 + -1 \times -4 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 4 \end{bmatrix} \quad (8, 4) \end{aligned}$$

(b) M_y

$$\begin{aligned} M_y: \begin{bmatrix} x' \\ y' \end{bmatrix} &= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 8 \\ -4 \end{bmatrix} \\ &= \begin{bmatrix} -1 \times 8 + 0 \times -4 \\ 0 \times 8 + 1 \times -4 \end{bmatrix} \\ &= \begin{bmatrix} -8 \\ -4 \end{bmatrix} \quad (-8, -4) \end{aligned}$$

② Find the image equation after

$y = (x-2)^2$ is reflected in y -axis

$$M_y: \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x' = -1 \times x + 0 \times y$$

$$= -x, \quad x = -x'$$

$$y' = 0 \times x + 1 \times y$$

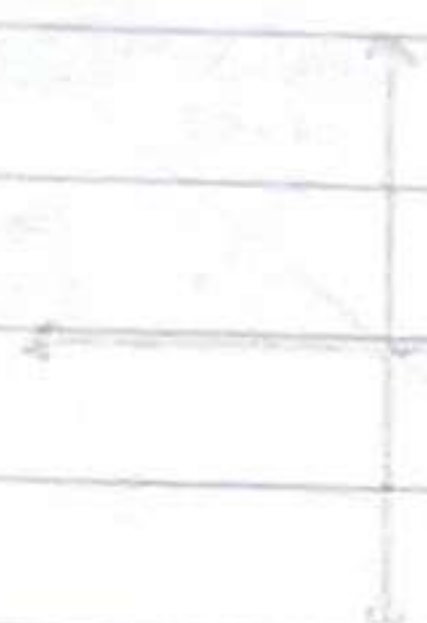
$$= y, \quad y = y'$$

$$\text{原} \quad y = (x-2)^2$$

$$y' = (-x' - 2)^2$$

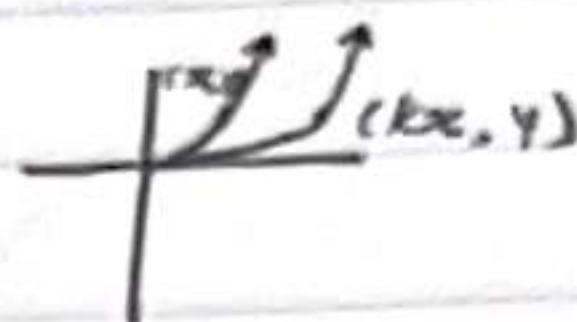
$$y = x^2 + 4x + 4 \quad 4 < 2$$

$$\text{image equation: } y = (x+2)^2$$



One-way dilation

- A dilation from the y-axis OR parallel to the x-axis can be represented by:



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} kx \\ y \end{bmatrix}$$

where k is a dilation factor.

$$(x', y') = (kx, y)$$

- A dilation from the x-axis OR parallel to the y-axis can be represented by:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ ky \end{bmatrix}$$

where k is a dilation factor

$$(x', y') = (x, ky)$$

Two-way dilation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k_1 x \\ k_2 y \end{bmatrix}$$

where k_1 and k_2 are the dilation factors in the x-axis and y-axis

★ Example Question:

- ① Find image of $(2, -5)$
if it dilates 3 parallel

to x-axis

$$D_x: \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \end{bmatrix} (6, -5)$$

- ② Find image equa of $y = \frac{1}{x+1}$
dilated 2 from y-axis

$$D_y: \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x = \frac{1}{2}x' \quad y = y'$$

$$\frac{1}{2}y' = \frac{1}{\frac{1}{2}x' + 1}$$

$$y' = \frac{1}{\frac{1}{2}x' + 1} \times \frac{2}{2}$$

$$y = 2\left(\frac{1}{2}x + 1\right)$$

$$= \frac{2}{x+2}$$

- ③ Pre-image equa: $x + 2y = 2$

$$\text{Image equa: } y = -\frac{x}{2} + 1$$

Is there an invariant point?

An invariant point is one that does not change when the graph is transformed. Therefore the point will be at $y' = y$

$$\text{sub } y = -\frac{x}{2} + 1 \text{ in } x + 2y = 2$$

$$x + 2\left(-\frac{x}{2} + 1\right) = 2$$

$$\frac{2x}{2} = 0$$

$$x = 0$$

$$\text{sub } x = 0 \text{ in } y = -\frac{x}{2} + 1$$

$$y = 0 + 1, (0, 1)$$

The invariant point is $(0, 1)$

Combinations of transformations

- To form the single transformation matrix T , the first matrix T_1 must be multiplied by the second matrix T_2 .

$$T = T_2 \times T_1$$

注意!! T_2 在前头

Transformation matrices

$$M_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

reflection in the x -axis (y, x)

$$M_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

reflection in the y -axis

$$M_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

reflection in the line $y=x$

$$D_y = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

dilation from the y -axis or parallel to x -axis

$$D_x = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

dilation from the x -axis or parallel to y -axis

$$D_{xy} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$$

dilation from parallel to both the x - and y -axis

$$+ \begin{bmatrix} a \\ b \end{bmatrix}$$

a represent movements in x -axis direction

b represent movements in y -axis direction

Example Question:

① Find the image point (x', y') where (x, y) undergoes:

Ⓐ reflection in y -axis

Ⓑ 4 units translate in positive direction of x -axis

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Image point $(-x+4, y)$

② Reverse the translation order in Q①, will the image different?

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right)$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x+4 \\ y \end{bmatrix}$$

$$= \begin{bmatrix} -x-4 \\ y \end{bmatrix}$$

Image point $(-x-4, y)$, the image is

different from Q①

* reflection 和 dilation 时, pre-image point 都要放在右边, 而不是看先后来。

A4 Spiral Notebook
120 Page 48/20

MADE IN CHINA

42-690-122

9 541106 026091

FOR AU / NZ: IMPORTED FOR KIMMY STORES IN AUSTRALIA
AND NEW ZEALAND

Relations, functions and transformation

Relations

- A mathematical relation is any set of ordered pairs

Domain and Range

- domain is the set of all the x -values of the ordered pairs
- range is the set of all the y -values of the ordered pairs
- Example For $A = \{(-2, 4), (1, 5), (3, 4)\}$
Domain is $\{-2, 1, 3\}$, Range is $\{4, 5\}$

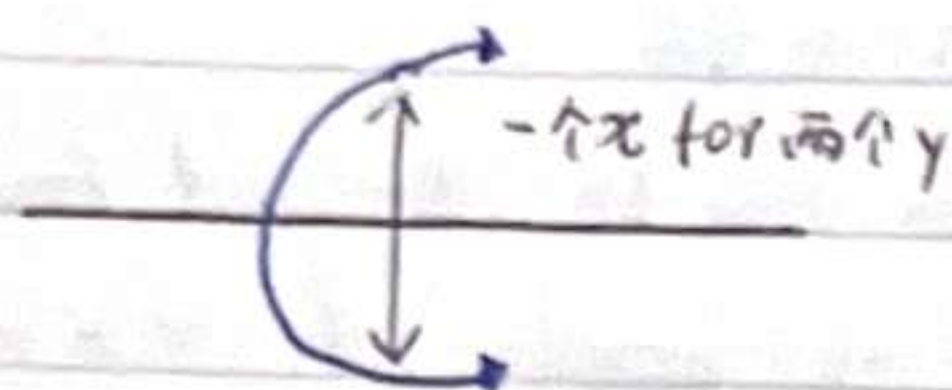
Functions

- A function is a set of ordered pairs in which every x -value is paired to a unique y -value.
- To test: any vertical line which cuts the graph of a function will only do once.

Type of correspondence

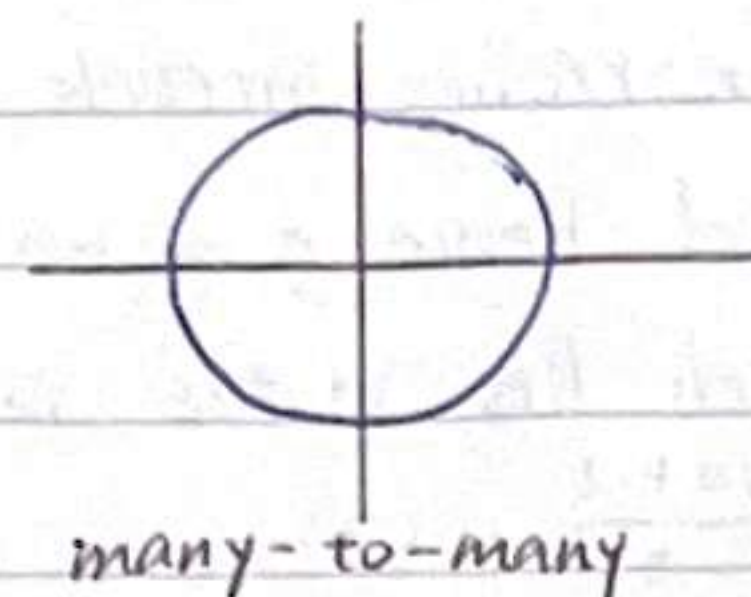
- one-to-one correspondence
- one-to-many correspondence
- many-to-one correspondence
- many-to-many correspondence

Example 图例



one-to-many

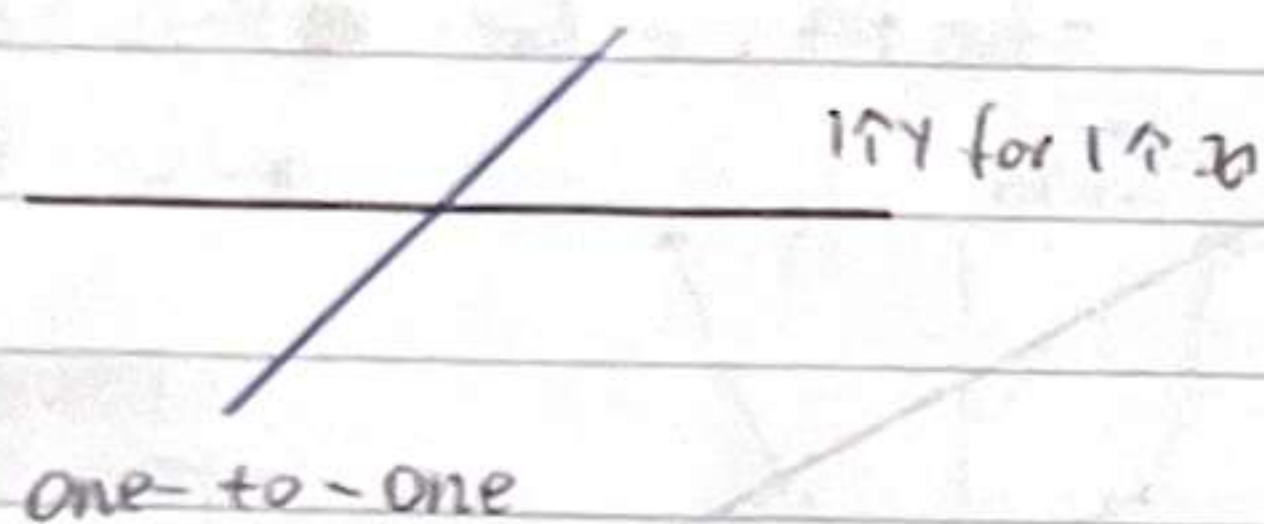
[one x -value pair to many y -values]



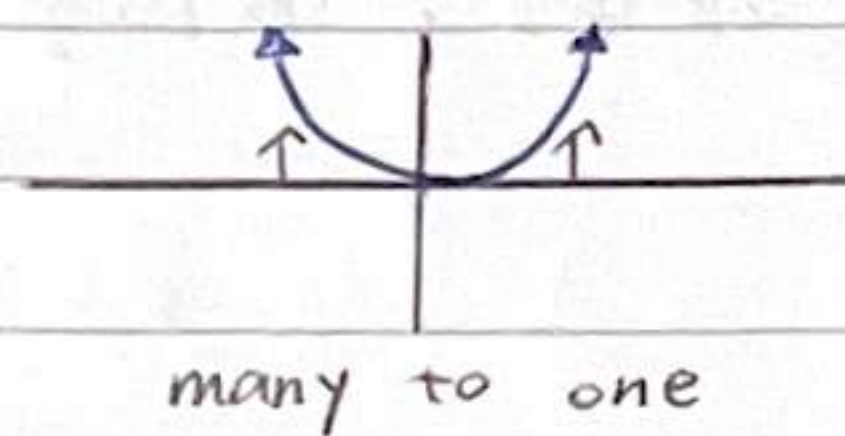
many-to-many

[two x -values for one y -value]

[two y -values for one x -value]



one-to-one



many to one

[two x get one y]

0 Function notation

- The mapping $x \rightarrow x^2$ is written formally as:

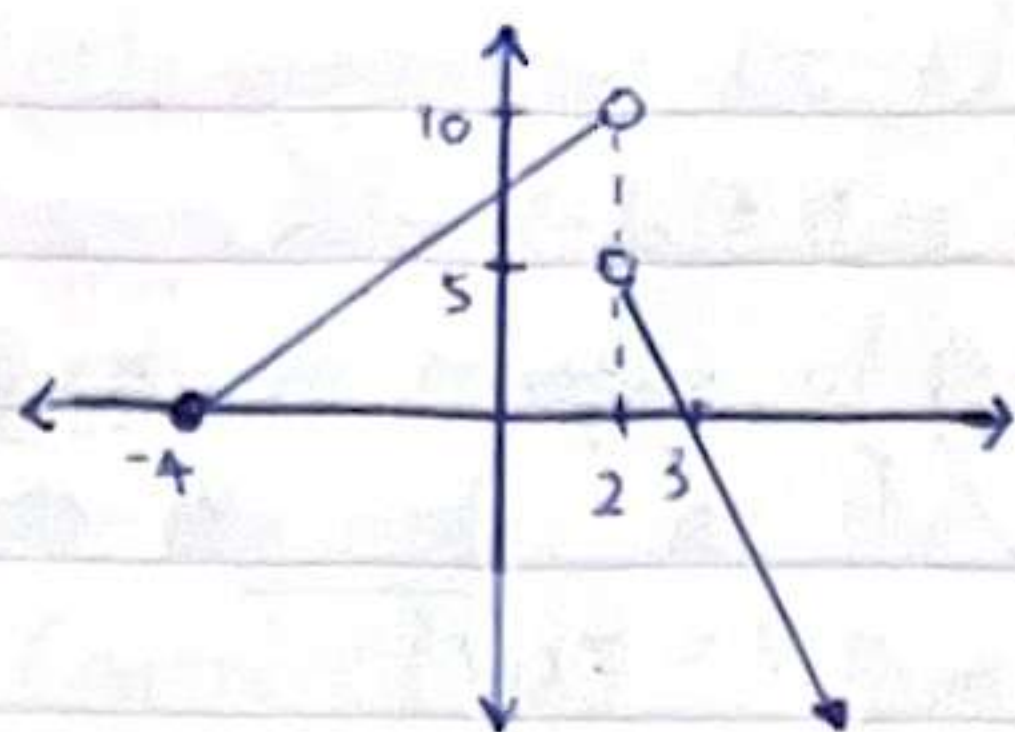
$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

$\downarrow$ name of function $\downarrow$ domain of f $\downarrow$ codomain
 $\downarrow$ the set of all the available y -values.

$\downarrow$ rule for, or equa of f

0 Example Question ★:

① State domain and range



Domain $[-4, 2) \cup (2, \infty)$

Range: $(-\infty, 10)$

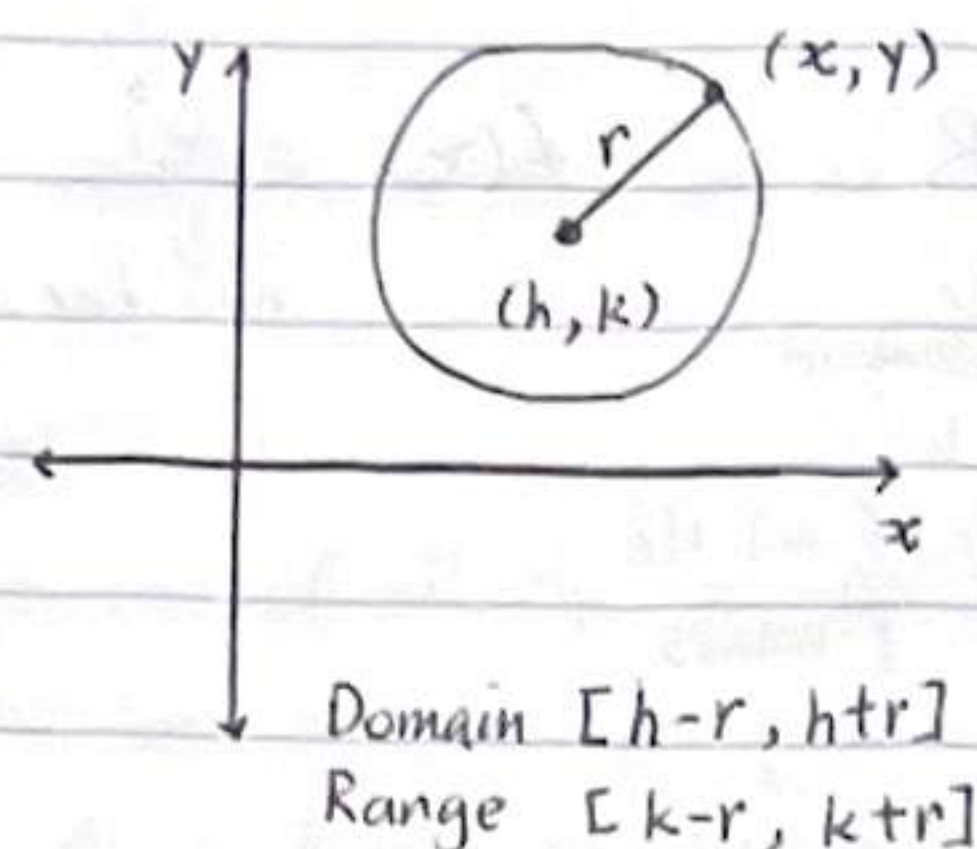
② $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$

Find the image of -3

$$f(-3) = -3$$

Circle

Equation



Distance between two points:

$$\sqrt{(x-h)^2 + (y-k)^2} = r$$

Equation of a circle:

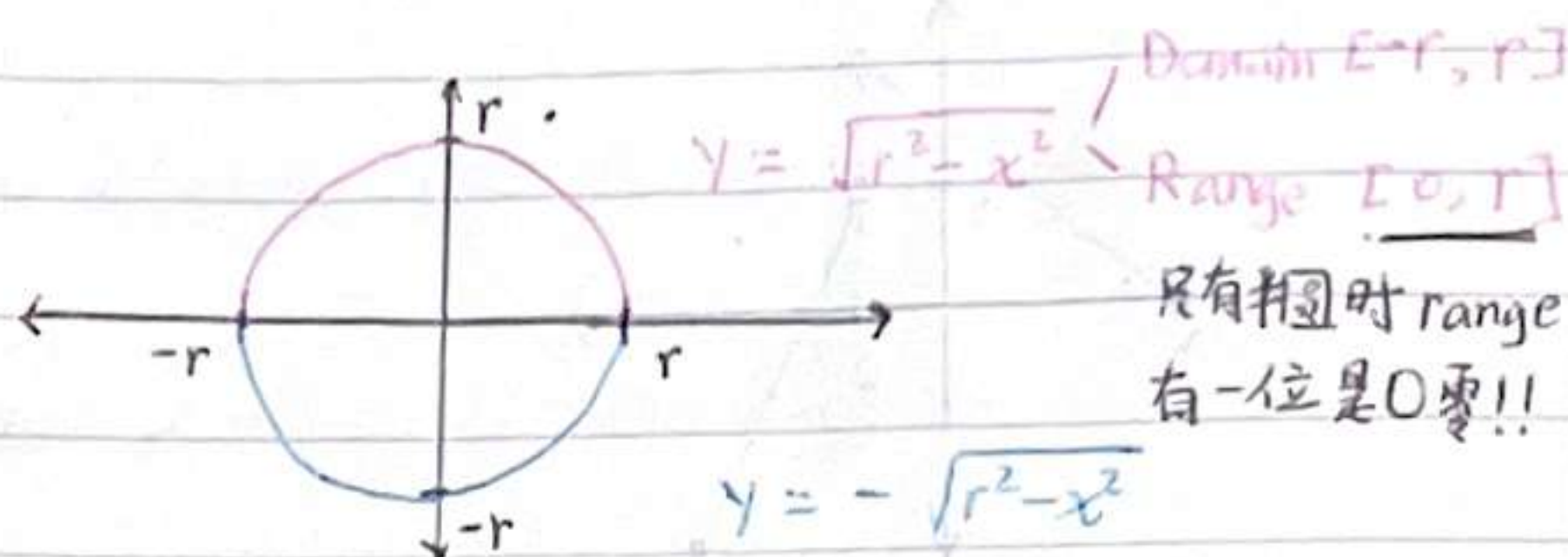
$$(x-h)^2 + (y-k)^2 = r^2$$

Semi circle

- The equation of the circle $x^2 + y^2 = r^2$ can be rearrange to make y the subject.

$$y^2 = r^2 - x^2$$

$$y = \pm \sqrt{r^2 - x^2}$$



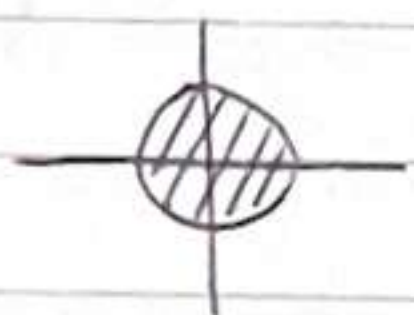
- For circle with centre (h, k) , the equation of the semicircle

- Above the x -axis $y = \sqrt{r^2 - (x-h)^2} + k$

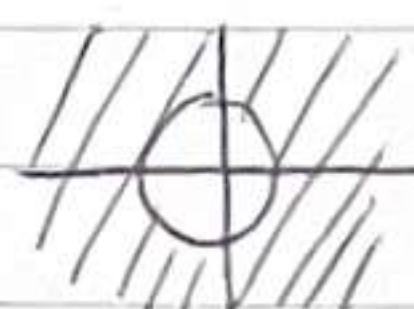
- Below the x -axis $y = -\sqrt{r^2 - (x-h)^2} + k$

Region

$$- x^2 + y^2 \leq r^2$$



$$- x^2 + y^2 \geq r^2$$



Example Question ★:

① State centre, radius, domain and range for $16x^2 + 16y^2 = 81$, SKETCH.

$$x^2 + y^2 = \frac{81}{16}$$

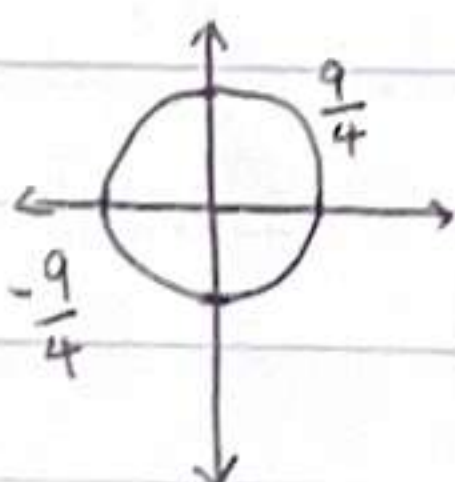
$$r = \sqrt{\frac{81}{16}}$$

$$= \frac{9}{4}$$

Centre: $(0, 0)$ Radius $\frac{9}{4}$

Domain: $[-\frac{9}{4}, \frac{9}{4}]$

Range: $[-\frac{9}{4}, \frac{9}{4}]$



② A circle with centre $(-5, 0)$ passes through a point $(2, 3)$, determine the equation

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x+5)^2 + y^2 = r^2$$

Substitute $(2, 3)$

$$(2+5)^2 + 3^2 = r^2$$

Expand

$$x^2 + 10x + 25 + y^2 = 58$$

$$x^2 + y^2 + 10x - 33 = 0$$

③ Find points of intersection $Y=2x$

$$(x-2)^2 + (y-2)^2 = 1$$

Substitute: $(x-2)^2 + (2x-2)^2 = 1$

$$(x^2 - 4x + 4) + (4x^2 - 8x + 4) = 1$$

$$x^2 + 4x^2 - 4x - 8x + 4 + 4 = 1$$

$$5x^2 - 12x + 7 = 0$$

$$(5x-7)(x-1) = 0$$

$$x = \frac{7}{5}, 1$$

用 x 求 y : $(\frac{7}{5}, \frac{14}{5})$ $(1, 2)$

↳ point of intersect

④ Find m so that $Y=mx-3$ is tangent of $x^2 + y^2 = 4$

Substitute: $x^2 + (mx-3)^2 = 4$

$$x^2(1+m^2) - 6mx + 5 = 0$$

Calculate Δ : $\Delta = 16m^2 - 20$ (简化后)

For line to be a tangent; $\Delta = 0$

$$16m^2 - 20 = 0$$

$$m^2 = \frac{5}{4}$$

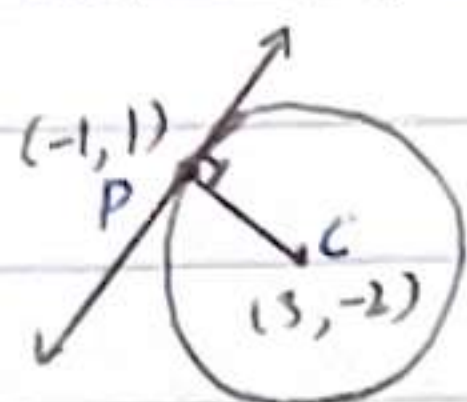
$$m = \pm \frac{\sqrt{5}}{2}$$

⑤ Circle equation $(x-3)^2 + (y+2)^2 = 25$

known that $(-1, 1)$ lies on the circle,

Find the equation of the tangent drawn to the circle at the point $(-1, 1)$

(Centre $(3, -2)$ Point of contact $(-1, 1)$)



Gradient of CP: $\frac{1-(-2)}{-1-3} = -\frac{3}{4}$

* the tangent is perpendicular to CP, so

its gradient is $-1 \times -\frac{4}{3} = \frac{4}{3}$

The equation: $Y = mx + c$

$$1 = -\frac{4}{3} + c$$

$$c = \frac{7}{3}$$

$$[Y = \frac{4}{3}x + \frac{7}{3}] \times 3$$

$$3Y = 4x + 7$$

⑥ $Y = kx + 2$, $x^2 + y^2 + 5x - 4y + 10 = 0$

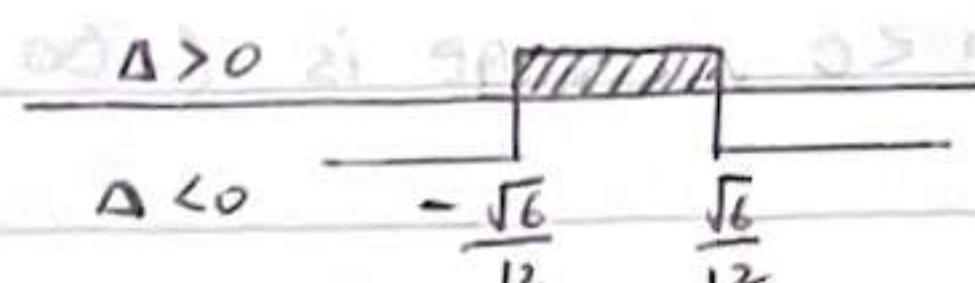
Find values of k if:

$$\Delta = \pm \frac{\sqrt{6}}{12}$$

(a) One intersection

$$k = \pm \frac{\sqrt{6}}{12}$$

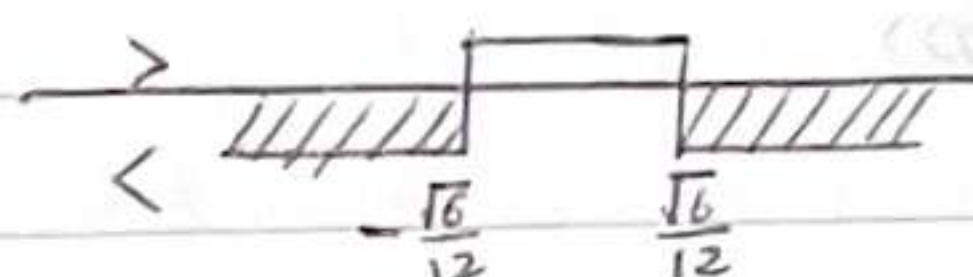
(b) two intersections $\Delta > 0$



$$k > -\frac{\sqrt{6}}{12} \quad k < \frac{\sqrt{6}}{12}$$

$$-\frac{\sqrt{6}}{12} < k < \frac{\sqrt{6}}{12}$$

(c) no intersection $\Delta < 0$

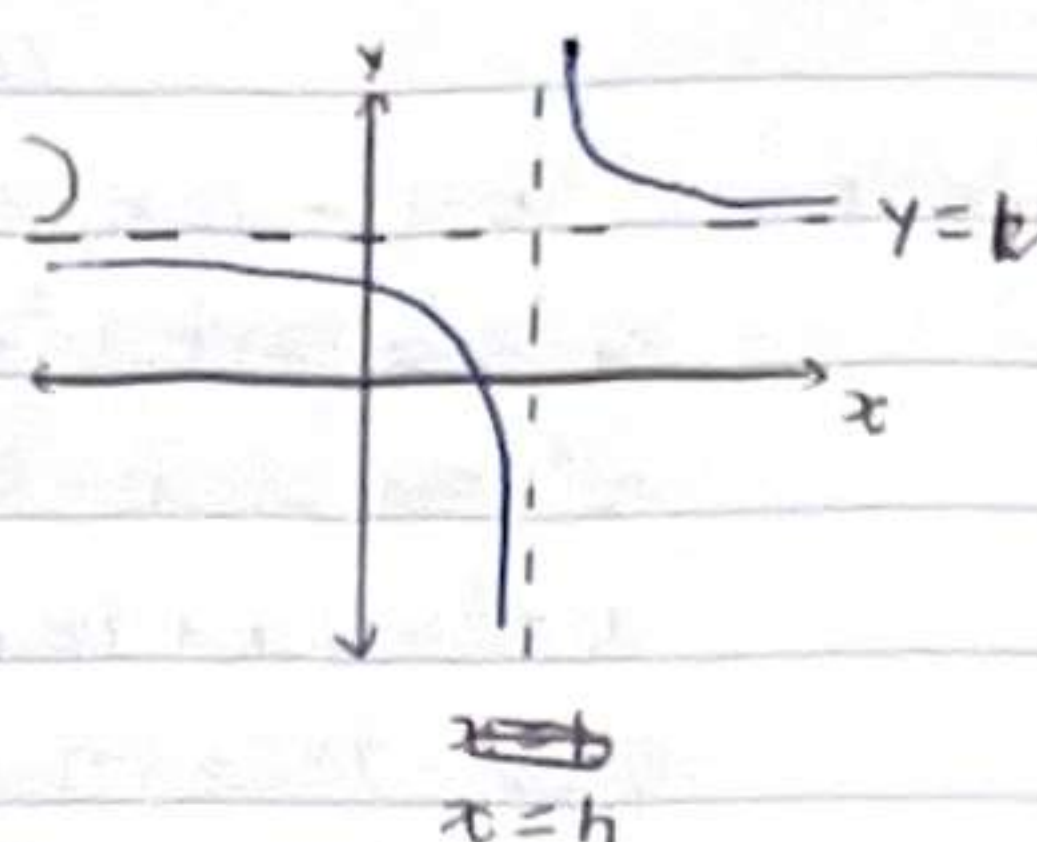


$$k < -\frac{\sqrt{6}}{12} \quad \text{or} \quad k > \frac{\sqrt{6}}{12}$$

Hyperbola & Truncus

General form of Hyperbola

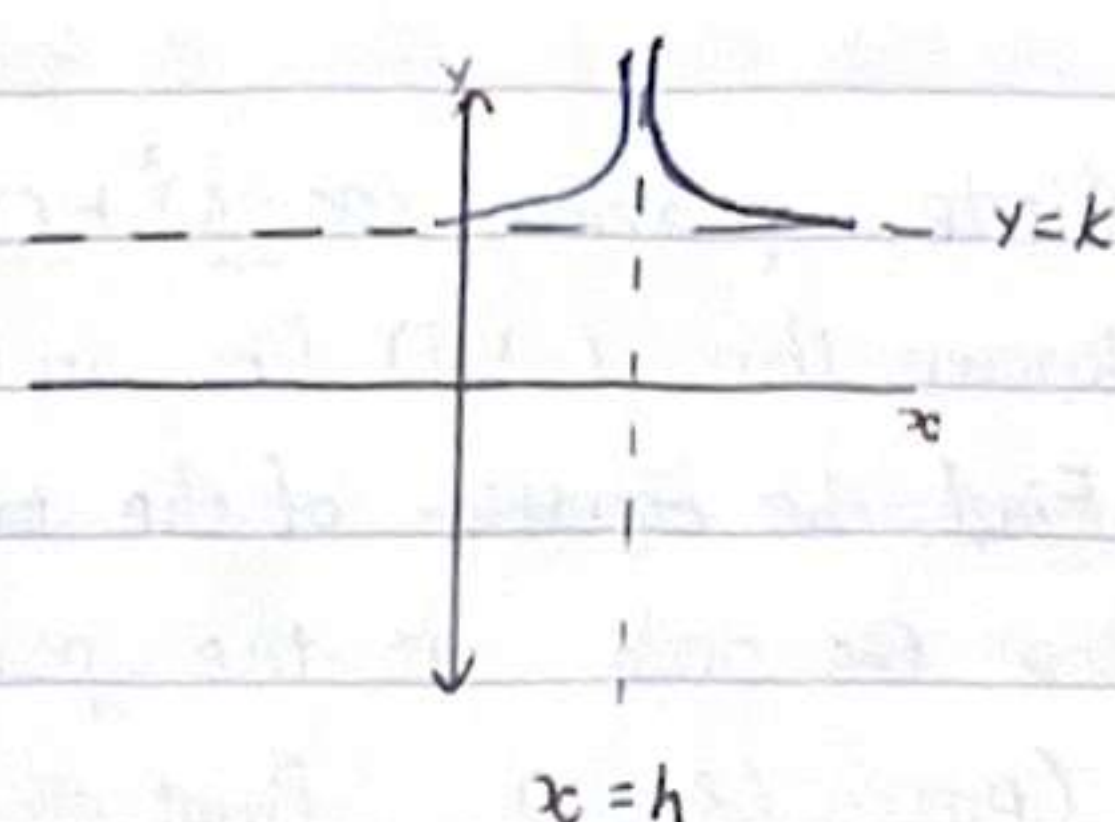
$$y = \frac{a}{x-h} + k$$



- Vertical asymptote has the equation $x=h$
- Horizontal asymptote has the equation $y=k$
- Domain is $\mathbb{R} \setminus \{h\}$
- Range is $\mathbb{R} \setminus \{k\}$

General form of Truncus

$$y = \frac{a}{(x-h)^2} + k$$



- Domain is $\mathbb{R} \setminus \{h\}$
- If $a > 0$, range is (k, ∞)
- If $a < 0$, range is $(-\infty, k)$

Example questions:

- ① In a truncus, Asymptote: $x = -3$
 $y = 5$
~~the~~ state the domain and range.
 $(5, \infty)$

- ③ Number of cattle at a time t years is modelled by $P = 30 + \frac{100}{2+t}$
 结合现实: $P \geq 0$



- ② Identify equa of the asymptotes, sketch and state the domain and range.

$$y = -2 - \frac{1}{(2-x)^2}$$

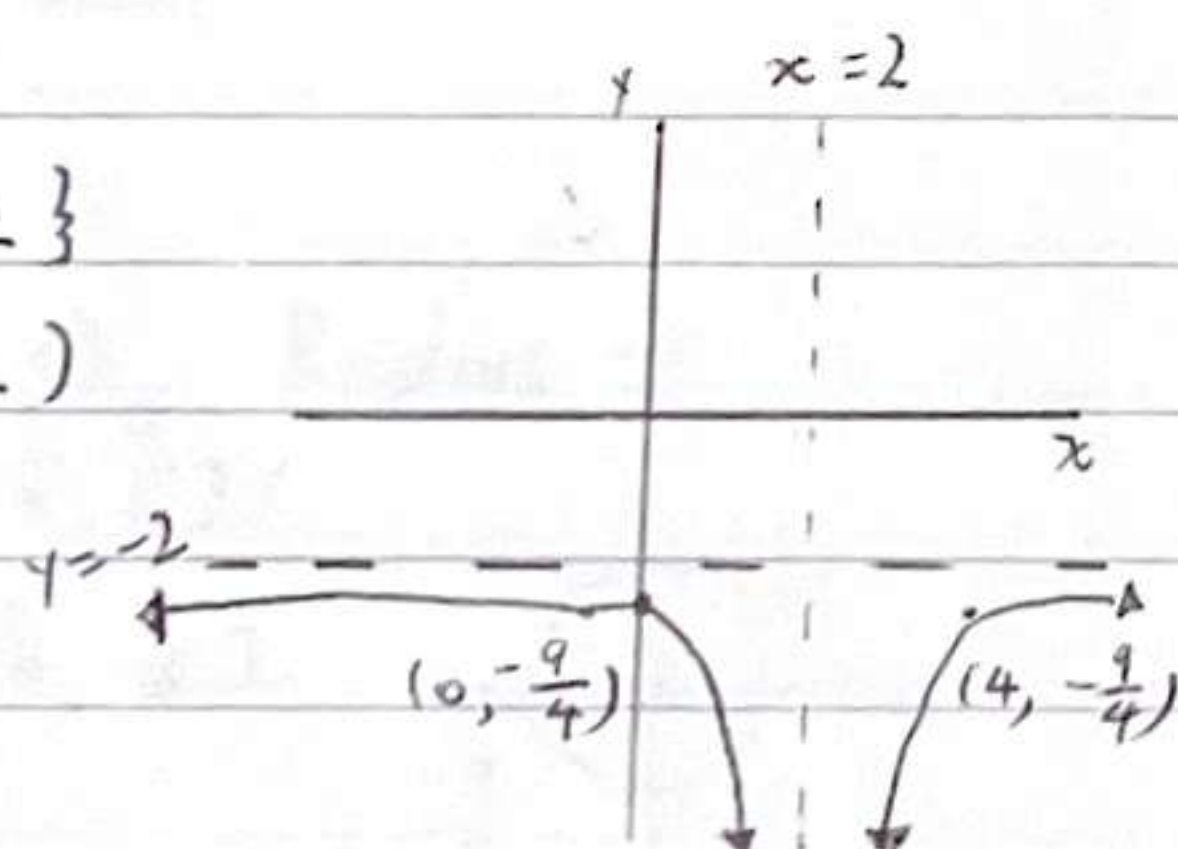
$$\text{Asymptote: } (2-x)^2 = 0 \Rightarrow x = 2, y = -2$$

$$y \text{ -in: if } x=0, y = -\frac{9}{4} \quad (0, -\frac{9}{4})$$

No x -in

Domain $\mathbb{R} \setminus \{2\}$

Range $(-\infty, -2)$



Add: $(4, -\frac{9}{4})$ is 对称 to $(0, -\frac{9}{4})$

- (a) State domain and range

Domain $\{t: t \geq 0\}$ 不能是-2但现实已限制

Range: 求原始: $t=0, P=80$

Range: $(30, 80)$

- (b) What is the minimum number the herd of cattle is expected to reach in this model?

The number of cattle will never go below 30.

The relation $y^2 = x$



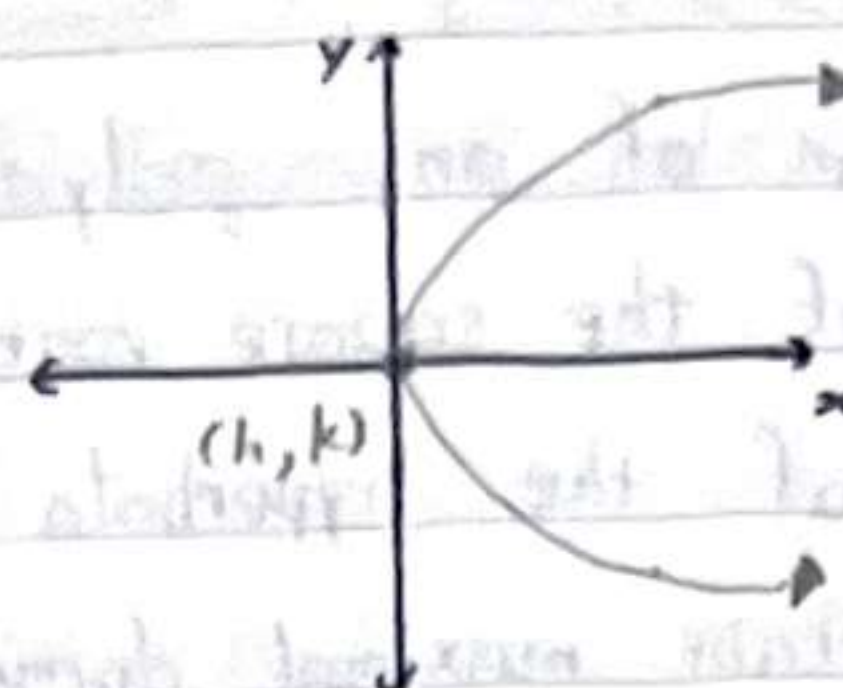
○ General form

$$(y-k)^2 = a(x-h)$$

- Axis of symmetry: $y=k$

- x -in: 设 $y=0$

- y -in: 设 $x=0$, 可以有兩個



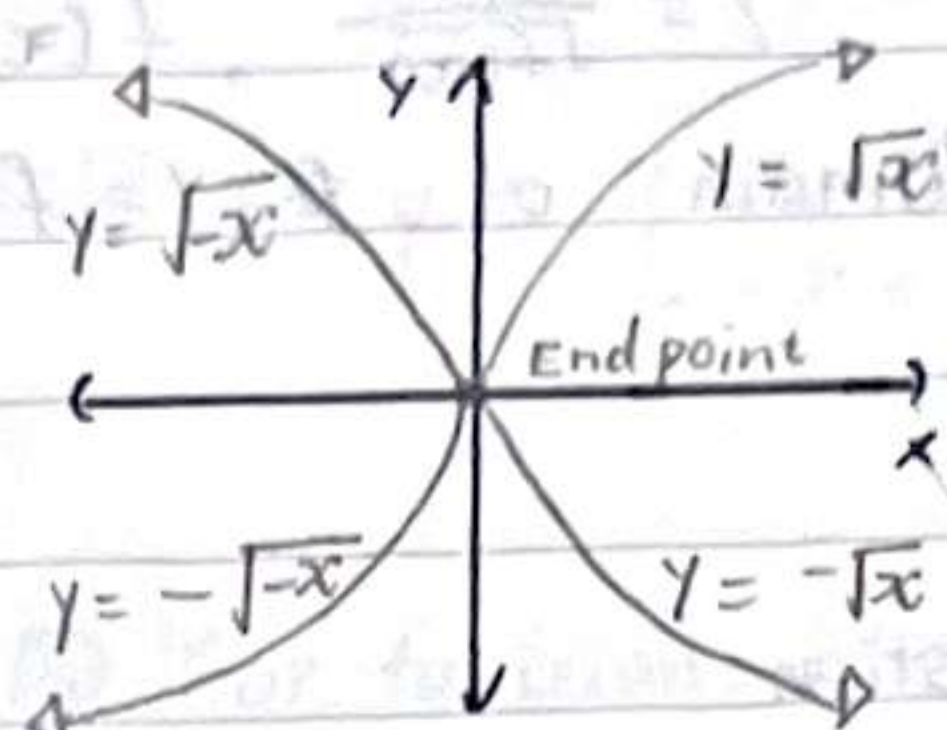
○ Square root function

$$y = a\sqrt{x-h} + k$$

- End point (h, k)

- Domain $[h, \infty)$

- Range $\begin{cases} a > 0, [k, \infty) \\ a < 0, (-\infty, k] \end{cases}$



$$y = a\sqrt{-(x-h)} + k$$

- End point (h, k)

- Domain $(-\infty, h]$

- Range $\begin{cases} a > 0, [k, \infty) \\ a < 0, (-\infty, k] \end{cases}$

* R^+ 正数
 R^- 负数 不包括 0

6.6

Other functions and relations

Maximal domain

- Maximal domain of any polynomial function is R
- the domain of the square root function $y = \sqrt{x}$ is $R^+ \cup \{0\}$
- the domain of the hyperbola $y = \frac{1}{x}$ is $R \setminus \{0\}$ as denominator cannot be zero.
- Hence, to obtain maximal domains:

$$y = \frac{g(x)}{f(x)}, f(x) \neq 0$$

$$y = \sqrt{f(x)}, f(x) \geq 0$$

$$y = \frac{g(x)}{\sqrt{f(x)}}, f(x) > 0$$

- The maximal domain of $y = f(x) \pm g(x)$ is $\underset{\substack{\downarrow \\ \text{domain of} \\ f(x)}}{df} \cap \underset{\substack{\downarrow \\ \text{domain of} \\ g(x)}}{dg}$

- Example Question:

Find the maximal domain

(a) $\frac{1}{x^2-9}$

$$x^2-9 \neq 0$$

$$x \neq \pm 3$$

$$\text{MDomain } R \setminus \{\pm 3\}$$

(b) $y = \frac{x-4}{x^2+9}$

$$x^2+9 \neq 0$$

$$x^2 \neq -9$$

$$\text{MDomain: } R$$

(c) $y = \frac{1}{\sqrt{x^3-8}}$

$$x^3-8 > 0$$

$$x^3 > 8$$

$$x > 2 \quad \text{MDomain: } (2, \infty)$$

Equation of the inverse

- To obtain equation of the inverse: interchange x and y in the equation of the original and make y the subject
- If the inverse of a function f is itself a function, then the inverse function is denoted f^{-1} .
* for f^{-1} to exist, f must be a one-to-one function.

- The inverse ~~function~~ of equation of $f(x) = \sqrt{x}$ is $f^{-1}(x) = x^2, x \geq 0$

写成 mapping notation: $f: [0, \infty) \rightarrow R, f(x) = \sqrt{x}$
 $\rightarrow$ real number

Inverse ~~function~~: $f^{-1}: [0, \infty) \rightarrow R, f^{-1}(x) = x^2$

Domain of f^{-1} : $d_{f^{-1}}$

Range of f^{-1} : $R_{f^{-1}}$

- The inverse of the cubic function $y = x^3$ is $f: R \rightarrow R, f(x) = \sqrt[3]{x}$

∞ The many to one (2个x get 1个y) function has an inverse with a one to many correspondence.

- If there is an intersection between two equations (one with its reverse), the intersection will always occur at the mirror line $y=x$. For example: $y = 2-x^2$, inverse: $x = \pm \sqrt{2-y}$, Find intersection
 $x = 2-x^2$ (solve for)

Example Question

① State the maximal domains.

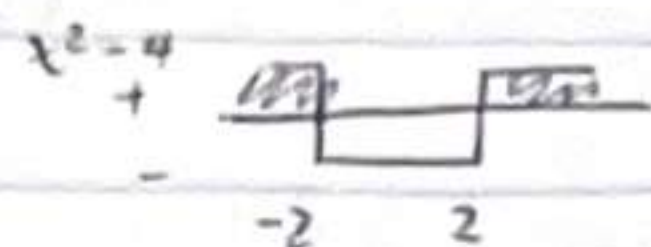
$$y = \sqrt{x^2 - 4}$$

$$f(x) \geq 0$$

$$x^2 - 4 \geq 0$$

$$x^2 \geq 4$$

$$x \geq \pm 2$$



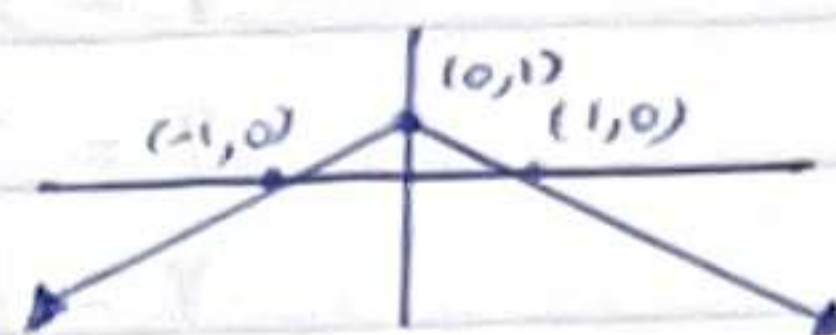
$$x < -2 \cup x > 2$$

$$x \leq -2 \cup x \geq 2$$

⑤ Hybrid functions

Example: $y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

⑥ Form a rule.



Equation $(-1,0)(0,1)$: $m = \frac{1-0}{0-(-1)} = 1$

$$y = x + 1$$

Equation $(0,1)(1,0)$: $m = -1$

$$y = -x + 1$$

Rule is $y = \begin{cases} x+1, & x \leq 0 \\ -x+1, & x > 0 \end{cases}$

② For function, $f: [0, 3) \rightarrow \mathbb{R}$, $f(x) = 2x - 2$

(a) Domain and range of f

Domain $[0, 3)$

End point: $x=0, y=-2$ (close)

$x=3, y=4$ (open)

Range is $[-2, 4)$

(b) State the f^{-1} 's domain and range

$$df^{-1} = rf = [-2, 4)$$

$$rf^{-1} = df = [0, 3)$$

(c) Form rules of inverse

调换 y 和 x , 原: $y = 2x - 2$

换: $x = \frac{y+2}{2}$

$$y \text{ as the 主角: } f^{-1}(x) = \frac{x+2}{2}$$

Inverse function is

$$f^{-1}: [-2, 4) \rightarrow \mathbb{R}, f^{-1}(x) = \frac{x+2}{2}$$

③ Find the rule for the inverse $y = \sqrt{2x+1}$

$$x = \frac{y^2 - 1}{2}$$

$$y = \frac{x^2 - 1}{2}, x \geq 0$$

④ Given $f: [a, \infty) \rightarrow \mathbb{R}$, $f(x) = x^2 - 4x + 9$

find the smallest value of a , so that f^{-1} exist

For f^{-1} exist, the parabola must be one-to-one

Factorise: $f(x) = (x-2)^2 + 5$ TP $(2, 5)$

so the domain is $[2, \infty)$, the function

is 1 to 1.

⑥ For function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} -x-1, & x < -1 \\ \sqrt{1-x^2}, & -1 \leq x \leq 1 \\ x+1, & x > 1 \end{cases}$

(a) calculate value of

(i) $f(0)$ $0 < -1$ 所以 $-x-1$, value $= -1$

(ii) $f(-2)$ $-2 < -1$ value $= 1$

(b) Show the function is not continuous at $x=1$

The two branches around $x=1$ are

$y = \sqrt{1-x^2}$, $-1 \leq x \leq 1$ and $y = x+1$, $x > 1$.

For $y = \sqrt{1-x^2}$, there is a closed endpoint.

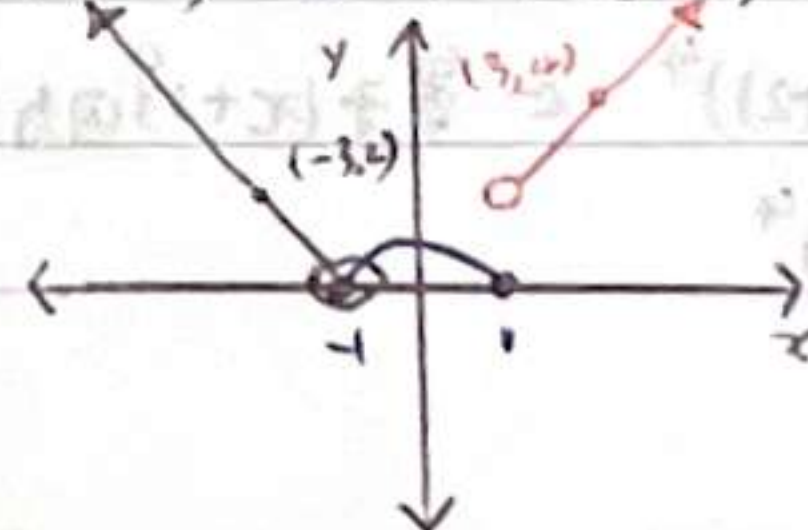
$(1,0)$ but for $y = x+1$, there is an open endpoint $(1,2)$. The two branches do not join. Hence the function is not continuous at $x=1$ as there will be a break in its graph.

(c) Sketch the graph and state its correspondence.

N $-x-1$, $x=-1$ $y=0$, $x=-3$ $y=2$

M $\sqrt{1-x^2}$, $x=-1$ $y=0$, $x=1$ $y=0$

N $x+1$, $x=1$ $y=2$, $x=3$ $y=4$



many-to-one

Transformation of Functions

○ Horizontal

$$y = x^2 \rightarrow y = (x-h)^2$$

$$y = \frac{1}{x} \rightarrow y = \frac{1}{x-h}$$

$$y = \sqrt{x} \xrightarrow{\text{to right}} y = \sqrt{x-h}$$

○ Vertical

$$y = x^2 \rightarrow y = x^2 + k$$

$$y = \frac{1}{x} \rightarrow y = \frac{1}{x} + k$$

$$y = \sqrt{x} \xrightarrow{\text{upward}} y = \sqrt{x} + k$$

○ Reflect

$$x\text{-axis} \rightarrow -f(x)$$

* 在原 equa touch x-in 的地方反转

$$y\text{-axis} \rightarrow f(-x)$$

* 在原 equa touch y-in 的地方反转

○ Dilation

$$x\text{-axis} \rightarrow af(x)$$

* 在 equa 外

$$y\text{-axis} \rightarrow f(ax)$$

* 在 equa 内

Factor of $\frac{1}{2}$: $y = (2x)^2 = 4x^2$

Factor of 2: $y = (\frac{1}{2}x)^2$

$$= \frac{x^2}{4}$$

○ Example Q ★:

① For $y = x^4$, display the equa after:

(a) reflected in the y-axis and move 2 units to left

$$\begin{aligned} y &= x^4 \\ &= (-x)^4 \leftarrow \text{等于 } x^4 \text{ 因为 } -x \times -x = x^2 \\ &= x^4 \\ &= (x+2)^4 \end{aligned}$$

② For $y = \frac{1}{x}$, show dilate from y-axis with factor of $\frac{1}{2}$ then move 3 units to left

$$\begin{aligned} y &= \frac{1}{x} \\ &= \frac{1}{2x} \\ &= \frac{1}{2(x+3)} \leftarrow \text{横移时, units 紧贴 } x \end{aligned}$$

③ For $y = \frac{1}{x}$, dilate by 3 in y-axis, reflect in y-axis

$$\begin{aligned} y &= \frac{1}{x} \\ y &= \frac{1}{(\frac{x}{3})} \leftarrow \text{dilate, 紧贴 } x \text{ in y-axis 时} \\ &= \frac{3}{x} \\ &= \frac{3}{(-x)} \leftarrow \text{reflect, in y-axis 时 紧贴 } x \\ &= -\frac{3}{x} \end{aligned}$$

(b) move 2 units upwards and reflected in x-axis

$$\begin{aligned} y &= x^4 \\ y &= (x-2)^4 \\ y &= ((-x)-2)^4 \\ &= (-(x+2))^4 \leftarrow \text{等于 } (x+2)^4 \text{ 因为负数相乘是正} \\ &= (x+2)^4 \end{aligned}$$

④ Give the coordinates of the image

of the point $(3, -4)$ if it is

(a) 2 units to left, 4 units down

$$(3-2, -4-4)$$

$$= (1, -8)$$

(b) reflected in the y-axis,

reflected in the x-axis

$$\text{in y-axis: } (-3, -4)$$

$$\text{in x-axis: } (-3, 4)$$

(c) dilated by a factor $\frac{1}{5}$ from the

x-axis parallel to the y-axis

$$(3, -4 \times \frac{1}{5})$$

$$= (3, -\frac{4}{5})$$

(d) dilated by a factor $\frac{1}{5}$ from the

y-axis parallel to the x-axis

$$(3, -4)$$

$$(3 \times \frac{1}{5}, -4)$$

$$(\frac{3}{5}, -4)$$

12.2

Average rate of change

① $x = t^3 + 4t^2 + 3t$

(a) calculate average rate of change of distance x over time $t \in [0, 2]$ and $[0.9, 1.1]$

$$f(0) = 0 \quad f(2) = 30 \quad \frac{30-0}{2-0} = 15$$

$$f(0.9) = 6.669 \quad f(1.1) = 9.471 \quad \frac{9.471 - 6.669}{1.1 - 0.9} = \frac{2.802}{0.2} = 14.01$$

(b) which of the average speed in part (a) would be the better estimate for the instantaneous speed at $t = 1$?Since $t = 1$ is closer to the range $[0.9, 1.1]$ thus $t \in [0.9, 1.1]$ would be the better estimate.

12.4

The derivative function

○ Find the gradient at a point by differentiating from **first principles**.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{(x+h) - x}$$

12.5

Differentiation of polynomials by rule

○ Rule

$$\text{If } f(x) = x^n, \quad f'(x) = nx^{n-1}$$

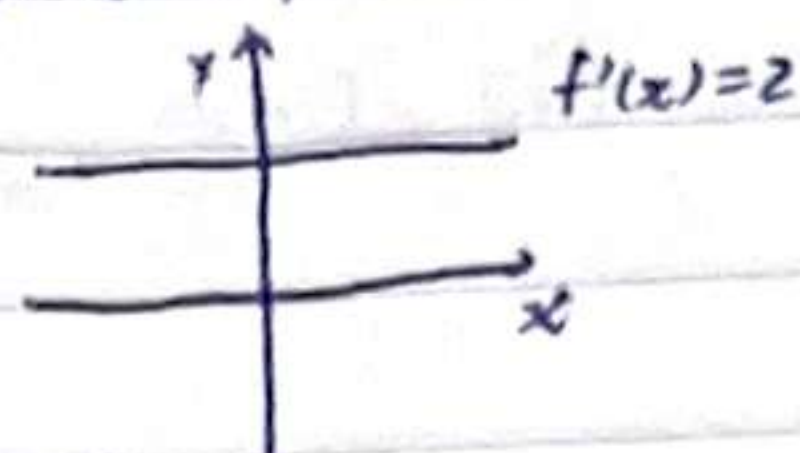
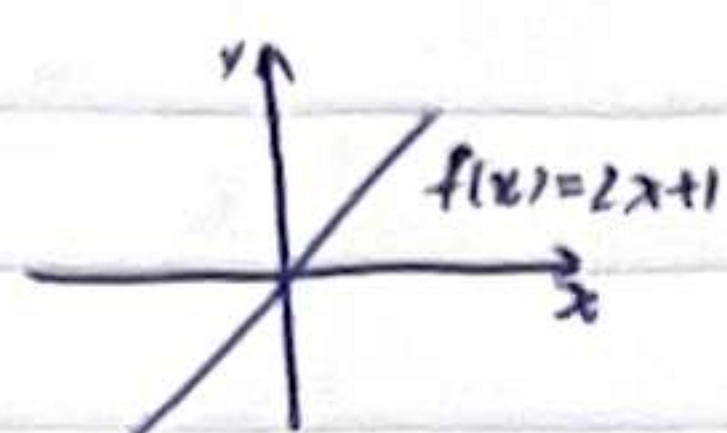
○ Example

$$f(x) = x^3 + x^2 + 1$$

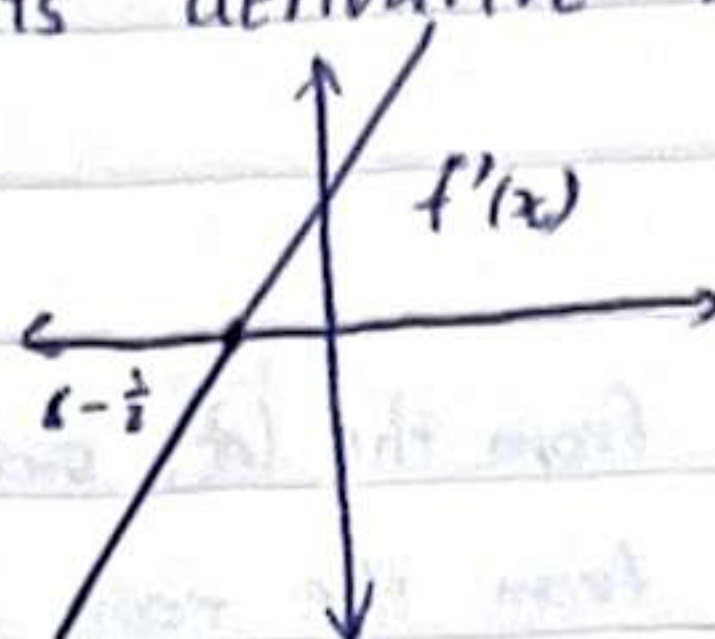
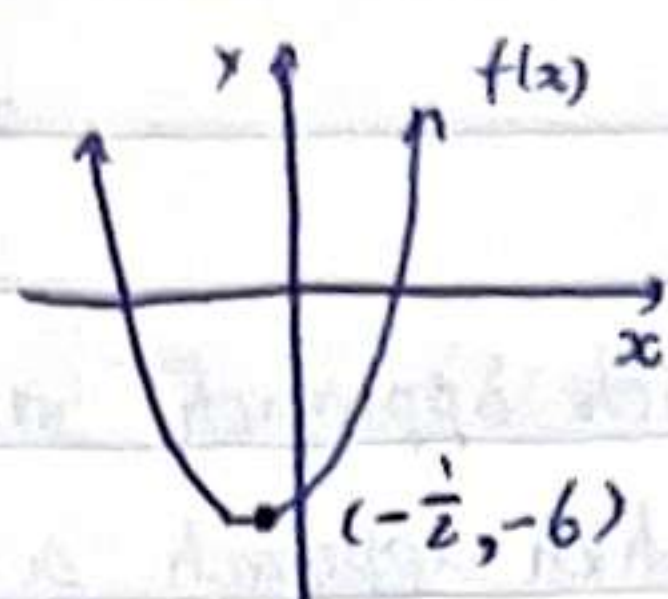
$$f'(x) = 3x^2 + 2x$$

o Sketch derivative

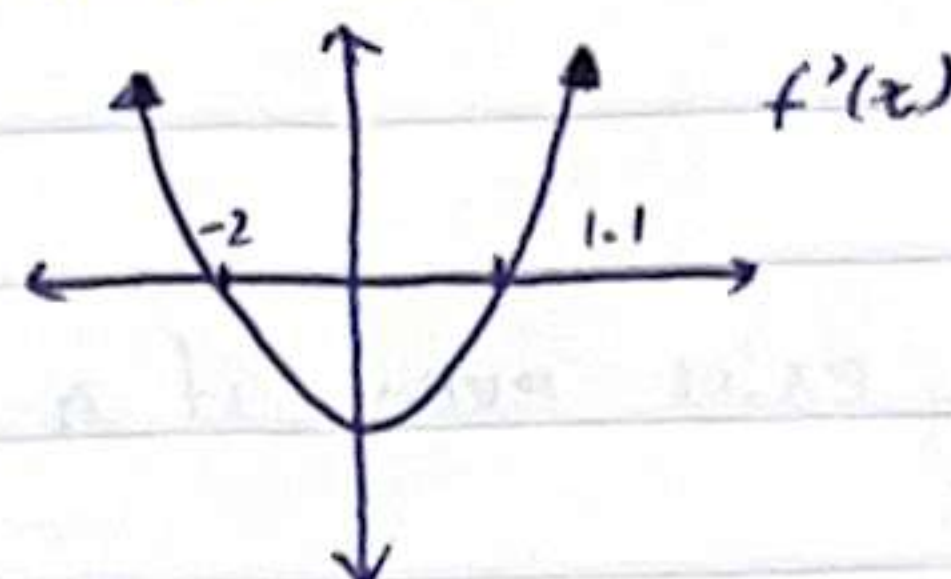
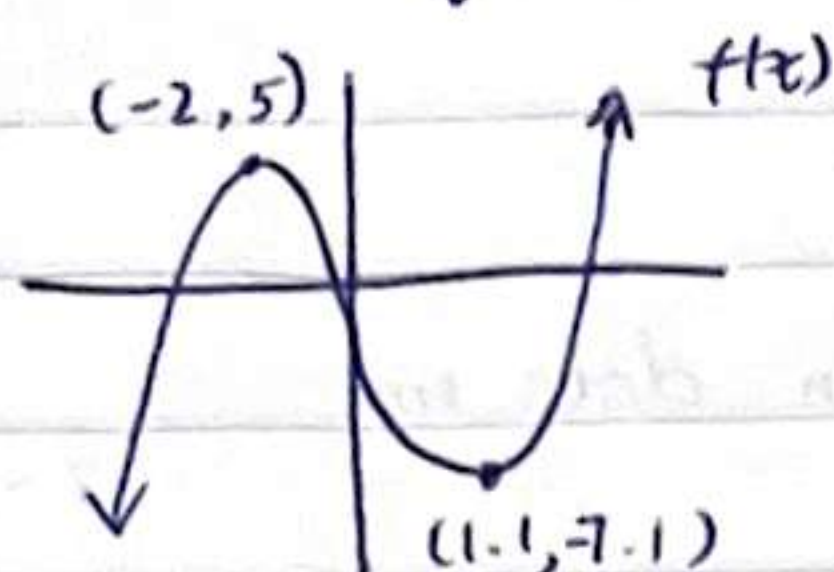
- If a function is a quadratic, its derivative will be linear



- If a function is a cubic, its derivative will be a quadratic

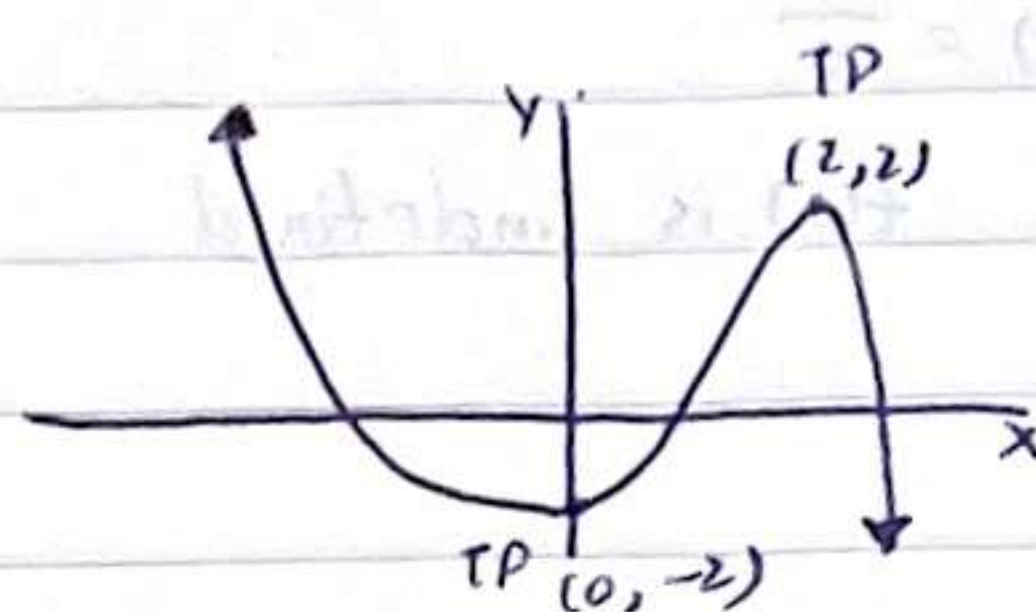


- If a function is a quartic, its derivative will be a cubic



• Example:

Sketch $\frac{dy}{dx}$ for



Answer: cubic $\rightarrow$ parabola

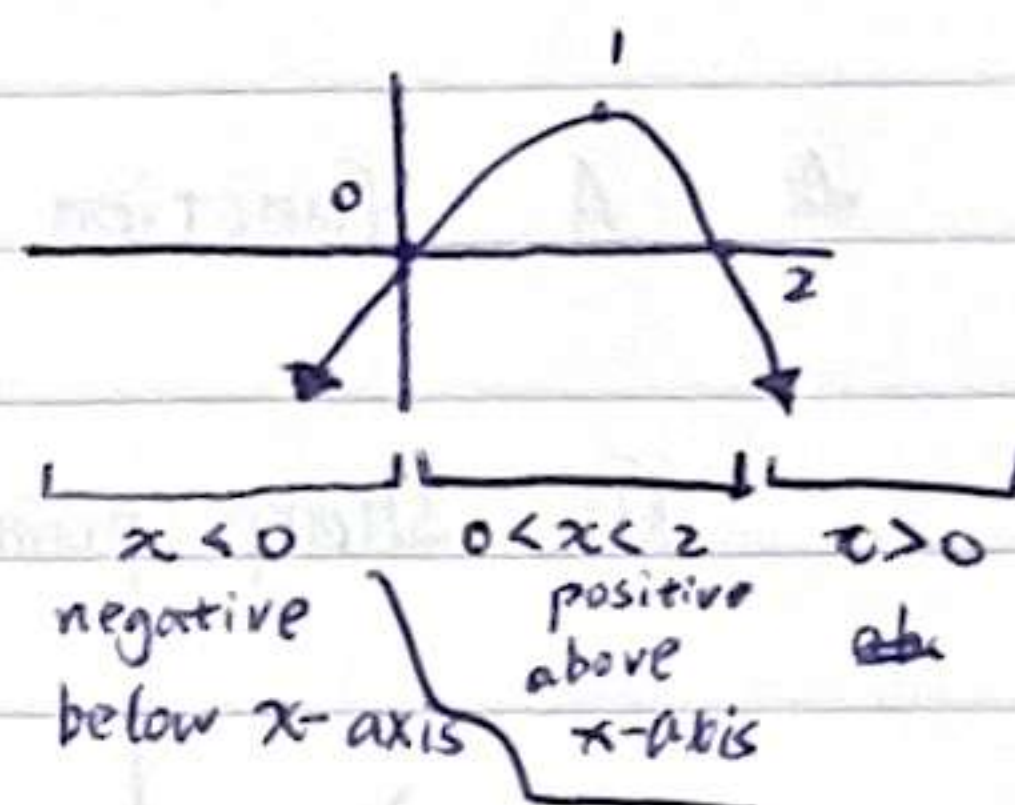
x -in: 0, 2

$x < 0$: negative $\searrow$

$0 < x < 2$: positive $\nearrow$

$x > 2$: negative $\searrow$

TP: $\frac{2+0}{2} = 1$



o CAS.

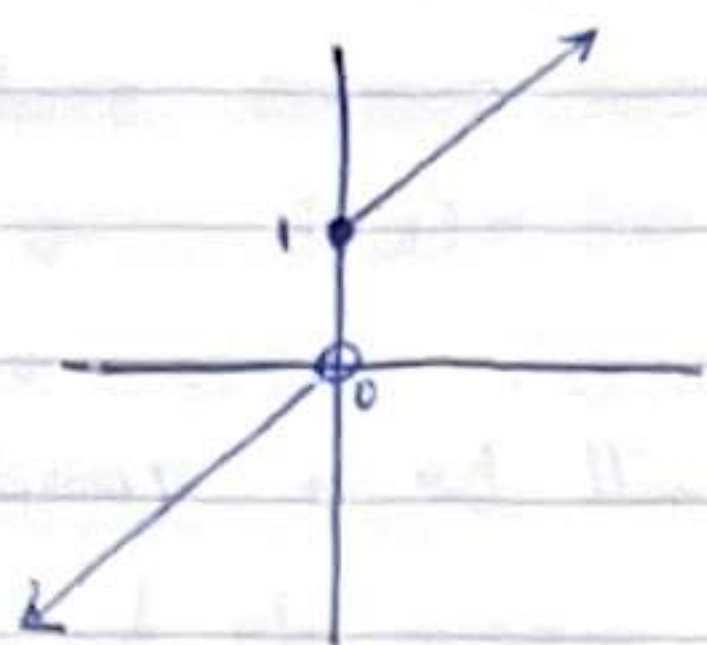
- Find the rate of change at (1, 2)

Iterative / calculation / diff / at value: 1

- Find equation of derivative / differentiate a function

Iterative / calculation / diff / Differentiation

★ A function can have two limits.



$$f(x) = \begin{cases} -x, & x < 0 \\ x+1, & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = 0$$

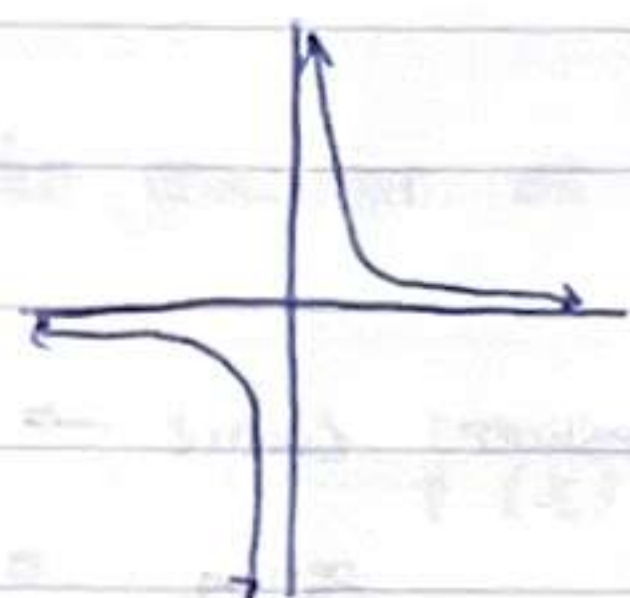
$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$\lim_{x \rightarrow 0^-} f(x)$: approaching from the left side, which approach a y-value of 0

$\lim_{x \rightarrow 0^+} f(x)$: approaching from the right side, which approach a y-value of 1.

Overall $\lim_{x \rightarrow 0} f(x)$ does not exist as $\lim_{x \rightarrow 0^-} \neq \lim_{x \rightarrow 0^+}$

★ A limit can exist even if a function does not



$$f(x) = \frac{1}{x}$$

$f(0)$ is undefined.

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

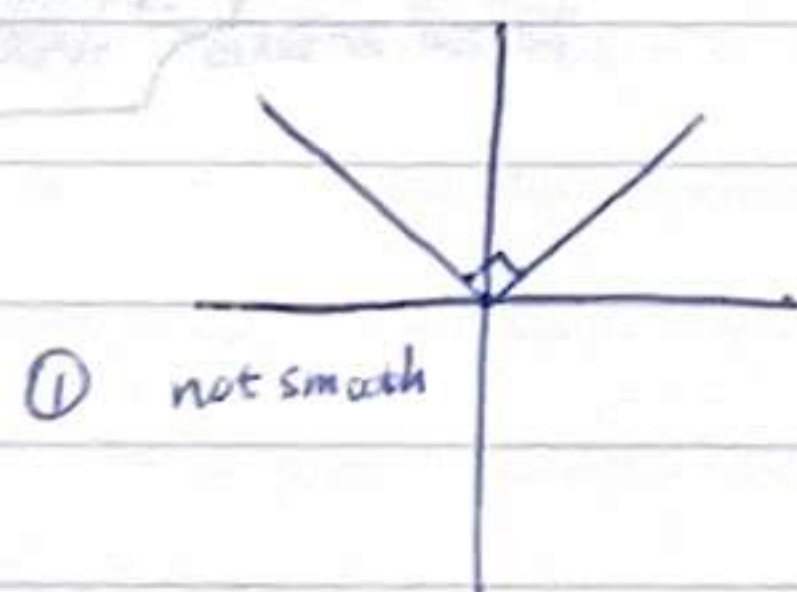
Overall $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist as limit 不相等

★ A function is NOT differentiable at

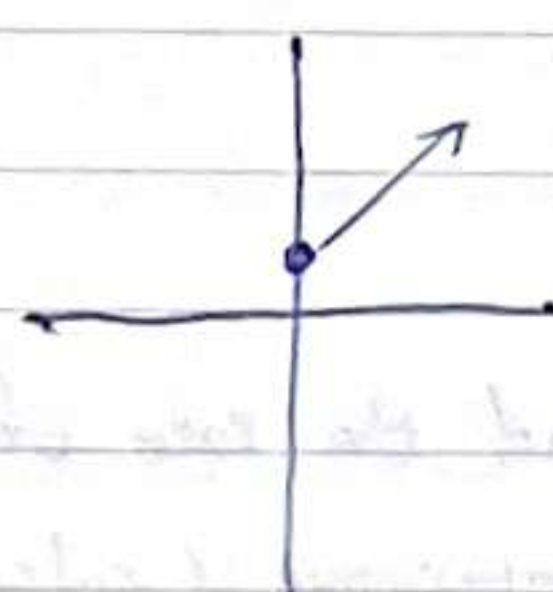
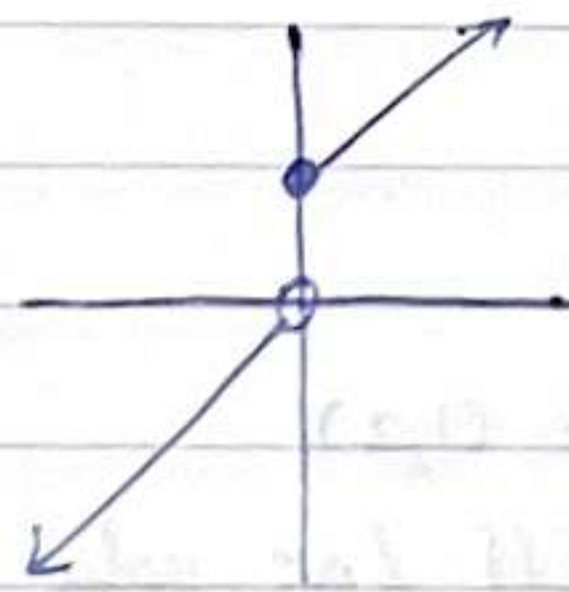
① sharp points (cusps)

② Jumps

③ Endpoints



① not smooth



②和③是 discontinuity

A function is only differentiable when it is both smooth & continuous.

13.3

Differentiate different type of function.

$$x^{-n} = \frac{1}{x^n}, \quad \frac{1}{x^n} = x^{-n}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x}, \quad \sqrt[n]{x} = x^{\frac{1}{n}}$$

- for $f(x) = x^{-n}$, $f'(x) = -n x^{-n-1}$

例: x^{-3}

$$f'(x) = -3x^{-4}$$

- 例:

$$\textcircled{1} y = \sqrt{x}$$

$$= x^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{1}{2} \times \frac{1}{x^{\frac{1}{2}}}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

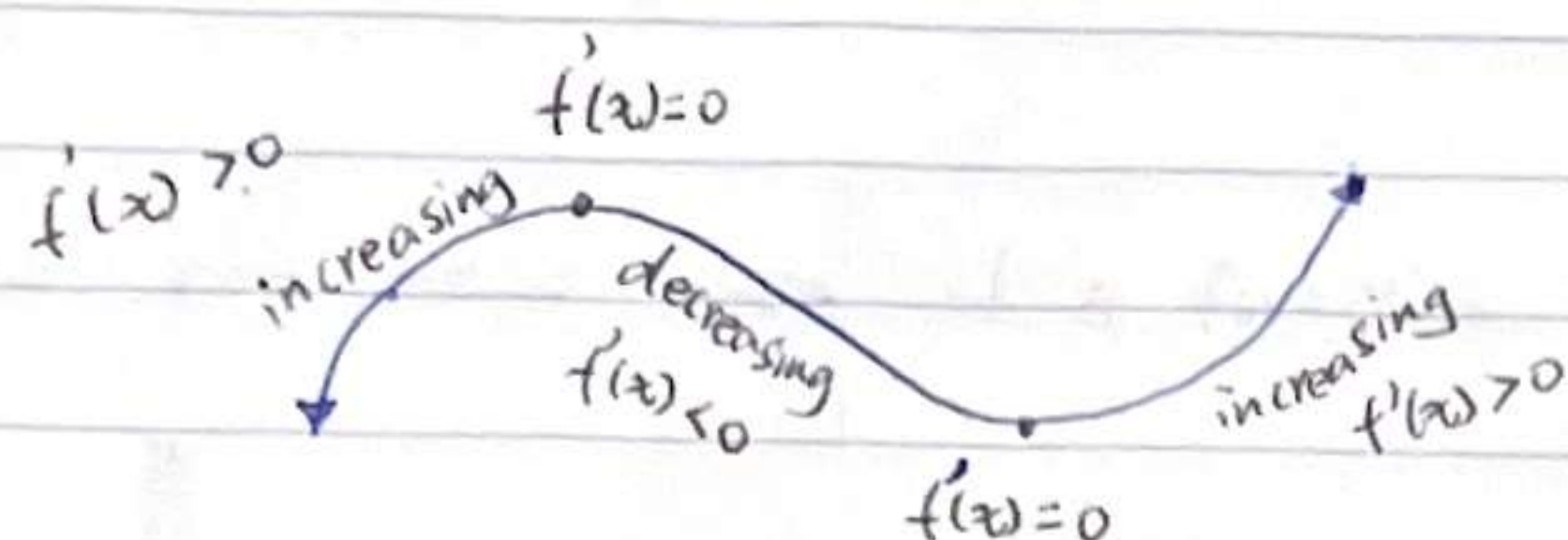
13.4.

Newton's Method of finding x -intercepts

- ① Make one guess of x -in, such as $x=2$
- ② Sub into eqn $f(x)$ to see if $y=0$
- ③ if not, get a new x value by $\text{New } x \text{ value} = x - \frac{f(x)}{f'(x)}$
- ④ Loop until $y=0$

Increasing and Decreasing function

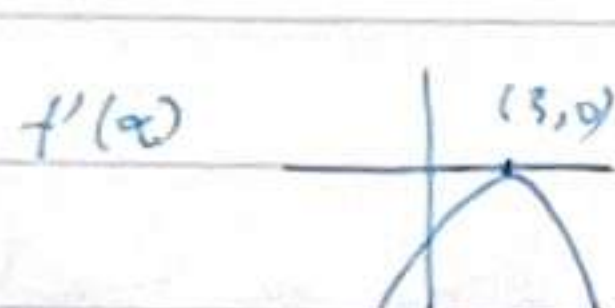
- A function is increasing where $f'(x)$ is positive and decreasing where $f'(x)$ is negative



$$\boxed{f(x) \text{ TP} = x\text{-in of } f'(x)}$$

$$\boxed{x\text{-in of } f'(x) = f(x) \text{ 的 TP}}$$

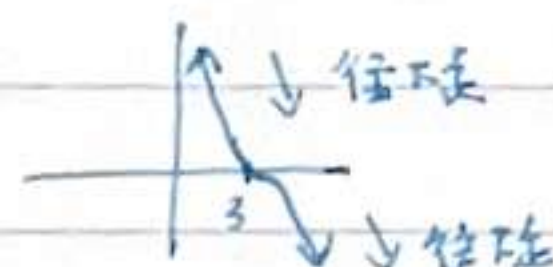
$f'(x) = 3$ 代表 $f(x)$ 是 Any line with gradient of 3 $\neq (1,3)$



TP之前是 negative (在 x -axis 下) $\searrow$ 往下走

TP之后也是 negative $\searrow$ 往下走

代表 $f(x)$ 是 Inverted cubic with PT 在 $x=3$



Curve sketching

O Stationary points

* At a stationary point on a curve $y=f(x)$, $f'(x)=0$

- Three types of stationary points

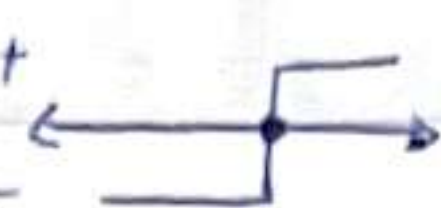
1. (local) minimum turning point

2. (local) maximum turning point

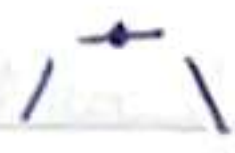
3. stationary point of inflection

- Minimum TP 

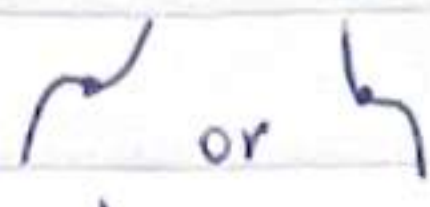
- slope of tangent  ← 这就是 sign test

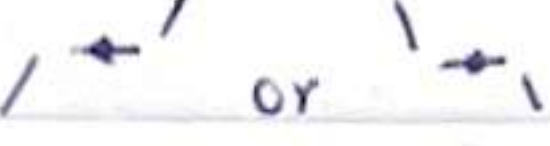
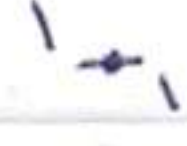
- sign diagram of $f'(x)$ 

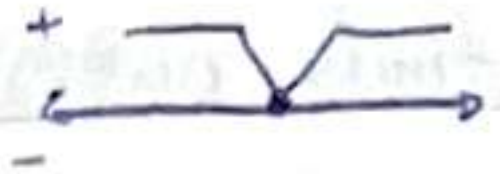
- Maximum TP 

- slope of tangent 

- sign diagram of gradient function $f'(x)$ 

- Stationary point of Inflection  or 

- slope of tangent  or 

- sign diagram of gradient function $f'(x)$  or 

- Global maximum point / Absolute Max point 指全图最高的点

- Local maximum point 指周围有别的点衬托的高点, 如: 

- 一个图可以没有 local max point

13.6

Application Question

○ Use calculus to calculate the greatest area

① when $f'(x) = 0$, find the value of x

② substitute x into $f(x)$, get greatest area.

13.7

Speed, Velocity & Acceleration

○ $f(x)$

- This finds out displacement 点 (particle) 目前的位置 according to t (time)
- write as $x = 3t^2 - 6t$
- if $x = 0$, the particle is at origin
- if $x > 0$, the particle is ~~at~~ to the right from origin

○ $f'(x)$

- This finds out the velocity (sometimes equal to speed)
- write as 6 m/s

○ $f''(x)$

- This finds out the acceleration
- $f''(x)$ is further differentiate the $f'(x)$, write as m/s^2

○ Average

$$\text{Average velocity} = \frac{\text{change in displacement}}{\text{change in time}} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$\text{Average acceleration} = \frac{\text{change in velocity}}{\text{change in time}} = \frac{V_2 - V_1}{t_2 - t_1}$$

0 从 $f'(x)$ 变 $f(x)$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

- Example

calculate indefinite integrals

$$\begin{aligned} \int \frac{5x^6 + 3x^3}{4x^2} dx \\ &= \frac{5x^6}{4x^2} + \frac{3x^3}{4x^2} \\ &= \frac{5}{4}x^4 + \frac{3}{4}x \\ &= \frac{5}{4} \times \frac{x^5}{5} + \frac{3}{4} \times \frac{x^2}{2} \\ &= \frac{5}{28}x^5 + \frac{3}{8}x^2 + C \end{aligned}$$

0 CAS

方法1: (a) Interactive / Calculation / $\int$ (第三个)

(b) 选 Indefinite integral

(c) Type Expression: of $f'(x)$

方法2: (a) Keyboard / Math 2 / $\int$ (第三行第四个)

(b) 出来这个: $\int \square d\square$ 这里填 variable 'x'
 $\square \uparrow$
 这里填 Equa of $f'(x)$

注意!!! 填 Expression 时不要省略括号

如: $x^{\frac{3}{2}}$ 要 Type: $x^{(3/2)}$

再比如: $\frac{(x+4)x}{2x^4}$ 要 Type: $[(x+4)x] \div (2x^4)$

用CAS算完别忘记加 $+ C$

14.4

Application of anti-differentiation

Displacement $\rightarrow$ Velocity $\rightarrow$ acceleration $x(t)$ $v(t)$ $a(t)$

14.5

Definite integral

Definite integral: $\int_a^b$ Equaaaa... dx
 $\uparrow$
 通常是 $v(t)$

例: $\int_0^2 3x^2 dx$

① 先 ~~求~~ antidifferentiate: $\int_0^2 3x^2 dx = [x^3]_0^2$

② 套 b - 套 a

$= (2)^3 - (0)^3$

③ 得出 value

$= 8 \leftarrow$ 得出 distance

B.2

Probability

$$\frac{n(R)}{n(\text{total})} = \Pr(R)$$

Additional formula: 适用于 both independent and dependent events

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

* 1 不是质数 (prime number)

B.3

Conditional Probability

(The probability of obtaining two Heads given at least one Head has occurred is called conditional probability.)

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

↑ It can be written as $P(A|B)$ where A is the event of obtaining two heads and B is the event of at least one Head.

B.4

Independence

if two events A and B are independent, then:

$$\Pr(A) \times \Pr(B) = \Pr(A \cap B)$$

$$\Pr(A|B) = \Pr(A)$$

Counting techniques

Example Question: 5 students choose from 7 boys and 4 girls to form a committee.

(a) How many committees can be formed?

$$\text{Total} = 7 + 4 = 11$$

→ choose 5 people from 11

can write as: $\binom{11}{5}$ = read as "11 choose 5"

$$= {}^{11}C_5$$

$$= \frac{11!}{5! \times (11-5)!}$$

$$= \frac{11!}{5! 6!} \leftarrow \text{CAS: Keyboard / Catalog } \rightarrow \text{Advance}$$

$$= 462 \quad / nCr(11, 5) / \text{'EXE'}$$

$${}^aC_b = \frac{a!}{b!(a-b)!}$$

(b) How many committees contains exactly 2 boys and 3 girls?

$$\text{男总数} \rightarrow {}^7C_2 \times {}^4C_3 = 84$$

(c) How many committees have at least 3 girls.

choose 3 from 4 girls

choose 4 from 4 girls

choose 2 from 7 boys

or choose 1 from 7 boys

= 5

$$\text{总而言之: } {}^7C_2 \times {}^4C_3 + {}^4C_4 \times {}^7C_1 =$$

(d) How many committees have both the youngest & oldest student?

What is the $\text{Pr}()$?

oldest youngest

X X 3 4 5

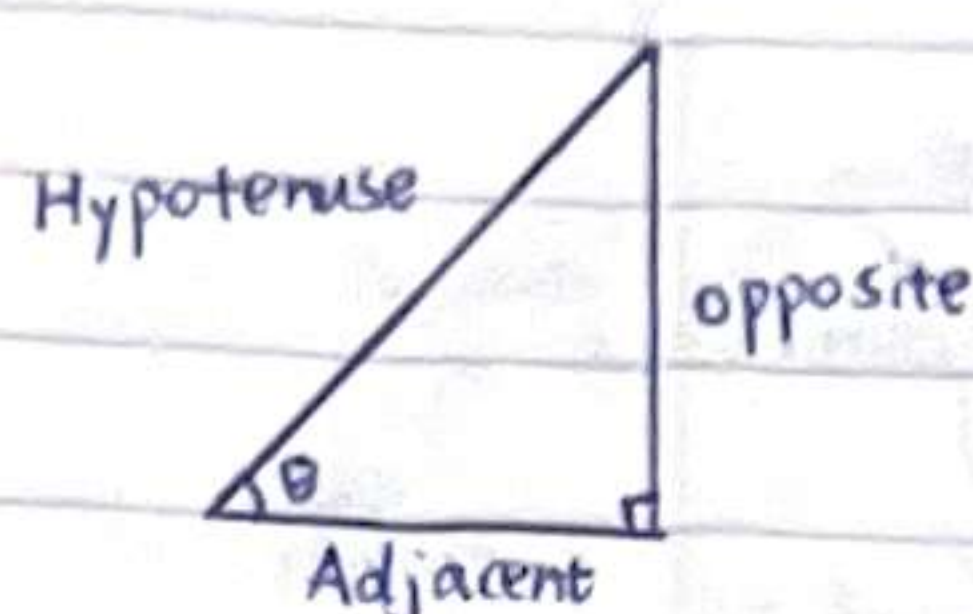
9 people left after (11-2) to choose 3

$${}^9C_3 = 84$$

$$\text{Pr}() = \frac{84}{462}$$

9.2

Trigonometric Ratios



SOH $\sin(\theta) = \frac{O}{H}$
 CAH $\cos(\theta) = \frac{A}{H}$
 TOA $\tan(\theta) = \frac{O}{A}$

只对直角三角形适用

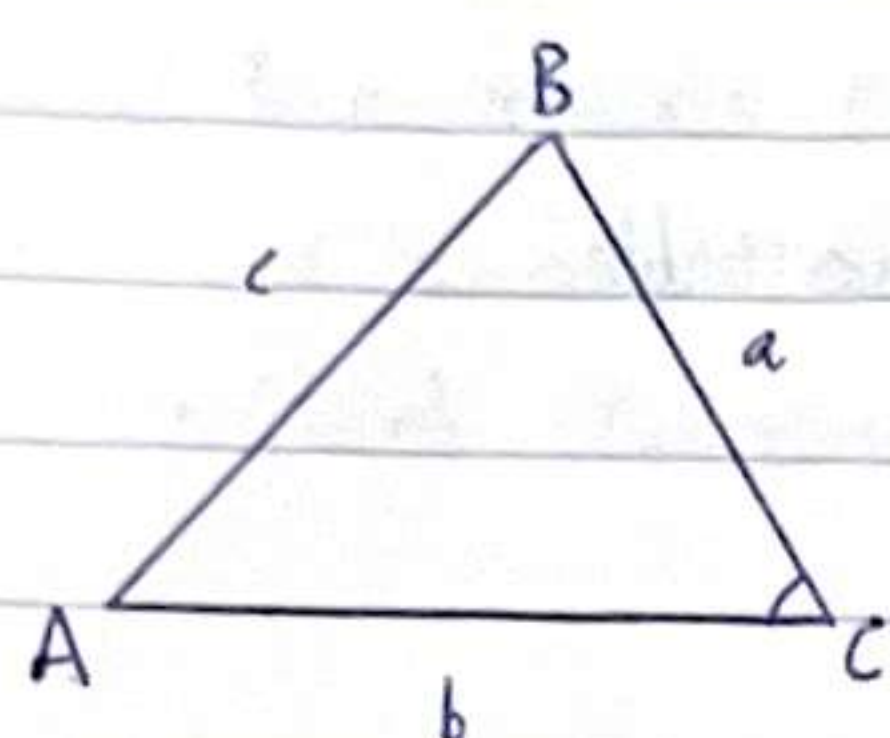
Pythagoras' theorem:

$$a^2 + b^2 = c^2$$

Exact value table

	0°	30°	45°	60°	90°	180°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
Sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
Tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	∞	0

undefined



适用于任何三角形

$$\text{Area} = \frac{1}{2} ab \sin(C)$$

必须是夹角 $\angle C$

9.3

Circular measure

radians 标志

$$\pi^c = 180^\circ$$

$$\text{Radians to degrees: } 2\pi^c \times \frac{180^\circ}{\pi^c} = 360^\circ$$

$$\text{Degrees to radians: } 360^\circ \times \frac{\pi^c}{180^\circ} = 2\pi^c$$



$$\text{Length of Arc} = r \theta$$

angle in radians

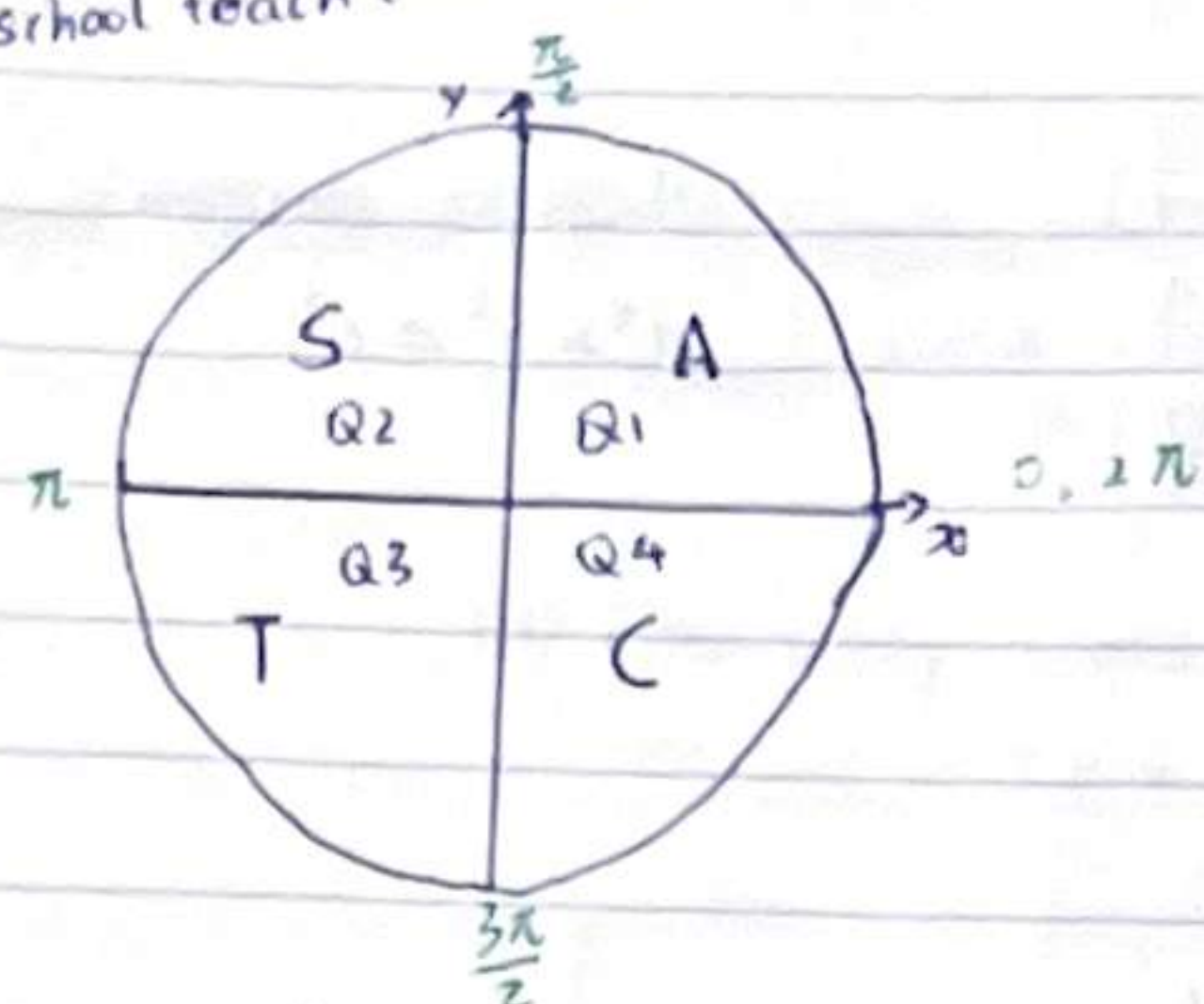
CAS: degree to radian

例: Convert $112^\circ 15'$ to Radian① Degree mode, 用 $\square\square\square$ 输入 $112^\circ 15' 0''$ 得 112.25② Rad mode, 输入 112.25° , 得 ≈ 1.959

9.4

'All school teach chemistry'

Unit circle definitions



$$x = \cos(\theta) \quad y = \sin(\theta)$$

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

For θ where $0 < \theta < \frac{\pi}{2}$

9.5

Symmetry properties

Find exact value for different angle:

- ① Convert degree to Radian
- ② 根据 Radian 判断 which quadrant is the angle locating at
- ③ 根据 quadrant 和 angle 的前缀 (sin, tan, cos), 对照 exact value table 得出一个数 (exact value).

例: $\cos(300^\circ)$

① $\cos 300^\circ \times \frac{\pi}{180} = \frac{5\pi}{3}$

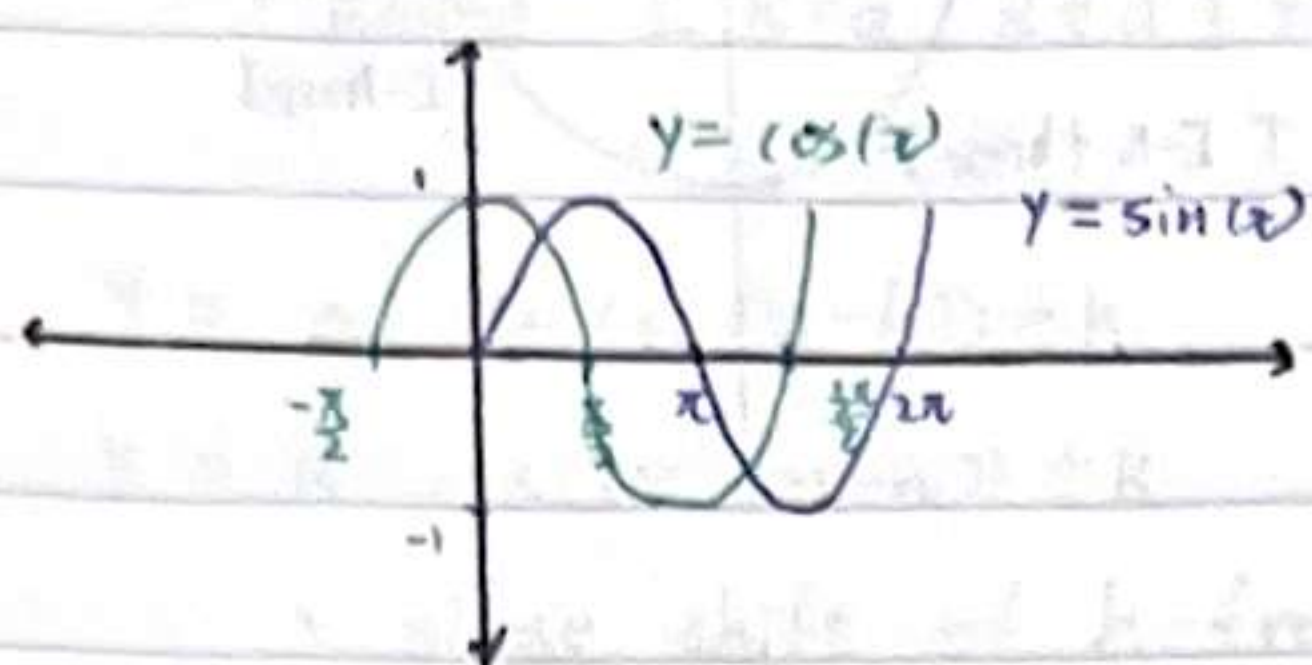
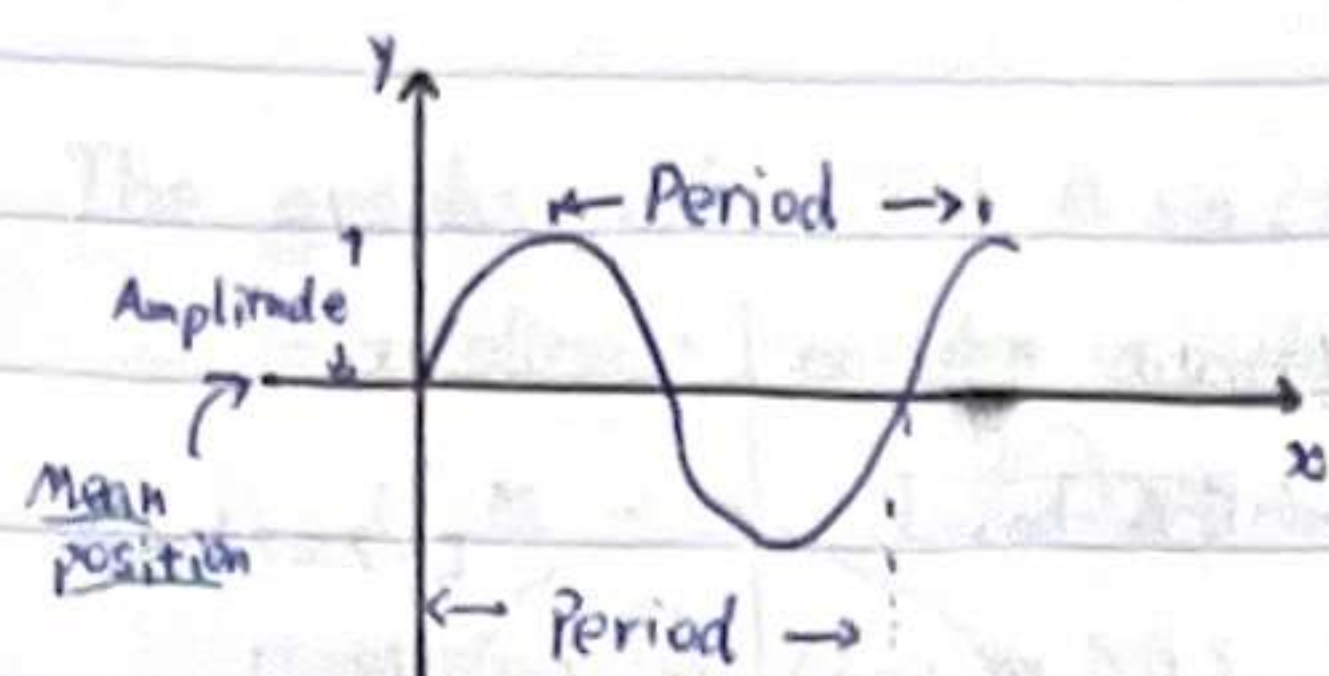
②  在 Q4, cos 是正的

③ $\cos(\frac{\pi}{3}) = \frac{1}{2}$

④ cos 是正的, $\frac{1}{2}$ 也是正的

CAS : Find exact value

① Rad mode 上 无论是 $\frac{\pi}{3}$ 还是 60° 都用 rad mode, 但 60° 必须加上 $^\circ$ 符号② Type $\sin(\frac{\pi}{3})$, $\cos(-30^\circ)$ 都可



$$\cos\left(x - \frac{\pi}{2}\right) = \sin(x)$$

$$\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$$

- The two functions are 'out of phase' by $\frac{\pi}{2}$ and have a phase difference or phase shift of $\frac{\pi}{2}$.
- The x-axis ($y=0$) is the 'equilibrium' or mean position.
- At x-axis (mean position) to a max or min point is called amplitude.
- Both functions: Amplitude: 1 unit

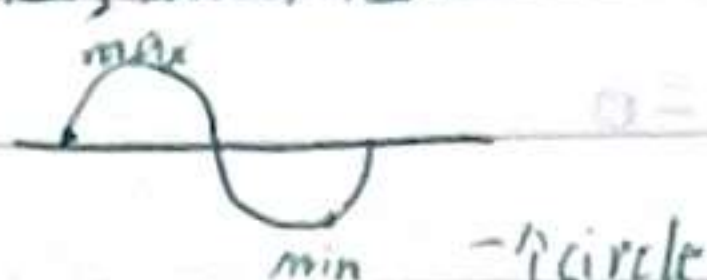
Period: 2π units

Type of correspondence: many to one

Domain: [图中最小的x值, 图中最大的x值]

Range: [最小y值, 最大y值]

- 一对 max 和 min 代表一个 circle, 如



例题: $\cos(x) - x^2 = 0$

(a) Determine the number of solutions to the equation

The number of solutions are determined by the number of intersections of the graph of $y = \cos(x)$ and $y = x^2$.